

On some equations concerning the Ramanujan unpublished papers and the Instantons. New possible mathematical connections with various parameters of Theoretical Physics, String Theory and Supersymmetry Breaking. III

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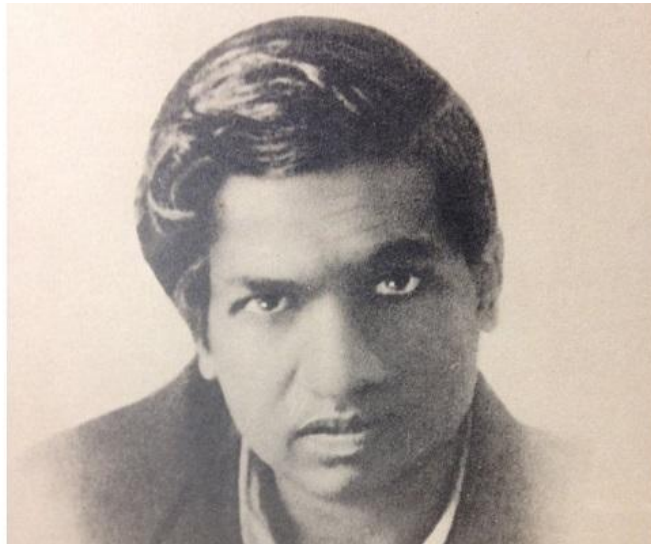
Abstract

In this research thesis (part III), we analyze some equations concerning the Ramanujan unpublished papers and the Instantons. We describe the new possible mathematical connections with various parameters of Theoretical Physics, String Theory and Supersymmetry Breaking.

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Srinivasa Ramanujan (1887-1920)



<https://www.moduscc.it/ramanujan-il-grande-matematico-indiano-13453-131115/>

Vesuvius landscape with gorse – Naples



<https://www.pinterest.it/pin/95068242114589901/>

From:

A. A. Karatsuba, “On the zeros of arithmetic Dirichlet series without Euler product,” *Izv. Ross. Akad. Nauk, Ser. Mat.* 57 (5), 3–14 (1993)

On the Zeros of the Davenport Heilbronn Function

S. A. Gritsenko - Received May 15, 2016 - ISSN 0081-5438, Proceedings of the Steklov Institute of Mathematics, 2017, Vol. 296, pp. 65–87.

We have:

Let

$$\kappa = \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}$$

and χ_1 be a character modulo 5 such that $\chi_1(2) = i$.

The Davenport–Heilbronn function $f(s)$ is defined by the equality

$$f(s) = \frac{1 - i\kappa}{2} L(s, \chi_1) + \frac{1 + i\kappa}{2} L(s, \bar{\chi}_1), \quad \text{where} \quad L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

The function $f(s)$ satisfies the Riemann-type functional equation

$$g(s) = g(1 - s), \quad \text{where} \quad g(s) = \left(\frac{\pi}{5}\right)^{-s/2} \Gamma\left(\frac{s+1}{2}\right) f(s),$$

but there is no Euler product for it.

$$(\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1) = \kappa$$

Input:

$$\frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}$$

Decimal approximation:

0.2840790438404122960282918323931261690910880884457375827591626661

...

0.28407904384.... = κ

Alternate forms:

$$\frac{1}{4} \left(\sqrt{10 - 2\sqrt{5}} - 2\sqrt{5} + \sqrt{5(10 - 2\sqrt{5})} - 2 \right)$$

$$\frac{1}{4} (1 + \sqrt{5}) \left(\sqrt{10 - 2\sqrt{5}} - 2 \right)$$

$$\frac{1}{2} \left(-1 - \sqrt{5} + \sqrt{2(5 + \sqrt{5})} \right)$$

Minimal polynomial:

$$x^4 + 2x^3 - 6x^2 - 2x + 1$$

Expanded forms:

$$\frac{\sqrt{10 - 2\sqrt{5}}}{\sqrt{5} - 1} - \frac{2}{\sqrt{5} - 1}$$

$$\frac{1}{4} \sqrt{10 - 2\sqrt{5}} + \frac{1}{4} \sqrt{5(10 - 2\sqrt{5})} + \frac{1}{2} (-1 - \sqrt{5})$$

For $((((\sqrt{10-2\sqrt{5}}-2))(\sqrt{5}-1)))) = 8\pi G$; $G = 0.011303146014$

Indeed:

$$(((\sqrt{10-2\sqrt{5}}-2))(\sqrt{5}-1)))/(8\pi)$$

Input:

$$\frac{\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}}{8\pi}$$

Result:

$$\frac{\sqrt{10-2\sqrt{5}}-2}{8(\sqrt{5}-1)\pi}$$

Decimal approximation:

0.0113031460140052147973750129442035744685760313920017808594909667
...

0.01130314.... = g (gravitational coupling constant)

Property:

$$\frac{-2 + \sqrt{10-2\sqrt{5}}}{8(-1+\sqrt{5})\pi} \text{ is a transcendental number}$$

Alternate forms:

$$\frac{\sqrt{10-2\sqrt{5}}-2\sqrt{5}+\sqrt{5(10-2\sqrt{5})}-2}{32\pi}$$

$$-\frac{1 + \sqrt{5} - \sqrt{2(5 + \sqrt{5})}}{16\pi}$$

$$\frac{-1 - \sqrt{5} + \sqrt{2(5 + \sqrt{5})}}{16\pi}$$

Expanded forms:

$$-\frac{1}{16\pi} - \frac{\sqrt{5}}{16\pi} + \frac{\sqrt{10 - 2\sqrt{5}}}{32\pi} + \frac{\sqrt{5(10 - 2\sqrt{5})}}{32\pi}$$

$$\frac{\sqrt{10 - 2\sqrt{5}}}{8(\sqrt{5} - 1)\pi} - \frac{1}{4(\sqrt{5} - 1)\pi}$$

Series representations:

$$\frac{\sqrt{10 - 2\sqrt{5}} - 2}{(8\pi)(\sqrt{5} - 1)} = \frac{-2 + \sqrt{9 - 2\sqrt{5}} \sum_{k=0}^{\infty} \binom{\frac{1}{2}}{k} (9 - 2\sqrt{5})^{-k}}{8\pi \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} \right)}$$

$$\frac{\sqrt{10 - 2\sqrt{5}} - 2}{(8\pi)(\sqrt{5} - 1)} = \frac{-2 + \sqrt{9 - 2\sqrt{5}} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (9 - 2\sqrt{5})^{-k}}{k!}}{8\pi \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{4}\right)^k \left(-\frac{1}{2}\right)_k}{k!} \right)}$$

$$\frac{\sqrt{10-2\sqrt{5}}-2}{(8\pi)(\sqrt{5}-1)} = \frac{-2+\sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (10-2\sqrt{5}-z_0)^k z_0^{-k}}{k!}}{8\pi \left(-1+\sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (5-z_0)^k z_0^{-k}}{k!}\right)}$$

for (not ($z_0 \in \mathbb{R}$ and $-\infty < z_0 \leq 0$))

We note that:

$$(((\sqrt{10-2\sqrt{5}}-2))/(\sqrt{5}-1))*((2i(\sqrt{5}-1)t + \sqrt{5}-1)/(2(\sqrt{2(5-\sqrt{5})}-2))) - 2)))$$

Input:

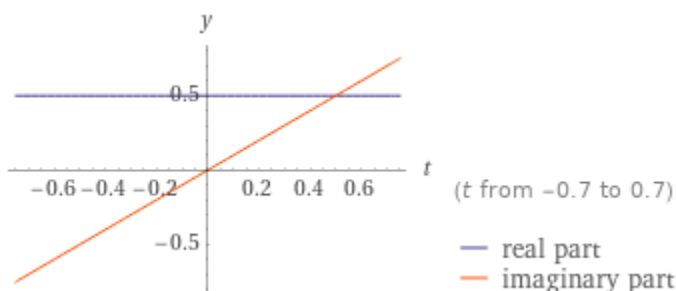
$$\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1} \times \frac{2i(\sqrt{5}-1)t + \sqrt{5}-1}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)}$$

i is the imaginary unit

Exact result:

$$\frac{\left(\sqrt{10-2\sqrt{5}}-2\right)\left(2i\left(\sqrt{5}-1\right)t + \sqrt{5}-1\right)}{2\left(\sqrt{5}-1\right)\left(\sqrt{2\left(5-\sqrt{5}\right)}-2\right)}$$

Plot:



Alternate form assuming $t > 0$:

$$\begin{aligned} & \frac{i \sqrt{10 - 2\sqrt{5}} t}{\sqrt{2(5 - \sqrt{5})} - 2} - \frac{2 i t}{\sqrt{2(5 - \sqrt{5})} - 2} + \\ & \frac{\sqrt{5(10 - 2\sqrt{5})}}{2(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} - \frac{\sqrt{10 - 2\sqrt{5}}}{2(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} - \\ & \frac{\sqrt{5}}{(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} + \frac{1}{(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} \end{aligned}$$

Alternate forms:

$$\frac{1}{8} (1 + \sqrt{5}) \left(2 i \sqrt{2(3 - \sqrt{5})} t + \sqrt{5} - 1 \right)$$

$$\frac{1}{2} (1 + 2 i t)$$

$$\frac{1}{2} + i t$$

$1/2 + it$ = real part of every nontrivial zero of the Riemann zeta function

Derivative:

$$\frac{d}{dt} \left(\frac{\left(\sqrt{10 - 2\sqrt{5}} - 2 \right) (2 i (\sqrt{5} - 1) t + \sqrt{5} - 1)}{(\sqrt{5} - 1) \left(2 \left(\sqrt{2(5 - \sqrt{5})} - 2 \right) \right)} \right) = i$$

Indefinite integral:

$$\int \frac{(\sqrt{10-2\sqrt{5}}-2)(2i(\sqrt{5}-1)t+\sqrt{5}-1)}{(\sqrt{5}-1)\left(2\left(\sqrt{2(5-\sqrt{5})}-2\right)\right)} dt = \frac{t}{2} + \frac{it^2}{2} + \text{constant}$$

And again:

$$(((\sqrt{10-2\sqrt{5}}-2))/(2x))) * ((2i(\sqrt{5}-1)t+\sqrt{5}-1)/(2(\sqrt{2(5-\sqrt{5})}-2))) = (1/2+it)$$

Input:

$$\frac{\sqrt{10-2\sqrt{5}}-2}{2x} \times \frac{2i(\sqrt{5}-1)t+\sqrt{5}-1}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)} = \frac{1}{2} + it$$

i is the imaginary unit

Exact result:

$$\frac{(\sqrt{10-2\sqrt{5}}-2)(2i(\sqrt{5}-1)t+\sqrt{5}-1)}{4\left(\sqrt{2(5-\sqrt{5})}-2\right)x} = \frac{1}{2} + it$$

Alternate form assuming t and x are real:

$$\frac{\sqrt{5}-1}{x} = 2$$

Alternate form:

$$\frac{(\sqrt{5} - 1)(1 + 2it)}{4x} = \frac{1}{2} + it$$

Alternate form assuming t and x are positive:

$$2x + 1 = \sqrt{5}$$

Expanded forms:

$$\begin{aligned} & \frac{i\sqrt{5(10-2\sqrt{5})}t}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \frac{i\sqrt{10-2\sqrt{5}}t}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \frac{i\sqrt{5}t}{\left(\sqrt{2(5-\sqrt{5})}-2\right)x} + \\ & \frac{it}{\left(\sqrt{2(5-\sqrt{5})}-2\right)x} + \frac{\sqrt{5(10-2\sqrt{5})}}{4\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \frac{\sqrt{10-2\sqrt{5}}}{4\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \\ & \frac{\sqrt{5}}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} + \frac{1}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} = \frac{1}{2} + it \end{aligned}$$

$$\frac{i\sqrt{5}t}{2x} - \frac{it}{2x} + \frac{\sqrt{5}}{4x} - \frac{1}{4x} = \frac{1}{2} + it$$

Solutions:

$$t = \frac{i}{2}, \quad x \neq 0$$

$$x = \frac{\sqrt{5}}{2} - \frac{1}{2}$$

Input:

$$\frac{\sqrt{5}}{2} - \frac{1}{2}$$

Decimal approximation:

0.6180339887498948482045868343656381177203091798057628621354486227

...

$$0.6180339887\dots = \frac{1}{\phi}$$

Solution for the variable x:

$$x = \frac{-2i\sqrt{5}t + 2it - \sqrt{5} + 1}{-2 - 4it}$$

Implicit derivatives:

$$\frac{\partial x(t)}{\partial t} = \frac{2(-1 + \sqrt{5} - 2x)x}{(-1 + \sqrt{5})(-i + 2t)}$$

$$\frac{\partial t(x)}{\partial x} = \frac{(-1 + \sqrt{5})(-i + 2t)}{2(-1 + \sqrt{5} - 2x)x}$$

From:

HOW INSTANTONS SOLVE THE U(1) PROBLEM - *G. 't HOOFT* - Institute for Theoretical Physics, Princetonplein 5, P.O. Box 80.006, 3508 TA Utrecht, The Netherlands - PHYSICS REPORTS (Review Section of Physics Letters) 142, No. 6 (1986) 357—387. North-Holland, Amsterdam - Received 18 April 1986

We have that:

The rest of the procedure to sum all contributions of the eigenmodes of the operators $\mathcal{M}_A$, $\mathcal{M}_\psi$, $\mathcal{M}_{gh}$ and possible scalar contributions $\mathcal{M}_s$, and the contributions of the collective coordinates is all as explained in [19]. The result is now the effective Lagrangian (for the case that the color gauge group is SU(2)):

$$\begin{aligned} \mathcal{L}^{\text{eff}}(z) = & 2^{10+3N^f} \pi^{6+2N^f} g^{-8} \int \rho^{3N^f-5} d\rho \exp\left\{-\frac{8\pi^2}{[g_R^D(\mu)]^2}\right. \\ & + \log(\mu\rho) \left[\frac{22}{3} - \frac{1}{6} \sum_t N^s(t) C(t) - \frac{2}{3} N^f \right] + A - \sum_t N^s(t) A(t) - N^f B \Big\} \\ & \times \left\langle \prod_{s=1}^{N^f} (\bar{\psi}_s \omega)(\bar{\omega} \psi_s) \right\rangle + \text{h.c.} , \end{aligned} \quad (\text{B13})$$

where N^f is the number of fermions in the doublet representation, ρ is a scalar parameter for the instanton, μ is an arbitrary mass unit enabling us to obtain a renormalization group invariant expression, and $N^s(t)$ is the number of scalar field representations with color t .

Defining coefficients $\alpha(t)$ as in table 1, we have now:

$$\begin{aligned} A &= -\alpha(1) + \frac{11}{3} (\log 4\pi - \gamma) + \frac{1}{3} = 7.053\,991\,03 \\ A(t) &= -\alpha(t) + \frac{1}{12} (\log 4\pi - \gamma) C(t) \\ A(1/2) &= 0.308\,690\,69 \\ A(1) &= 1.094\,576\,62 \\ A(3/2) &= 2.481\,356\,10 \\ B &= -2\alpha(1/2) + \frac{1}{3} (\log 4\pi - \gamma) = 0.359\,522\,90 . \end{aligned} \quad (\text{B14})$$

Table 1

t	$C(t)$	$\alpha(t)$
0	0	0
1/2	1	$2R - \frac{1}{3} \log 2 - 17/72$
1	4	$8R + \frac{1}{3} \log 2 - 16/9$
3/2	10	$20R + 4 \log 3 - \frac{1}{3} \log 2 - 265/36$

$$R = \frac{1}{12} (\log 2\pi + \gamma) + \frac{1}{2\pi^2} \sum_s \frac{\log s}{s^2} = 0.248\,754\,477.$$

For:

$$A = 7.05399103 \quad B = 0.35952290 \quad N^s(t) = 8 \quad N^f = 2$$

$$A(t) = 0.30869069 \quad C(t) = 1 \quad \rho = 1 \quad n \geq 0, \quad n = 2; \quad \mu = 16; \quad a_1 = 10$$

$$g_B^2(n) = 768^2 \quad g = 0.30282212$$

From:

$$\begin{aligned} \mathcal{L}^{\text{eff}}(z) = & 2^{10+3N^f} \pi^{6+2N^f} g^{-8} \int \rho^{3N^f-5} d\rho \exp\left\{-\frac{8\pi^2}{[g_R^D(\mu)]^2}\right. \\ & \left. + \log(\mu\rho) \left[\frac{22}{3} - \frac{1}{6} \sum_t N^s(t) C(t) - \frac{2}{3} N^f \right] + A - \sum_t N^s(t) A(t) - N^f B \right\} \\ & \times \left\langle \prod_{s=1}^{N^f} (\bar{\psi}_s \omega)(\bar{\omega} \psi_s) \right\rangle + \text{h.c.}, \end{aligned} \quad (\text{B13})$$

$$\begin{aligned} & 2^{10+3N^f} \pi^{6+2N^f} g^{-8} \int \rho^{3N^f-5} d\rho \exp\left\{-\frac{8\pi^2}{[g_R^D(\mu)]^2}\right. \\ & \left. + \log(\mu\rho) \left[\frac{22}{3} - \frac{1}{6} \sum_t N^s(t) C(t) - \frac{2}{3} N^f \right] + A - \sum_t N^s(t) A(t) - N^f B \right\} \\ & \times \left\langle \prod_{s=1}^{N^f} (\bar{\psi}_s \omega)(\bar{\omega} \psi_s) \right\rangle + \text{h.c.}, \end{aligned}$$

$$(2^{16} * \pi^{10} * (0.30282212)^{-8}) \int \text{Exp}(((((-8\pi^2)/(768^2))+\ln(16) (22/3-1/6*8*1-2/3*2)+7.05399103-(8*0.30869069-2*0.35952290))))]x$$

Input interpretation

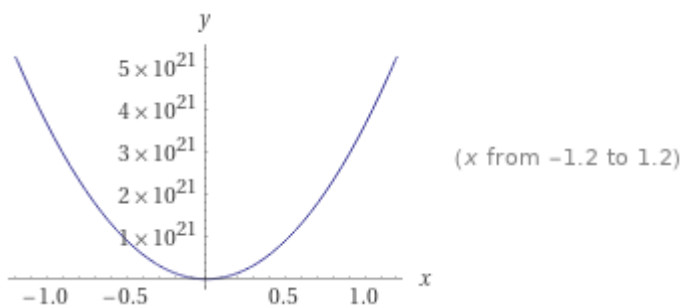
$$\frac{2^{16} \pi^{10}}{0.30282212^8} \int \exp\left(\frac{-8 \pi^2}{768^2} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 - \frac{2}{3} \times 2\right) + 7.05399103 - (8 \times 0.30869069 + 2 \times (-0.35952290))\right) x dx$$

$\log(x)$ is the natural logarithm

Result

$$3.62996 \times 10^{21} x^2$$

Plot



Alternate form assuming x is real

$$3.62996 \times 10^{21} x^2 + 0$$

Indefinite integral assuming all variables are real

$$1.20999 \times 10^{21} x^3 + \text{constant}$$

$$3.62996 \times 10^{21} (1.2^2) * x = y$$

Input interpretation

$$3.62996 \times 10^{21} \times 1.2^2 x = y$$

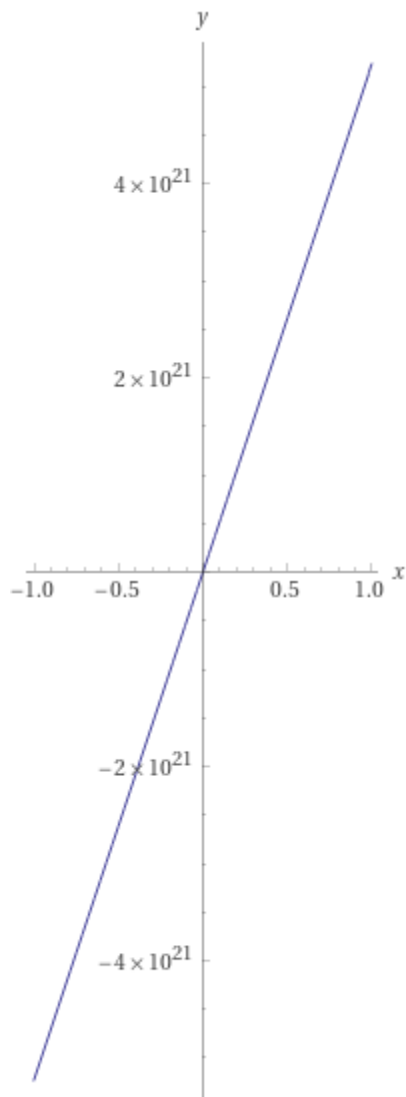
Result

$$5.22714 \times 10^{21} x = y$$

Geometric figure

line

Implicit plot



Alternate forms

$$x = 1.91309 \times 10^{-22} y$$

$$5.22714 \times 10^{21} x - y = 0$$

Alternate form assuming x and y are real

$$5.22714 \times 10^{21} x + 0 = y$$

Real solution

$$y \approx 5.22714 \times 10^{21} x + 0$$

Solution

$$y = 5\,227\,142\,400\,000\,000\,000\,000\,x$$

Integer solution

$$x = 0, \quad y = 0$$

Partial derivatives

$$\frac{\partial}{\partial x}(5\,227\,142\,400\,000\,000\,000\,x) = 5\,227\,142\,400\,000\,000\,000$$

$$\frac{\partial}{\partial y}(5\,227\,142\,400\,000\,000\,000\,000\,x) = 0$$

Considering:

$$\left\langle \prod_{s=1}^{N_f} (\bar{\psi}_s \omega)(\bar{\omega} \psi_s) \right\rangle + \text{h.c.}$$

= 1/2 , and x = 1.2 , from:

$$3.62996 \times 10^{21} x^2$$

we obtain:

$$3.62996 \times 10^{21} (1.2^2) * 0.5$$

Input interpretation

$$3.62996 \times 10^{21} \times 1.2^2 \times 0.5$$

Result

2613571200000000000000

Scientific notation

2.6135712×10^{21}

$2.6135712 * 10^{21}$

Indeed:

$((((1.2^2 * 1/2) (2^{16} * \pi^{10} * (0.30282212)^{-8}) \text{ integrate } [\text{Exp}((((-8\pi^2)/(768^2)) + \ln(16) (22/3 - 1/6 * 8 * 1 - 2/3 * 2) + 7.05399103 - (8 * 0.30869069 - 2 * 0.35952290)))))]x))$

Input interpretation

$$\left(1.2^2 \times \frac{1}{2}\right) \times \frac{2^{16} \pi^{10}}{0.30282212^8} \int \exp\left(\frac{-8\pi^2}{768^2} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 - \frac{2}{3} \times 2\right) + 7.05399103 - (8 \times 0.30869069 + 2 \times (-0.35952290))\right) x \, dx$$

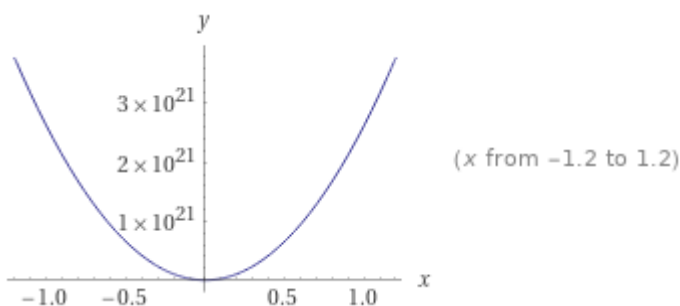
$\log(x)$ is the natural logarithm

Result

$2.61357 \times 10^{21} x^2$

$2.61357 * 10^{21}$

Plot



Alternate form assuming x is real

$2.61357 \times 10^{21} x^2 + 0$

Indefinite integral assuming all variables are real

$$8.71191 \times 10^{20} x^3 + \text{constant}$$

From which:

$$\left((e^{(1+2e+1/\pi-2\pi)} \pi^{(2+2e)} \csc(e \pi)) \left(((1.2^{2*1/2}) (2^{16} * \pi^{10} * (0.30282212)^{-8}) \int \exp\left(\frac{-8\pi^2}{768^2}\right) + \ln(16) \left(\frac{22}{3} - \frac{1}{6} * 8 * 1 - \frac{2}{3} * 2\right) + 7.05399103 - (8 * 0.30869069 - 2 * 0.35952290)) \right) dx \right)$$

where

$$e^{1+2e+1/\pi-2\pi} \pi^{2+2e} \csc(e \pi) \approx 10309.307017$$

Input interpretation

$$\left(e^{1+2e+1/\pi-2\pi} \pi^{2+2e} \csc(e \pi) \left(\left(1.2^2 \times \frac{1}{2} \right) \times \frac{2^{16} \pi^{10}}{0.30282212^8} \int \exp\left(\frac{-8 \pi^2}{768^2} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 - \frac{2}{3} \times 2\right) + 7.05399103 - (8 \times 0.30869069 + 2 \times (-0.35952290)) \right) x dx \right) \right)$$

$\csc(x)$ is the cosecant function

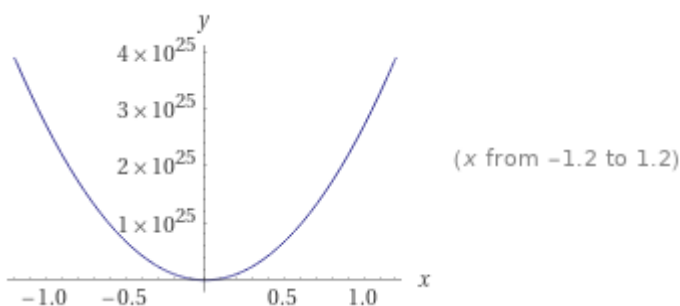
$\log(x)$ is the natural logarithm

Result

$$2.69441 \times 10^{25} x^2$$

$2.69441 * 10^{25} \approx c^3 = 2.6944002417 \times 10^{25}$, where c is the speed of light 299792458

Plot



Alternate form assuming x is real

$$2.69441 \times 10^{25} x^2 + 0$$

Indefinite integral assuming all variables are real

$$8.98137 \times 10^{24} x^3 + \text{constant}$$

$((((1.2^2 \times \frac{1}{2})(2^{16} * \text{Pi}^{10} * (0.30282212)^{-8}) \text{integrate} [\text{Exp}((((- 8\text{Pi}^2)/(768^2)) + \ln(16) (22/3 - 1/6 * 8 * 4 - 2/3 * 2) + 7.05399103 - (8 * 1.09457662 - 2 * 0.35952290)))))]x))$

Input interpretation

$$\left(1.2^2 \times \frac{1}{2}\right) \times \frac{2^{16} \pi^{10}}{0.30282212^8} \int \exp\left(\frac{-8 \pi^2}{768^2} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 4 - \frac{2}{3} \times 2\right) + 7.05399103 - (8 \times 1.09457662 + 2 \times (-0.35952290))\right) x \, dx$$

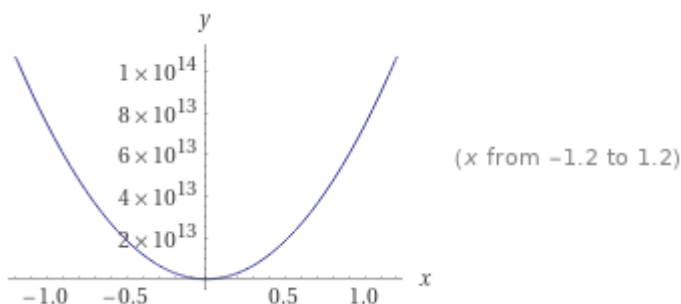
$\log(x)$ is the natural logarithm

Result

$$7.41835 \times 10^{13} x^2$$

$$7.41835 \times 10^{13}$$

Plot



Alternate form assuming x is real

$$7.41835 \times 10^{13} x^2 + 0$$

Indefinite integral assuming all variables are real

$$2.47278 \times 10^{13} x^3 + \text{constant}$$

(((1.2^2*1/2)(2^16 * Pi^10 * (0.30282212)^-8) integrate [Exp((((-8Pi^2)/(768^2))+ln(16) (22/3-1/6*8*10-2/3*2)+7.05399103-(8*2.48135610-2*0.35952290))))]x))

Input interpretation

$$\left(1.2^2 \times \frac{1}{2}\right) \times \frac{2^{16} \pi^{10}}{0.30282212^8} \int \exp\left(\frac{-8 \pi^2}{768^2} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 10 - \frac{2}{3} \times 2\right) + 7.05399103 - (8 \times 2.48135610 + 2 \times (-0.35952290))\right) x \, dx$$

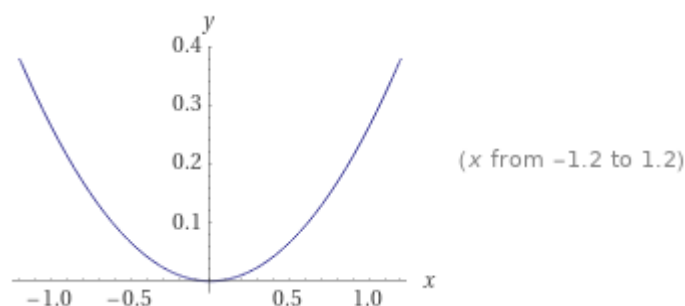
$\log(x)$ is the natural logarithm

Result

$$0.262532 x^2$$

0.262532

Plot



Alternate form assuming x is real

$$0.262532 x^2 + 0$$

Indefinite integral assuming all variables are real

$$0.0875106 x^3 + \text{constant}$$

We have the following six results:

2.61357*10²¹ ; 7.41835*10¹³ ; 0.262532 and 0.061126894 ; 0.38469373 ;
1.2168198...

For the following expressions:

$$2R - \frac{1}{8} \log 2 - 17/72$$

$$8R + \frac{1}{3} \log 2 - 16/9$$

$$20R + 4 \log 3 - \frac{5}{3} \log 2 - 265/36$$

From:

$$A(t) = -\alpha(t) + \frac{1}{12} (\log 4\pi - \gamma) C(t)$$

$$-(2(0.248754477) - 1/6 \ln(2) - 17/72) + 1/12 \ln(4\pi - 0.5772156649) \times 1$$

Input interpretation

$$-\left(2 \times 0.248754477 - \frac{1}{6} \log(2) - \frac{17}{72}\right) + \frac{1}{12} \log(4\pi - 0.5772156649) \times 1$$

$\log(x)$ is the natural logarithm

Result

0.061126894...

0.061126894...

Alternative representations

$$\begin{aligned} & -\left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72}\right) + \frac{1}{12} \log(4\pi - 0.577216) = \\ & -0.497509 + \frac{1}{6} \log(a) \log_a(2) + \frac{1}{12} \log(a) \log_a(-0.577216 + 4\pi) + \frac{17}{72} \end{aligned}$$

$$\begin{aligned}
& -\left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72}\right) + \frac{1}{12} \log(4\pi - 0.577216) = \\
& -0.497509 + \frac{\log_e(2)}{6} + \frac{1}{12} \log_e(-0.577216 + 4\pi) + \frac{17}{72}
\end{aligned}$$

$$\begin{aligned}
& -\left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72}\right) + \frac{1}{12} \log(4\pi - 0.577216) = \\
& -0.497509 - \frac{\text{Li}_1(-1)}{6} - \frac{1}{12} \text{Li}_1(1.57722 - 4\pi) + \frac{17}{72}
\end{aligned}$$

Series representations

$$\begin{aligned}
& -\left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72}\right) + \frac{1}{12} \log(4\pi - 0.577216) = \\
& -0.261398 + 0.333333 i \pi \left[\frac{\arg(2-x)}{2\pi} \right] + \\
& 0.166667 i \pi \left[\frac{\arg(-0.577216 + 4\pi - x)}{2\pi} \right] + 0.25 \log(x) + \\
& \sum_{k=1}^{\infty} \frac{(-1)^k \left(-0.166667 (2-x)^k - 0.0833333 (-0.577216 + 4\pi - x)^k \right) x^{-k}}{k} \text{ for } \\
& x < 0
\end{aligned}$$

$$\begin{aligned}
& -\left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72}\right) + \frac{1}{12} \log(4\pi - 0.577216) = \\
& -0.261398 + 0.333333 i \pi \left[-\frac{-\pi + \arg\left(\frac{2}{z_0}\right) + \arg(z_0)}{2\pi} \right] + \\
& 0.166667 i \pi \left[-\frac{-\pi + \arg\left(\frac{4(-0.144304+\pi)}{z_0}\right) + \arg(z_0)}{2\pi} \right] + 0.25 \log(z_0) + \\
& \sum_{k=1}^{\infty} \frac{(-1)^k \left(-0.166667 (2-z_0)^k - 0.0833333 (-0.577216 + 4\pi - z_0)^k \right) z_0^{-k}}{k}
\end{aligned}$$

$$\begin{aligned}
& -\left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72}\right) + \frac{1}{12} \log(4\pi - 0.577216) = -0.261398 + \\
& \frac{1}{6} \left\lfloor \frac{\arg(2 - z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \frac{1}{12} \left\lfloor \frac{\arg(-0.577216 + 4\pi - z_0)}{2\pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \\
& \frac{\log(z_0)}{4} + \frac{1}{6} \left\lfloor \frac{\arg(2 - z_0)}{2\pi} \right\rfloor \log(z_0) + \frac{1}{12} \left\lfloor \frac{\arg(-0.577216 + 4\pi - z_0)}{2\pi} \right\rfloor \log(z_0) + \\
& \sum_{k=1}^{\infty} -\frac{(-1)^k (2(2 - z_0)^k + (-0.577216 + 4\pi - z_0)^k) z_0^{-k}}{12k}
\end{aligned}$$

Integral representations

$$\begin{aligned}
& -\left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72}\right) + \frac{1}{12} \log(4\pi - 0.577216) = \\
& -0.261398 + \int_1^2 \frac{0.107384 + \pi(-0.166667 + 0.25t) - 0.098576t}{(0.644304 + \pi(-1 + t) - 0.394304t)t} dt
\end{aligned}$$

$$\begin{aligned}
& -\left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72}\right) + \frac{1}{12} \log(4\pi - 0.577216) = \\
& -0.261398 + \int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{0.0833333(-1.57722 + 4\pi)^{-s} (0.5 + (-1.57722 + 4\pi)^s) \Gamma(-s)^2 \Gamma(1+s)}{i\pi \Gamma(1-s)} \\
& ds \text{ for } -1 < \gamma < 0
\end{aligned}$$

$$-(8(0.248754477)+1/3 \ln(2) -16/9) + 1/12 \ln(4\pi-0.5772156649)*4$$

Input interpretation

$$-\left(8 \times 0.248754477 + \frac{1}{3} \log(2) - \frac{16}{9}\right) + \frac{1}{12} \log(4\pi - 0.5772156649) \times 4$$

$\log(x)$ is the natural logarithm

Result

0.38469373...

0.38469373

Alternative representations

$$-\left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{12} \log(4\pi - 0.577216) 4 =$$

$$-1.99004 - \frac{1}{3} \log(a) \log_a(2) + \frac{16}{9} + \frac{4}{12} \log(a) \log_a(-0.577216 + 4\pi)$$

$$-\left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{12} \log(4\pi - 0.577216) 4 =$$

$$-1.99004 - \frac{\log_e(2)}{3} + \frac{16}{9} + \frac{4}{12} \log_e(-0.577216 + 4\pi)$$

$$-\left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{12} \log(4\pi - 0.577216) 4 =$$

$$-1.99004 + \frac{\text{Li}_1(-1)}{3} + \frac{16}{9} - \frac{4}{12} \text{Li}_1(1.57722 - 4\pi)$$

Series representations

$$-\left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{12} \log(4\pi - 0.577216) 4 = -0.212258 -$$

$$0.666667 i \pi \left[\frac{\arg(2-x)}{2\pi} \right] + 0.666667 i \pi \left[\frac{\arg(-0.577216 + 4\pi - x)}{2\pi} \right] +$$

$$\sum_{k=1}^{\infty} \frac{(-1)^k (0.333333 (2-x)^k - 0.333333 (-0.577216 + 4\pi - x)^k) x^{-k}}{k} \text{ for } x <$$

$$0$$

$$-\left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{12} \log(4\pi - 0.577216) 4 =$$

$$-0.212258 - 0.666667 i \pi \left[-\frac{-\pi + \arg\left(\frac{2}{z_0}\right) + \arg(z_0)}{2\pi} \right] +$$

$$0.666667 i \pi \left[-\frac{-\pi + \arg\left(\frac{4(-0.144304+\pi)}{z_0}\right) + \arg(z_0)}{2\pi} \right] +$$

$$\sum_{k=1}^{\infty} \frac{(-1)^k (0.333333 (2-z_0)^k - 0.333333 (-0.577216 + 4\pi - z_0)^k) z_0^{-k}}{k}$$

$$\begin{aligned}
& -\left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{12} \log(4\pi - 0.577216) 4 = \\
& -0.212258 - \frac{1}{3} \left[\frac{\arg(2 - z_0)}{2\pi} \right] \log\left(\frac{1}{z_0}\right) + \frac{1}{3} \left[\frac{\arg(-0.577216 + 4\pi - z_0)}{2\pi} \right] \log\left(\frac{1}{z_0}\right) - \\
& \frac{1}{3} \left[\frac{\arg(2 - z_0)}{2\pi} \right] \log(z_0) + \frac{1}{3} \left[\frac{\arg(-0.577216 + 4\pi - z_0)}{2\pi} \right] \log(z_0) + \\
& \sum_{k=1}^{\infty} \frac{(-1)^k \left((2 - z_0)^k - (-0.577216 + 4\pi - z_0)^k \right) z_0^{-k}}{3k}
\end{aligned}$$

Integral representations

$$\begin{aligned}
& -\left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{12} \log(4\pi - 0.577216) 4 = \\
& -0.212258 + \int_1^2 \frac{-0.214768 + 0.333333\pi}{(0.644304 + \pi(-1+t) - 0.394304t)t} dt
\end{aligned}$$

$$\begin{aligned}
& -\left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{12} \log(4\pi - 0.577216) 4 = \\
& -0.212258 + \int_{-i\infty+\gamma}^{i\infty+\gamma} - \\
& \frac{0.166667(-1.57722 + 4\pi)^{-s} (-1 + (-1.57722 + 4\pi)^s) \Gamma(-s)^2 \Gamma(1+s)}{i\pi \Gamma(1-s)} \\
& ds \text{ for } -1 < \gamma < 0
\end{aligned}$$

$$-(20(0.248754477)+4 \ln(3) - 5/3 \ln(2) -265/36) + 1/12 \ln(4\text{Pi}-0.5772156649)*10$$

Input interpretation

$$-\left(20 \times 0.248754477 + 4 \log(3) - \frac{5}{3} \log(2) - \frac{265}{36}\right) + \frac{1}{12} \log(4\pi - 0.5772156649) \times 10$$

$\log(x)$ is the natural logarithm

Result

1.2168198...

1.2168198...

Alternative representations

$$\begin{aligned} & -\left(20 \times 0.248754 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) + \frac{1}{12} \log(4 \pi - 0.577216) 10 = \\ & -4.97509 + \frac{5}{3} \log(a) \log_a(2) - 4 \log(a) \log_a(3) + \\ & \frac{10}{12} \log(a) \log_a(-0.577216 + 4 \pi) + \frac{265}{36} \end{aligned}$$

$$\begin{aligned} & -\left(20 \times 0.248754 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) + \frac{1}{12} \log(4 \pi - 0.577216) 10 = \\ & -4.97509 + \frac{5 \log_e(2)}{3} - 4 \log_e(3) + \frac{10}{12} \log_e(-0.577216 + 4 \pi) + \frac{265}{36} \end{aligned}$$

$$\begin{aligned} & -\left(20 \times 0.248754 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) + \frac{1}{12} \log(4 \pi - 0.577216) 10 = \\ & -4.97509 + 4 \operatorname{Li}_1(-2) - \frac{5 \operatorname{Li}_1(-1)}{3} - \frac{10}{12} \operatorname{Li}_1(1.57722 - 4 \pi) + \frac{265}{36} \end{aligned}$$

Series representations

$$\begin{aligned} & -\left(20 \times 0.248754 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) + \frac{1}{12} \log(4 \pi - 0.577216) 10 = \\ & 2.38602 + \frac{10}{3} i \pi \left[\frac{\arg(2-x)}{2 \pi} \right] - 8 i \pi \left[\frac{\arg(3-x)}{2 \pi} \right] + \\ & \frac{5}{3} i \pi \left[\frac{\arg(-0.577216 + 4 \pi - x)}{2 \pi} \right] - \frac{3 \log(x)}{2} + \\ & \sum_{k=1}^{\infty} - \frac{(-1)^k (10 (2-x)^k - 24 (3-x)^k + 5 (-0.577216 + 4 \pi - x)^k) x^{-k}}{6 k} \quad \text{for } x < 0 \end{aligned}$$

$$\begin{aligned}
& -\left(20 \times 0.248754 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) + \frac{1}{12} \log(4 \pi - 0.577216) 10 = \\
& 2.38602 + 3.33333 i \pi \left[-\frac{-\pi + \arg\left(\frac{2}{z_0}\right) + \arg(z_0)}{2 \pi} \right] - \\
& 8 i \pi \left[-\frac{-\pi + \arg\left(\frac{3}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + \\
& 1.66667 i \pi \left[-\frac{-\pi + \arg\left(\frac{4(-0.144304 + \pi)}{z_0}\right) + \arg(z_0)}{2 \pi} \right] - 1.5 \log(z_0) + \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k \\
& (-1.66667 (2 - z_0)^k + 4 (3 - z_0)^k - 0.833333 (-0.577216 + 4 \pi - z_0)^k) z_0^{-k}
\end{aligned}$$

$$\begin{aligned}
& -\left(20 \times 0.248754 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) + \frac{1}{12} \log(4 \pi - 0.577216) 10 = \\
& 2.38602 + \frac{5}{3} \left[\frac{\arg(2 - z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) - 4 \left[\frac{\arg(3 - z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + \\
& \frac{5}{6} \left[\frac{\arg(-0.577216 + 4 \pi - z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) - \frac{3 \log(z_0)}{2} + \frac{5}{3} \left[\frac{\arg(2 - z_0)}{2 \pi} \right] \log(z_0) - \\
& 4 \left[\frac{\arg(3 - z_0)}{2 \pi} \right] \log(z_0) + \frac{5}{6} \left[\frac{\arg(-0.577216 + 4 \pi - z_0)}{2 \pi} \right] \log(z_0) + \\
& \sum_{k=1}^{\infty} -\frac{(-1)^k (10 (2 - z_0)^k - 24 (3 - z_0)^k + 5 (-0.577216 + 4 \pi - z_0)^k) z_0^{-k}}{6 k}
\end{aligned}$$

Integral representations

$$\begin{aligned}
& -\left(20 \times 0.248754 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) + \frac{1}{12} \log(4 \pi - 0.577216) 10 = \\
& 2.38602 + \int_1^{-0.577216 + 4 \pi} \frac{0.48017 + 0.833333 \pi^2 + \pi (-1.28217 - 0.958333 t) + 0.461208 t - 0.1875 t^2}{(-0.644304 + \pi + 0.25 t) (-0.894304 + \pi + 0.5 t) t} dt
\end{aligned}$$

$$-\left(20 \times 0.248754 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) + \frac{1}{12} \log(4 \pi - 0.577216) 10 =$$

$$2.38602 + \int_{-i \infty + \gamma}^{i \infty + \gamma} \frac{1}{i \pi \Gamma(1-s)} e^{-2.07944 s} (-0.394304 + \pi)^{-s}$$

$$(0.416667 \times 2^s + 0.833333 \times 8^s (-0.394304 + \pi)^s - 2(-1.57722 + 4 \pi)^s)$$

$$\Gamma(-s)^2 \Gamma(1+s) ds \text{ for } -1 < \gamma < 0$$

$\Gamma(x)$ is the gamma function

From the following obtained results

2.61357×10^{21} ; 7.41835×10^{13} ; 0.262532 ; 0.061126894 ; 0.38469373 ;
1.2168198

after some calculations, we obtain:

$[(2.61357 \times 10^{21} / 7.41835 \times 10^{13}) \times 0.262532 \times 0.061126894 \times 0.38469373 \times$
 $1.2168198]^{1/3}$

Input interpretation

$$\sqrt[3]{\frac{2.61357 \times 10^{21}}{7.41835 \times 10^{13}} \times 0.262532 \times 0.061126894 \times 0.38469373 \times 1.2168198}$$

Result

64.2038...

[64.2038....](#)

$27(((2.61357 \times 10^{21} / 7.41835 \times 10^{13}) \times 0.262532 \times 0.061126894 \times 0.38469373 \times$
 $1.2168198]^{1/3})) - 5 - \frac{1}{2}$

Input interpretation

$$27 \sqrt[3]{\frac{2.61357 \times 10^{21}}{7.41835 \times 10^{13}} \times 0.262532 \times 0.061126894 \times 0.38469373 \times 1.2168198} - 5 - \frac{1}{2}$$

Result

1728.00...

1728

This result is very near to the mass of candidate glueball **$f_0(1710)$ scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

$$27[(2.61357 \times 10^{21} / 7.41835 \times 10^{13}) * 0.262532 * 0.061126894 * 0.38469373 * 1.2168198]^{1/3 - 5 + 1/2}$$

Input interpretation

$$27 \sqrt[3]{\frac{2.61357 \times 10^{21}}{7.41835 \times 10^{13}} \times 0.262532 \times 0.061126894 \times 0.38469373 \times 1.2168198 - 5 + \frac{1}{2}}$$

Result

1729.00...

1729 as above

$$(((27[(2.61357 \times 10^{21} / 7.41835 \times 10^{13}) * 0.262532 * 0.061126894 * 0.38469373 * 1.2168198]^{1/3 - 5 + 1/2})))^{1/15}$$

Input interpretation

$$\left(27 \sqrt[3]{\frac{2.61357 \times 10^{21}}{7.41835 \times 10^{13}} \times 0.262532 \times 0.061126894 \times 0.38469373 \times 1.2168198 - 5 + \frac{1}{2}} \right)^{(1/15)}$$

Result

1.643815...

$$1.643815\dots \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934\dots \text{ (trace of the instanton shape)}$$

$$(((2.61357 \times 10^{21} / 7.41835 \times 10^{13}) \times 0.262532 \times 0.061126894 \times 0.38469373 \times 1.2168198)^{1/3})^{2-26+276}$$

Input interpretation

$$\sqrt[3]{\frac{2.61357 \times 10^{21}}{7.41835 \times 10^{13}} \times 0.262532 \times 0.061126894 \times 0.38469373 \times 1.2168198}^{2-26+276}$$

Result

4372.13...

$$4372.13\dots \approx 4372$$

where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

We have that:

Finally, the spinors ω are normalized by

$$\sum_{\alpha} \omega_{\alpha} \bar{\omega}_{\alpha} = \frac{1}{2}(1 + \gamma_5) \quad (\text{B15})$$

and required to be smeared in color space, such that for instance

$$\langle \omega_{\alpha} \bar{\omega}_{\beta} \rangle = \frac{1}{4} \delta_{\alpha\beta} (1 + \gamma_5) \quad (\text{B16})$$

and, in the case $L = N^f = 2$:

$$\left\langle \prod_{s=1}^2 (\bar{\psi}_s \omega)(\bar{\omega} \psi_s) \right\rangle = \frac{1}{24} (2\delta_{\alpha_1}^{\beta_1} \delta_{\alpha_2}^{\beta_2} - \delta_{\alpha_1}^{\beta_2} \delta_{\alpha_2}^{\beta_1}) \epsilon^{st} \bar{\psi}_1^{\alpha_1} (1 + \gamma_5) \psi_s^{\beta_1} \bar{\psi}_2^{\alpha_2} (1 + \gamma_5) \psi_t^{\beta_2} \quad (\text{B17})$$

where s and t are flavor indices and α_i, β_i color indices.

The ρ integral may seem to diverge in most interesting cases ($N^f > 1$). Note however that it would be natural to choose

$$\mu = 1/\rho \quad (\text{B18})$$

and substitute g^{-8} by the running value $g(\mu)^{-8}$. At large ρ one might take $g \propto \rho$ and thus improve the convergence. Of course the infrared end of the integral is quite uncertain because in our perturbative procedure the effects of confinement etc. have not been taken into account. This inhibits a precise evaluation of the amplitude. A rough estimate (for a color SU(2) theory) is obtained if we take at large ρ

$$g^2(1/\rho) \rightarrow 16\pi\rho^2\sigma \quad (\text{B19})$$

where σ is the string constant. Then quarks with color charge 1/2 at a distance ρ from each other feel a force

$$\sigma = \frac{1}{4} g^2 / 4\pi\rho^2. \quad (\text{B20})$$

Our integral becomes, in the case $N^f = 2$,

$$\mathcal{Q}^{\text{eff}} = 2^{16} \pi^{10} e^{A-2B} \mathcal{Q}_1 \int \frac{\rho d\rho}{g^8(1/\rho)} \exp - 8\pi^2/g^2(1/\rho). \quad (\text{B21})$$

From (B19) we get

$$\rho d\rho \cong g dg / 16\pi\sigma \quad (\text{B22})$$

and using $x = 1/g^2$ the integral in (B21) is

$$\frac{1}{32\pi\sigma} \int_0^{\infty} dx x^2 e^{-8\pi^2 x} = 2^{-13} \pi^{-7} \sigma^{-1} \quad (\text{B23})$$

so that

$$\mathcal{L}^{\text{eff}} = 8\pi^3 e^{A-2B} \sigma^{-1} \mathcal{L}_1 \quad (\text{B24})$$

where $\mathcal{L}_1$ is the Lagrangian (B17).

This result is uncomfortably large, but then the approximations used here (eq. B19) could at best only be expected to yield the order of magnitude of the expected interaction, which is clearly a strong one. Note that we used a minimal subtraction scheme that included the term $\log 4\pi$ in (B9). If we left it out then (B24) would be reduced by a factor $(4\pi)^{-3} = 2^{-6}\pi^{-3}$. This is just to illustrate how sensitively the amplitude obtained depends upon the assumptions.

For:

$$A = 7.05399103 \quad B = 0.35952290$$

$$\sigma = \frac{1}{4} g^2 / 4\pi \rho^2 .$$

From:

$$\mathcal{L}^{\text{eff}} = 8\pi^3 e^{A-2B} \sigma^{-1} \mathcal{L}_1$$

we obtain:

$$2.6135712 \times 10^{21} = x * ((8\pi^3 * e^{(7.05399103 - 2 * 0.35952290)} * 1 / [(0.25 * 0.30282212^2) / (4\pi)]))$$

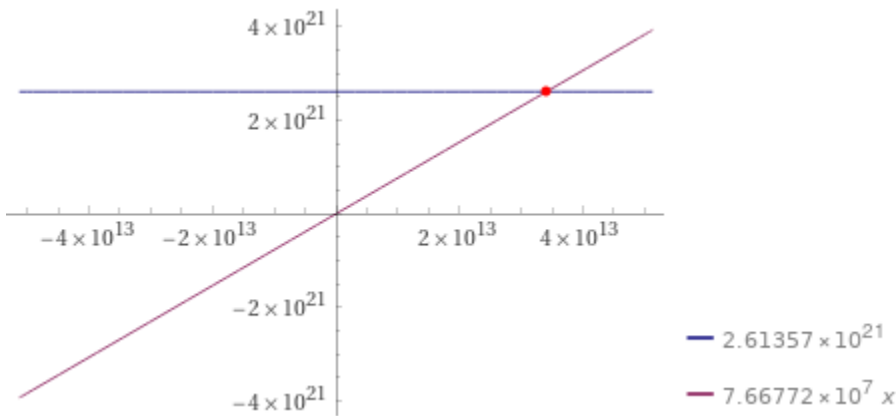
Input interpretation

$$2.6135712 \times 10^{21} = x \left(8\pi^3 e^{7.05399103 + 2 \times (-0.35952290)} \times \frac{1}{\frac{0.25 \times 0.30282212^2}{4\pi}} \right)$$

Result

$$2.61357 \times 10^{21} = 7.66772 \times 10^7 x$$

Plot



Alternate form

$$2.61357 \times 10^{21} - 7.66772 \times 10^7 x = 0$$

Alternate form assuming x is real

$$2.61357 \times 10^{21} = 7.66772 \times 10^7 x + 0$$

Solution

$$x \approx 3.40854 \times 10^{13}$$

$$3.40854 \times 10^{13}$$

From the following expression:

$$2^{14} \pi^6 g^{-8} \frac{d^4 z d\rho}{\rho^5} \exp \left\{ -\frac{8\pi^2}{g^2(\mu_0)} + \ln(\mu_0 \rho) \left[\frac{22}{3} - \frac{1}{6} \sum_t N^s(t) C(t) \right] - \alpha(1) - \sum_t N^s(t) \alpha(t) \right\}.$$

$$\frac{1}{0.30282212^8} 2^{14} \pi^6 \exp \left(\frac{-8\pi^2}{0.30282212^2 \times 16} + \right.$$

$$\log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 \right) - \left(8 \times 0.248754477033784 + \frac{1}{3} \log(2) - \frac{16}{9} \right) -$$

$$\left. 8 \left(2 \times 0.248754477033784 - \frac{1}{6} \log(2) - \frac{17}{72} \right) \right)$$

that is equal to $3.17769... \times 10^{-6}$, multiplying by the ratio between the two previous results, i.e. $2.6135712 \times 10^{21} / 3.40854 \times 10^{13}$

we obtain:

$$(2.6135712 \times 10^{21} / 3.40854 \times 10^{13}) 2^{14} \pi^6 (0.30282212)^{-8} (((\exp[(-8\pi^2)/(0.30282212^2 \times 16) + \ln(16)(\frac{22}{3} - \frac{1}{6} \times 8 \times 1) - (8 \times 0.2487544770337 + \frac{1}{3} \ln(2) - 16/9) - 8((2 \times 0.2487544770337 - \frac{1}{6} \ln(2) - \frac{17}{72})))]))$$

Input interpretation

$$\frac{2.6135712 \times 10^{21}}{3.40854 \times 10^{13}} \times \frac{2^{14} \pi^6}{0.30282212^8} \exp\left(\frac{-8\pi^2}{0.30282212^2 \times 16} + \log(16)\left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1\right) - \left(8 \times 0.2487544770337 + \frac{1}{3} \log(2) - \frac{16}{9}\right) - 8\left(2 \times 0.2487544770337 - \frac{1}{6} \log(2) - \frac{17}{72}\right)\right)$$

$\log(x)$ is the natural logarithm

Result

243.656...

243.656....

From which:

$$1/144 ((2.6135712 \times 10^{21} / 3.40854 \times 10^{13}) 2^{14} \pi^6 (0.30282212)^{-8} (((\exp[(-8\pi^2)/(0.30282212^2 \times 16) + \ln(16)(\frac{22}{3} - \frac{1}{6} \times 8 \times 1) - (8 \times 0.24875447703 + \frac{1}{3} \ln(2) - 16/9) - 8((2 \times 0.24875447703 - \frac{1}{6} \ln(2) - \frac{17}{72})))])) - 11)$$

Input interpretation

$$\frac{1}{144} \left(\frac{2.6135712 \times 10^{21}}{3.40854 \times 10^{13}} \times \frac{2^{14} \pi^6}{0.30282212^8} \exp\left(\frac{-8\pi^2}{0.30282212^2 \times 16} + \log(16)\left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1\right) - \left(8 \times 0.24875447703 + \frac{1}{3} \log(2) - \frac{16}{9}\right) - 8\left(2 \times 0.24875447703 - \frac{1}{6} \log(2) - \frac{17}{72}\right)\right) - 11 \right)$$

$\log(x)$ is the natural logarithm

Result

1.6156697089079701604064552449755795124464472046026208168259974984

...

1.6156697089.... result that is a very good approximation to the value of the golden ratio 1.618033988749...

$\frac{1}{4}((2.6135712 \times 10^{21} / 3.40854 \times 10^{13}) 2^{14} \pi^6 (0.30282212)^{-8} (((\exp[(-8\pi^2)/(0.30282212^2 \times 16) + \ln(16)(\frac{22}{3} - \frac{1}{6} \times 8 \times 1) - (8 \times 0.24875447 + \frac{1}{3} \ln(2) - \frac{16}{9}) - 8((2 \times 0.24875447 - \frac{1}{6} \ln(2) - \frac{17}{72})))])) + 13 + 1 - 1.65578)$

Input interpretation

$$\frac{1}{4} \left(\frac{2.6135712 \times 10^{21}}{3.40854 \times 10^{13}} \times \frac{2^{14} \pi^6}{0.30282212^8} \exp \left(\frac{-8 \pi^2}{0.30282212^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 \right) - \left(8 \times 0.24875447 + \frac{1}{3} \log(2) - \frac{16}{9} \right) - 8 \left(2 \times 0.24875447 - \frac{1}{6} \log(2) - \frac{17}{72} \right) \right) + 13 + 1 - 1.65578 \right)$$

$\log(x)$ is the natural logarithm

Result

64.0002...

64.0002... $\approx 64 = 8^2$

$(\frac{1}{4}((2.6135712e+21 / 3.40854e+13) 2^{14} \pi^6 (0.30282212)^{-8} (((\exp[(-8\pi^2)/(0.30282212^2 \times 16) + \ln(16)(\frac{22}{3} - \frac{1}{6} \times 8) - (8 \times 0.24875447 + \frac{1}{3} \ln(2) - \frac{16}{9}) - 8((2 \times 0.24875447 - \frac{1}{6} \ln(2) - \frac{17}{72})))])) + 13 - 0.65578))^2 + 276$

Input interpretation

$$\left(\frac{1}{4} \left(\frac{2.6135712 \times 10^{21}}{3.40854 \times 10^{13}} \times \frac{2^{14} \pi^6}{0.30282212^8} \exp \left(\frac{-8 \pi^2}{0.30282212^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \right) - \left(8 \times 0.24875447 + \frac{1}{3} \log(2) - \frac{16}{9} \right) - 8 \left(2 \times 0.24875447 - \frac{1}{6} \log(2) - \frac{17}{72} \right) \right) + 13 - 0.65578 \right) \right)^2 + 276$$

$\log(x)$ is the natural logarithm

Result

4372.02...

4372.02.... ≈ 4372

where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

27(1/4((2.6135712e+21 / 3.40854e+13) 2^14*Pi^6(0.30282212)^-8 (((exp[(-8Pi^2)/(0.30282212^2*16)+ln(16)(22/3-1/6*8)-(8*0.24875447+1/3 ln(2)-16/9)-8((2*0.24875447-1/6 ln(2) -17/72)))])))+13-0.65578))+1

Input interpretation

$$\begin{aligned} 27 \left(\frac{1}{4} \left(\frac{2.6135712 \times 10^{21}}{3.40854 \times 10^{13}} \times \frac{2^{14} \pi^6}{0.30282212^8} \exp \left(\frac{-8 \pi^2}{0.30282212^2 \times 16} + \right. \right. \right. \\ \left. \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \right) - \left(8 \times 0.24875447 + \frac{1}{3} \log(2) - \frac{16}{9} \right) - \right. \\ \left. \left. \left. 8 \left(2 \times 0.24875447 - \frac{1}{6} \log(2) - \frac{17}{72} \right) \right) + 13 - 0.65578 \right) \right) + 1 \end{aligned}$$

$\log(x)$ is the natural logarithm

Result

1729.00...

1729

This result is very near to the mass of candidate glueball **$f_0(1710)$ scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

$$(27(1/4((2.6135712e+21/3.40854e+13) 2^{14} \pi^6 (0.30282212)^{-8}(((\exp[(-8\pi^2)/(0.30282212^2 \times 16) + \ln(16)(22/3 - 1/6 \times 8) - (8 \times 0.24875447 + 1/3 \ln(2) - 16/9) - 8((2 \times 0.24875447 - 1/6 \ln(2) - 17/72))) + 13 - 0.65578)) + 1)^{1/15})$$

Input interpretation

$$\left(27 \left(\frac{1}{4} \left(\frac{1}{0.30282212^8} \frac{2.6135712 \times 10^{21}}{3.40854 \times 10^{13}} \times 2^{14} \pi^6 \exp \left(\frac{-8 \pi^2}{0.30282212^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \right) - \left(8 \times 0.24875447 + \frac{1}{3} \log(2) - \frac{16}{9} \right) - 8 \left(2 \times 0.24875447 - \frac{1}{6} \log(2) - \frac{17}{72} \right) \right) + 13 - 0.65578 \right) \right) + 1 \right)^{(1/15)}$$

$\log(x)$ is the natural logarithm

Result

1.6438155278833673876880818388205368107692784455352832846422747411

...

$$1.643815527.... \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934... \text{ (trace of the instanton shape)}$$

$$32/(e^3 \log^7(2)) (((((2.6135712e+21/3.40854e+13) 2^{14} \pi^6 (0.30282212)^{-8} \\ ((\exp[(-8\pi^2)/(0.30282212^2 \times 16)] + \ln(16)(\frac{22}{3} - \frac{1}{6} \times 8) - (8 \times 0.24875447 + \frac{1}{3} \ln(2) - \\ 16/9) - 8((2 \times 0.24875447 - \frac{1}{6} \ln(2) - \frac{17}{72})))))))))^3)$$

Input interpretation

$$\frac{32}{e^3 \log^7(2)} \left(\frac{2.6135712 \times 10^{21}}{3.40854 \times 10^{13}} \times \frac{2^{14} \pi^6}{0.30282212^8} \right. \\ \left. \exp \left(\frac{-8 \pi^2}{0.30282212^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \right) - \left(8 \times 0.24875447 + \right. \right. \right. \\ \left. \left. \left. \frac{1}{3} \log(2) - \frac{16}{9} \right) - 8 \left(2 \times 0.24875447 - \frac{1}{6} \log(2) - \frac{17}{72} \right) \right) \right)^3$$

$\log(x)$ is the natural logarithm

Result

$$2.99793490418114258703732113475484798319058034418803737786574... \times 10^8$$

$$2.9979349041... \times 10^8 \approx c$$

From:

Lost Notebook and Other Unpublished Papers - By Srinivasa Ramanujan -

Publication date: 1920-08-14

We have that:

$$\left(\frac{1}{\sqrt{e}}\right)^5 + \left(\frac{\sqrt{e-1}}{2}\right)^5 = \frac{1}{\sqrt{x}} \sqrt{\frac{(1-x)(1-x^2)}{1-x^2(1-x^2)}} \dots$$

$$\times \left\{ (1 - \sqrt{\frac{1}{2}}x + x^2)(1 - \sqrt{\frac{1}{2}}x + x^2) \dots \right\}^5$$

$$\left(\frac{1}{\sqrt{t}}\right)^5 + \left(\frac{\sqrt{5}-1}{2}\sqrt{t}\right)^5 = \frac{1}{\sqrt{x}} \sqrt{\frac{(1-x)(1-x^2)}{(1-x^5)(1-x^{10})}} \times \frac{1}{\left\{\left(1-\frac{\sqrt{5}-1}{2}x+x^2\right)\left(1-\frac{\sqrt{5}-1}{2}x^2+x^4\right)\right\}^6}$$

From the right-hand side, for $x = 2$:

$$1/\sqrt{2} * \sqrt{((1-2)(1-2^2))/((1-2^5)(1-2^{10}))} * 1/[(1-((\sqrt{5}-1)/2)*2+2^2)(1-((\sqrt{5}-1)/2)*2^2+2^4)]^6$$

Input

$$\frac{1}{\sqrt{2}} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}} \times \frac{1}{\left(\left(1-\left(\frac{1}{2}(\sqrt{5}-1)\right) \times 2+2^2\right)\left(1-\left(\frac{1}{2}(\sqrt{5}-1)\right) \times 2^2+2^4\right)\right)^6}$$

Result

$$\frac{1}{31 \sqrt{22} (6-\sqrt{5})^6 (17-2(\sqrt{5}-1))^6}$$

Decimal approximation

$$2.5725613215538447021103030635750823938819176873597145350338... \times 10^{-13}$$

2.57256132...*10⁻¹³ (possible anti-instanton value)

Alternate forms

$$\frac{\sqrt{11} (29421 + 13144 \sqrt{5}) 2^{-1/2}}{536143015838069981}$$

$$\frac{1}{27512614111 \sqrt{22} (\sqrt{5}-4)^6}$$

$$\sqrt{\left(\frac{1\,729\,418\,921}{52\,263\,515\,169\,443\,810\,634\,732\,131\,464\,243\,702} + \frac{12\,474\,504\sqrt{5}}{842\,959\,922\,087\,803\,397\,334\,389\,217\,165\,221}\right)}$$

From the left-hand side:

$$\left(\frac{1}{\sqrt{t}}\right)^5 + \left(\left(\frac{\sqrt{5}-1}{2}\right)\sqrt{t}\right)^5 = 2.5725613215538447021103030635750823938819176873597145350338 \times 10^{-13}$$

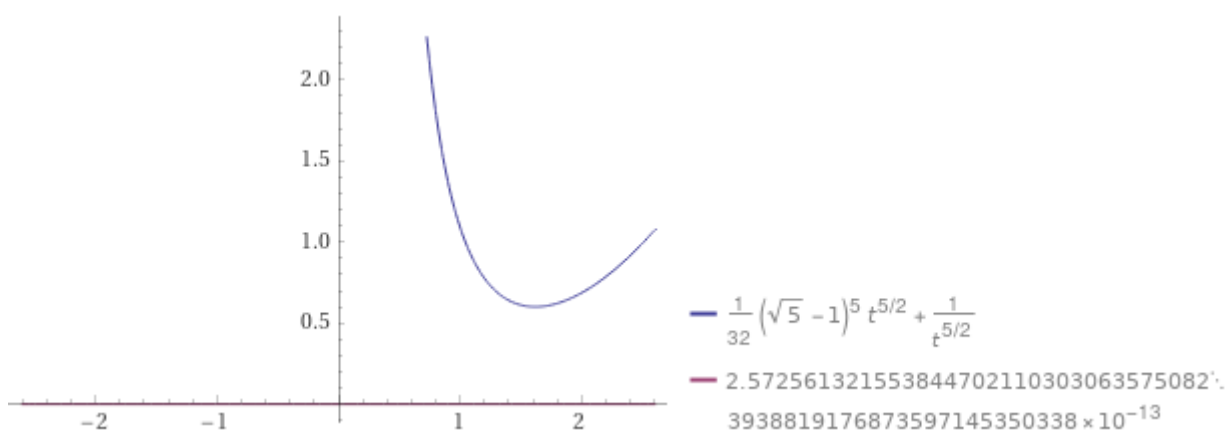
Input interpretation

$$\left(\frac{1}{\sqrt{t}}\right)^5 + \left(\left(\frac{1}{2}(\sqrt{5}-1)\right)\sqrt{t}\right)^5 = 2.5725613215538447021103030635750823938819176873597145350338 \times 10^{-13}$$

Result

$$\frac{1}{32}(\sqrt{5}-1)^5 t^{5/2} + \frac{1}{t^{5/2}} = 2.5725613215538447021103030635750823938819176873597145350338 \times 10^{-13}$$

Plot



Alternate form assuming t is real

$$\frac{1}{t^{5/2}}(1.000 + \\ 11.09016994374947424102293417182819058860154589902881431) = \\ 2.853014224674887501538999596841426031044753143302802418 \times 10^{-12}$$

Alternate forms

$$\frac{1}{2}(5\sqrt{5}-11)t^{5/2}+\frac{1}{t^{5/2}}=2.5725613215538447021103030635750823938819176873597145350338\times 10^{-13}$$

$$\frac{5\sqrt{5}t^5 - 11t^5 + 2}{2t^{5/2}} = 2.5725613215538447021103030635750823938819176873597145350338 \times 10^{-13}$$

Expanded form

$$\frac{5}{2} \sqrt{5} t^{5/2} - \frac{11 t^{5/2}}{2} + \frac{1}{t^{5/2}} = 2.5725613215538447021103030635750823938819176873597145350338 \times 10^{-13}$$

Solutions

$$t = -1.618033988749894848204586810614342306021167 - 2.77237818127982802077350286216 \times 10^{-13} i$$

(assuming a complex-valued square root)

$$t = -1.618033988749894848204586810614342306021167 + 2.77237818127982802077350286216 \times 10^{-13} i$$

(assuming a complex-valued square root)

$$t = -0.500000000000263668833494061358643799948092 - 1.538841768587541030087860287871077792834418 i$$

$$t = -0.500000000000263668833494061358643799948092 + 1.538841768587541030087860287871077792834418 i$$

$$t = 1.309016994375110380403166769119379605953864 - 0.951056516294929282010090350459129600311175 i$$

$$t = 1.309016994375110380403166769119379605953864 + 0.951056516294929282010090350459129600311175 i$$

$$(((1/32 (\sqrt{5} - 1)^5 (-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + 1/(-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2})))$$

Input interpretation

$$\frac{\frac{1}{32} (\sqrt{5} - 1)^5}{1} \frac{(-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + 1}{(-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2}}$$

i is the imaginary unit

Result

$$2.57256... \times 10^{-13} - 7.87071... \times 10^{-16} i$$

Polar coordinates

$$r = 2.57257 \times 10^{-13} \text{ (radius)}, \quad \theta = -0. \times 10^{-3} \text{ (angle)}$$

2.57257×10^{-13} as above

From which:

$$((1/ (((1/32 (\sqrt{5} - 1)^5 (-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + 1/(-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2}))))))$$

Input interpretation

$$1 / \left(\frac{\frac{1}{32} (\sqrt{5} - 1)^5 (-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + 1}{(-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2}} \right)$$

i is the imaginary unit

Result

$$3.88714... \times 10^{12} + 1.18926... \times 10^{10} i$$

Polar coordinates

$r = 3887158340733$ (radius), $\theta = 0. \times 10^{-3}$ (angle)
 $3.887158340733 \times 10^{12}$

and, after some calculations:

$$(\mathbf{11161 + 11468 + 1010 - 812 + 172 + 138} - 16) ((1/ (((1/32 (\sqrt{5} - 1)^5 (-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + 1/(-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2}))))))$$

where the values highlighted in red are the cubic roots of the following Ramanujan cubes:

Handwritten mathematical identities on a piece of paper:

$$135^3 + 138^3 = 172^3 - 1$$

$$11161^3 + 11468^3 = 14258^3 + 1$$

$$791^3 + 812^3 = 1010^3 - 1$$

Where $11161 + 11468 + 1010 - 812 + 172 + 138 - 16 = 23121$

$(23121) \left(\frac{1}{\left(\left(\left(\frac{1}{32} (\sqrt{5} - 1)^5 (-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + 1 / (-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} \right) \right) \right) \right)$

Input interpretation

$$23121 \times 1 / \left(\frac{1}{32} (\sqrt{5} - 1)^5 \right. \\ \left. (-1.618033988749894 + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + 1 / (-1.618033988749894 + \right. \\ \left. 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} \right)$$

i is the imaginary unit

Result

$$8.98746... \times 10^{16} + \\ 2.74970... \times 10^{14} i$$

Polar coordinates

$$r = 89874987996094332 \text{ (radius)}, \quad \theta = 0. \times 10^{-3} \text{ (angle)}$$

Scientific notation

$$8.9874987996094332 \times 10^{16} \\ 8.9874987996094332 \times 10^{16} \approx c^2 = 8.98755178... \times 10^{16}$$

$$64\ln((1/((((1/32(\sqrt{5}-1)^5(-1.618033988749894+2.77237818127982802077350286216\times 10^{-13}i)^{5/2}+1/(-1.618033988749894+2.77237818127982802077350286216\times 10^{-13}i)^{5/2})))))))-144+13+5$$

Input interpretation

$$64\log\left(1/\left(\frac{1}{32}(\sqrt{5}-1)^5(-1.618033988749894+2.77237818127982802077350286216\times 10^{-13}i)^{5/2}+1/(-1.618033988749894+2.77237818127982802077350286216\times 10^{-13}i)^{5/2}\right)\right)-144+13+5$$

$\log(x)$ is the natural logarithm

i is the imaginary unit

Result

$$1729.28\dots + \\ 0.195806\dots i$$

Polar coordinates

$$r = 1729.3 \text{ (radius)}, \quad \theta = 0.0001 \text{ (angle)}$$

$$1729.3$$

This result is very near to the mass of candidate glueball **$f_0(1710)$ scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

Polar forms

$$1729.3(\cos(0.0001) + i\sin(0.0001))$$

$$1729.3 e^{0.0001 i}$$

$$(1/27(64\ln((1/ (((1/32 (\sqrt{5} - 1)^5 (-\text{golden ratio} + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + 1/(-\text{golden ratio} + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2})))))))-144+13+5))^2-2\pi+276$$

Input interpretation

$$\left(\frac{1}{27} \left(64 \log \left(1 / \left(\frac{1}{32} (\sqrt{5} - 1)^5 (-\phi + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + \frac{1}{(-\phi + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2}} \right) \right) \right)^2 - 2\pi + 276 \right)$$

$\log(x)$ is the natural logarithm

ϕ is the golden ratio

i is the imaginary unit

Result

$$4371.77329884374740121057246430... - 2.60124860276096118014672443720... \times 10^{-11} i$$

Polar coordinates

$$r = 4371.77329884374740121057246438 \text{ (radius), } \theta = -5.950099478966448 \times 10^{-15} \text{ (angle)}$$

$$4371.7732988.... \approx 4372$$

where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

Polar forms

$$4371.77329884374740121057246438 \\ (\cos(-5.950099478966448 \times 10^{-15}) + i \sin(-5.950099478966448 \times 10^{-15}))$$

$$4371.77329884374740121057246438 e^{-5.950099478966448 \times 10^{-15} i}$$

$$(64 \ln((1/(((1/32 (\sqrt{5} - 1)^5 (-\text{golden ratio} + \\ 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + 1/(-\text{golden ratio} + \\ 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2})))))))-144+13+5)^{1/15}$$

Input interpretation

$$\left(64 \log \left(1 / \left(\frac{1}{32} (\sqrt{5} - 1)^5 \right. \right. \right. \\ \left. \left. \left. \frac{(-\phi + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2} + 1}{(-\phi + 2.77237818127982802077350286216 \times 10^{-13} i)^{5/2}} \right) \right) - \right. \\ \left. 144 + 13 + 5 \right)^{(1/15)}$$

$\log(x)$ is the natural logarithm

ϕ is the golden ratio

i is the imaginary unit

Result

$$1.6438327885745794492074548230632... - \\ 3.4746943482076080861798704380025... \times 10^{-16} i$$

Polar coordinates

$$r = 1.64383278857457944920745482306327 \\ (\text{radius}), \quad \theta = -2.1137760314542867 \times 10^{-16} \text{ (angle)}$$

$$1.643832788.... \approx \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934 ... \text{ (trace of the anti-instanton shape)}$$

Polar forms

1.64383278857457944920745482306327

$$(\cos(-2.1137760314542867 \times 10^{-16}) + i \sin(-2.1137760314542867 \times 10^{-16}))$$

$$1.64383278857457944920745482306327 e^{-2.1137760314542867 \times 10^{-16} i}$$

Now, we have that:

$$\left(\frac{1}{\sqrt{t}}\right)^5 - \left(\frac{\sqrt{5}+1}{2}\sqrt{t}\right)^5 = \frac{1}{\sqrt{x}} \sqrt{\frac{(1-x)(1-x^2)}{(1-x^5)(1-x^{10})}} \times \left\{ \left(1 + \frac{\sqrt{5}+1}{2}x + x^2\right) \left(1 + \frac{\sqrt{5}+1}{2}x^2 + x^4\right) \right\}^6$$

$$\left(\frac{1}{\sqrt{t}}\right)^5 - \left(\frac{\sqrt{5}+1}{2}\sqrt{t}\right)^5 = \frac{1}{\sqrt{x}} \sqrt{\frac{(1-x)(1-x^2)}{(1-x^5)(1-x^{10})}} \times \frac{1}{\left\{ \left(1 + \frac{\sqrt{5}+1}{2}x + x^2\right) \left(1 + \frac{\sqrt{5}+1}{2}x^2 + x^4\right) \right\}^6}$$

$$1/\sqrt{2} * \sqrt{((1-2)(1-2^2))/((1-2^5)(1-2^{10}))} *$$

$$1/[(1+((\sqrt{5}+1)/2)*2+2^2)(1+((\sqrt{5}+1)/2)*2^2+2^4)]^6$$

Input

$$\frac{1}{\sqrt{2}} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}} \times \frac{1}{\left(\left(1 + \left(\frac{1}{2}(\sqrt{5}+1)\right) \times 2 + 2^2\right) \left(1 + \left(\frac{1}{2}(\sqrt{5}+1)\right) \times 2^2 + 2^4\right) \right)^6}$$

Result

$$\frac{1}{31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6}$$

Decimal approximation

$$1.3176247951262521895900883785136111916210344002074660281415... \times 10^{-16}$$

1.31762479512...*10⁻¹⁶ (possible anti-instanton value)

Alternate forms

$$\frac{\sqrt{11} (29421 - 13144 \sqrt{5}) 2^{-1/2}}{536143015838069981}$$

$$\frac{1}{27512614111 \sqrt{22} (4 + \sqrt{5})^6}$$

$$\frac{1}{27512614111 \sqrt{22 (1729418921 + 773419248 \sqrt{5})}}$$

Minimal polynomial

$$870332118498272795633464800247580675588467314065575103524x^4 - 57599219203109912638213824520204x^2 + 1$$

And:

$$(1/\sqrt{t})^5 - (((((\sqrt{5}+1)/2))*\sqrt{t})))^5 = 1.3176247951262521895900883785136111916210344002074660281415 \times 10^{-16}$$

$$(1/\sqrt{-0.499999999999999960433953397 - 0.363271264002680440073091826570663319191 i})^5 - (((((\sqrt{5}+1)/2))*\sqrt{-0.499999999999999960433953397 - 0.363271264002680440073091826570663319191 i})))^5$$

Input interpretation

$$(1/(\sqrt{-0.499999999999999960433953397 + i \times (-0.363271264002680440073091826570663319191)}))^5 - \left(\left(\frac{1}{2} (\sqrt{5} + 1) \right) \sqrt{-0.499999999999999960433953397 + i \times (-0.363271264002680440073091826570663319191)} \right)^5$$

i is the imaginary unit

Result

$$1.317624795... \times 10^{-16} - 9.265224992... \times 10^{-28} i$$

Polar coordinates

$$r = 1.31762 \times 10^{-16} \text{ (radius), } \theta = -7.03176 \times 10^{-12} \text{ (angle)}$$

$$1.31762 \times 10^{-16}$$

From which:

$$299792458^2 / (1/((1/\sqrt{2} * \sqrt{((1-2)(1-2^2))}/((1-2^5)(1-2^{10})))) * 1/[(1+((\sqrt{5}+1)/2)*2+2^2)(1+((\sqrt{5}+1)/2)*2^2+2^4)]^6)))$$

Input

$$\frac{299792458^2}{1} \times \frac{1}{\left(\left(1 + \left(\frac{1}{2} (\sqrt{5} + 1) \right) \times 2 + 2^2 \right) \left(1 + \left(\frac{1}{2} (\sqrt{5} + 1) \right) \times 2^2 + 2^4 \right) \right)^6} \times \frac{1}{\sqrt{2} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}$$

Result

$$\frac{44937758936840882 \sqrt{\frac{2}{11}}}{31(6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6}$$

Decimal approximation

11.842221082517575110783382680453334883959805869903827081709688381

...

11.842221082.... result very near to the black hole entropy $\ln(139503) = 11.8458$

Alternate forms

$$\frac{44937758936840882 (29421 - 13144 \sqrt{5}) \sqrt{22}}{536143015838069981}$$

$$\frac{44937758936840882 \sqrt{\frac{2}{11}}}{27512614111 (4 + \sqrt{5})^6}$$

$$\frac{44937758936840882 \sqrt{\frac{2}{11(1729418921+773419248\sqrt{5})}}}{27512614111}$$

Minimal polynomial

$$\begin{aligned} &217583029624568198908366200061895168897116828516393775881x^4 - \\ &116315988725159233859130764664480085653218989852556347899150x^3 - \\ &216496x^2 + \\ &16311940630335764902440298498493781009449716884306739887988x - \\ &360919104 \end{aligned}$$

$$11.8422210825 \left(\frac{1}{\left(\frac{1}{\sqrt{2}} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}} \right)} \right) * \frac{1}{\left(1 + \left(\frac{\sqrt{5}+1}{2} \right)^2 + 2^2 \right) \left(1 + \left(\frac{\sqrt{5}+1}{2} \right)^2 + 2^4 \right)^6 \right)}$$

Input interpretation

$$11.8422210825 \times \frac{1}{\frac{1}{\sqrt{2}} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}} \times \frac{1}{\left(\left(1 + \left(\frac{1}{2} (\sqrt{5}+1) \right) \times 2 + 2^2 \right) \left(1 + \left(\frac{1}{2} (\sqrt{5}+1) \right) \times 2^2 + 2^4 \right)^6 \right)}}$$

Result

$$8.98755178735... \times 10^{16}$$

$$8.98755178735 * 10^{16} \approx c^2$$

$$48 \ln \left(\frac{1}{\left(\frac{1}{\sqrt{2}} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}} \right)} \right) * \frac{1}{\left(1 + \left(\frac{\sqrt{5}+1}{2} \right)^2 + 2^2 \right) \left(1 + \left(\frac{\sqrt{5}+1}{2} \right)^2 + 2^4 \right)^6 \right)} - 27$$

Input

$$48 \log \left(\frac{1}{\frac{1}{\sqrt{2}} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}} \times \frac{1}{\left(\left(1 + \left(\frac{1}{2} (\sqrt{5}+1) \right) \times 2 + 2^2 \right) \left(1 + \left(\frac{1}{2} (\sqrt{5}+1) \right) \times 2^2 + 2^4 \right)^6 \right)}} \right) - 27$$

$\log(x)$ is the natural logarithm

Exact result

$$48 \log \left(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6 \right) - 27$$

Decimal approximation

$$1728.1454769503361881543858341796040222785026446884357401395410493$$

...

$$1728.14547695....$$

This result is very near to the mass of candidate glueball **$f_0(1710)$ scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

Property

$-27 + 48 \log(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6)$ is a transcendental number

Alternate forms

$$48 \log(27512614111 \sqrt{22} (4 + \sqrt{5})^6) - 27$$

$$48 \log\left(27512614111 \sqrt{22 (1729418921 + 773419248 \sqrt{5})}\right) - 27$$

$$-27 + 24 \log(22) + 48 \log(31) + 288 \log((6 + \sqrt{5})(19 + 2\sqrt{5}))$$

$$3(-9 + 8 \log(21142) + 96 \log(31(4 + \sqrt{5})))$$

$$48\left(\frac{1}{2}(\log(2) + \log(11)) + \log(31) + 6 \log(6 + \sqrt{5}) + 6 \log(17 + 2(1 + \sqrt{5}))\right) - 27$$

Alternative representations

$$\begin{aligned}
 & 48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) 2^2 + 2^4 \right) \right)^6 \sqrt{2}}}} \right) - 27 = \\
 & -27 + 48 \log_e \left(\frac{1}{\frac{\sqrt{\frac{3}{(1-2^5)(1-2^{10})}}}{\left((6 + \sqrt{5}) (1 + 2^4 + 2(1 + \sqrt{5})) \right)^6 \sqrt{2}}} \right)
 \end{aligned}$$

$$\begin{aligned}
 & 48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) 2^2 + 2^4 \right) \right)^6 \sqrt{2}}} \right) - 27 = \\
 & -27 + 48 \log(a) \log_a \left(\frac{1}{\frac{\sqrt{\frac{3}{(1-2^5)(1-2^{10})}}}{\left((6 + \sqrt{5}) (1 + 2^4 + 2(1 + \sqrt{5})) \right)^6 \sqrt{2}}} \right)
 \end{aligned}$$

$$\begin{aligned}
& 48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) 2^2 + 2^4 \right) \right)^6 \sqrt{2}}}} \right) - 27 = \\
& -27 - 48 \operatorname{Li}_1 \left(1 - \frac{1}{\frac{\sqrt{\frac{3}{(1-2^5)(1-2^{10})}}}{\left((6 + \sqrt{5}) (1 + 2^4 + 2(1 + \sqrt{5})) \right)^6 \sqrt{2}}} \right)
\end{aligned}$$

Series representations

$$\begin{aligned}
& 48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) 2^2 + 2^4 \right) \right)^6 \sqrt{2}}} \right) - 27 = \\
& -27 + 48 \log \left(-1 + 27512614111 \sqrt{22} (4 + \sqrt{5})^6 \right) - \\
& 48 \sum_{k=1}^{\infty} \frac{\left(-\frac{1}{-1 + 27512614111 \sqrt{22} (4 + \sqrt{5})^6} \right)^k}{k}
\end{aligned}$$

$$\begin{aligned}
& 48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) 2^2 + 2^4 \right) \right)^6 \sqrt{2}}} \right) - 27 = \\
& -27 + 48 \log \left(-1 + 31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6 \right) - \\
& 48 \sum_{k=1}^{\infty} \frac{\left(-\frac{1}{-1 + 27512614111 \sqrt{22} (4 + \sqrt{5})^6} \right)^k}{k}
\end{aligned}$$

$$\begin{aligned}
& 48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) 2^2 + 2^4 \right) \right)^6 \sqrt{2}}}} \right) - 27 = \\
& -27 + 96 i \pi \left[\frac{\arg \left(27512614111 \sqrt{22} (4 + \sqrt{5})^6 - x \right)}{2 \pi} \right] + 48 \log(x) - \\
& 48 \sum_{k=1}^{\infty} \frac{(-1)^k \left(27512614111 \sqrt{22} (4 + \sqrt{5})^6 - x \right)^k x^{-k}}{k} \text{ for } x < 0
\end{aligned}$$

Integral representations

$$\begin{aligned}
& 48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) 2^2 + 2^4 \right) \right)^6 \sqrt{2}}}} \right) - 27 = \\
& -27 + 48 \int_1^{27512614111 \sqrt{22} (4 + \sqrt{5})^6} \frac{1}{t} dt
\end{aligned}$$

$$\begin{aligned}
& 48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) 2^2 + 2^4 \right) \right)^6 \sqrt{2}}}} \right) - 27 = \\
& -27 - \frac{24 i}{\pi} \int_{-i \infty + \gamma}^{i \infty + \gamma} \frac{\left(-1 + 27512614111 \sqrt{22} (4 + \sqrt{5})^6 \right)^{-s} \Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds \\
& \text{for } -1 < \gamma < 0
\end{aligned}$$

$$\left((48 \ln\left(\frac{1}{\left(\frac{1}{\sqrt{2}} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}\right)}\right) * \frac{1}{\left(1 + \left(\frac{\sqrt{5}+1}{2}\right) 2 + 2^2\right) \left(1 + \left(\frac{\sqrt{5}+1}{2}\right) 2^2 + 2^4\right)^6}\right) - 27 \right)^{1/15}$$

Input

$$\sqrt[15]{48 \log \left(\frac{1}{\frac{1}{\sqrt{2}} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}} \times \frac{1}{\left(1 + \left(\frac{1}{2}(\sqrt{5}+1)\right) 2 + 2^2\right) \left(1 + \left(\frac{1}{2}(\sqrt{5}+1)\right) 2^2 + 2^4\right)^6}} \right) - 27}$$

log(x) is the natural logarithm

Exact result

$$\sqrt[15]{48 \log \left(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6 \right) - 27}$$

Decimal approximation

1.6437610547722016332376756883072132097286472052573524941888036957

...

$$1.6437610547722... \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots \text{ (trace of the anti-instanton shape)}$$

Property

$$\sqrt[15]{-27 + 48 \log \left(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6 \right)}$$

is a transcendental number

Alternate forms

$$\sqrt[15]{48 \log \left(27512614111 \sqrt{22} (4 + \sqrt{5})^6 \right) - 27}$$

$$\sqrt[15]{48 \log \left(27512614111 \sqrt{22(1729418921 + 773419248\sqrt{5})} \right) - 27}$$

$$\sqrt[15]{3(-9 + 8 \log(22) + 16 \log(31) + 96 \log((6 + \sqrt{5})(19 + 2\sqrt{5})))}$$

$$\sqrt[15]{3(-9 + 8 \log(21142) + 96 \log(31(4 + \sqrt{5})))}$$

$$\sqrt[15]{48 \left(\frac{1}{2} (\log(2) + \log(11)) + \log(31) + 6 \log(6 + \sqrt{5}) + 6 \log(17 + 2(1 + \sqrt{5})) \right) - 27}$$

All 15th roots of $48 \log(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5})))^6 - 27$

$$e^0 \sqrt[15]{48 \log(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5})))^6 - 27} \approx 1.64376$$

(real, principal root)

$$e^{(2i\pi)/15} \sqrt[15]{48 \log(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5})))^6 - 27} \approx 1.50165 + 0.6686i$$

$$e^{(4i\pi)/15} \sqrt[15]{48 \log(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5})))^6 - 27} \approx 1.0999 + 1.2216i$$

$$e^{(2i\pi)/5} \sqrt[15]{48 \log(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5})))^6 - 27} \approx 0.5080 + 1.5633i$$

$$e^{(8i\pi)/15} \sqrt[15]{48 \log(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6) - 27} \\ \approx -0.17182 + 1.63476i$$

Alternative representations

$$\sqrt[15]{48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1+\frac{1}{2}(\sqrt{5}+1)2+2^2\right)\left(1+\frac{1}{2}(\sqrt{5}+1)2^2+2^4\right)\right)^6 \sqrt{2}}}} \right) - 27} = \\ \sqrt[15]{-27 + 48 \log_e \left(\frac{1}{\frac{\sqrt{\frac{3}{(1-2^5)(1-2^{10})}}}{\left((6+\sqrt{5})(1+2^4+2(1+\sqrt{5}))\right)^6 \sqrt{2}}}} \right)}$$

$$\sqrt[15]{48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1+\frac{1}{2}(\sqrt{5}+1)2+2^2\right)\left(1+\frac{1}{2}(\sqrt{5}+1)2^2+2^4\right)\right)^6 \sqrt{2}}}} \right) - 27} = \\ \sqrt[15]{-27 + 48 \log(a) \log_a \left(\frac{1}{\frac{\sqrt{\frac{3}{(1-2^5)(1-2^{10})}}}{\left((6+\sqrt{5})(1+2^4+2(1+\sqrt{5}))\right)^6 \sqrt{2}}}} \right)}$$

$$\sqrt[15]{48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1+\frac{1}{2}(\sqrt{5}+1)2+2^2 \right) \left(1+\frac{1}{2}(\sqrt{5}+1)2^2+2^4 \right) \right)^6 \sqrt{2}}}} \right) - 27 =}$$

$$\sqrt[15]{-27 - 48 \operatorname{Li}_1 \left(1 - \frac{1}{\frac{\sqrt{\frac{3}{(1-2^5)(1-2^{10})}}}{\left((6+\sqrt{5})(1+2^4+2(1+\sqrt{5})) \right)^6 \sqrt{2}}}} \right)}$$

Series representations

$$\sqrt[15]{48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1+\frac{1}{2}(\sqrt{5}+1)2+2^2 \right) \left(1+\frac{1}{2}(\sqrt{5}+1)2^2+2^4 \right) \right)^6 \sqrt{2}}}} \right) - 27 =}$$

$$\left(-27 + 48 \left(\log \left(-1 + 27512614111 \sqrt{22} (4 + \sqrt{5})^6 \right) - \sum_{k=1}^{\infty} \frac{\left(-\frac{1}{-1+27512614111 \sqrt{22} (4+\sqrt{5})^6} \right)^k}{k} \right) \right)^{(1/15)}$$

$$\sqrt[15]{48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) 2^2 + 2^4 \right) \right)^6 \sqrt{2}}}} \right) - 27 =$$

$$\left(-27 + 48 \left(\log \left(-1 + 31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6 \right) - \sum_{k=1}^{\infty} \frac{\left(-\frac{1}{-1+27512614111 \sqrt{22} (4+\sqrt{5})^6} \right)^k}{k} \right) \right)^{(1/15)}$$

$$\sqrt[15]{48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) 2^2 + 2^4 \right) \right)^6 \sqrt{2}}}} \right) - 27 =$$

$$\left(-27 + 48 \left(2 i \pi \left[\frac{\arg \left(27512614111 \sqrt{22} (4 + \sqrt{5})^6 - x \right)}{2 \pi} \right] + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k \left(27512614111 \sqrt{22} (4 + \sqrt{5})^6 - x \right)^k x^{-k}}{k} \right) \right)^{(1/15)}$$

15) for $x < 0$

Integral representations

$$\sqrt[15]{48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5}+1) 2+2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5}+1) 2^2+2^4 \right) \right)^6 \sqrt{2}}}} \right)} - 27 =$$

$$\sqrt[15]{-27 + 48 \int_1^{27512614111 \sqrt{22} (4+\sqrt{5})^6} \frac{1}{t} dt}$$

$$\sqrt[15]{48 \log \left(\frac{1}{\frac{\sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}}}{\left(\left(1 + \frac{1}{2} (\sqrt{5}+1) 2+2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5}+1) 2^2+2^4 \right) \right)^6 \sqrt{2}}}} \right)} - 27 = \left(-27 - \frac{24 i}{\pi} \right.$$

$$\left. \int_{-i \infty + \gamma}^{i \infty + \gamma} \frac{(-1 + 31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6)^{-s} \Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds \right)^{1/15} \text{ for } -1 < \gamma < 0$$

$$\left(\frac{1}{27} \left(48 \ln \left(\frac{1}{\left(\frac{1}{\sqrt{2}} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}} \right)} \right) \times \frac{1}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) \right) \times 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) \right) \times 2^2 + 2^4 \right)^6} \right) - 27 \right)^2 + 276 - \Phi$$

Input

$$\left(\frac{1}{27} \left(48 \log \left(\frac{1}{\frac{1}{\sqrt{2}} \sqrt{\frac{(1-2)(1-2^2)}{(1-2^5)(1-2^{10})}} \times \frac{1}{\left(\left(1 + \frac{1}{2} (\sqrt{5} + 1) \right) \times 2 + 2^2 \right) \left(1 + \frac{1}{2} (\sqrt{5} + 1) \right) \times 2^2 + 2^4 \right)^6}} \right) - 27 \right)^2 + 276 - \Phi$$

$\log(x)$ is the natural logarithm

Φ is the golden ratio conjugate

Exact result

$$-\Phi + 276 + \frac{1}{729} \left(48 \log \left(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6 \right) - 27 \right)^2$$

Exact form

$$-\phi + 277 + \frac{1}{81} \left(9 - 16 \log \left(27512614111 \sqrt{22} (4 + \sqrt{5})^6 \right) \right)^2$$

ϕ is the golden ratio

Decimal approximation

4372.0716635474708807064074958923092033620054638013767510619686818

...

4372.07166354747.... $\approx$ 4372

where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

Property

$$276 - \Phi + \frac{1}{729} \left(-27 + 48 \log \left(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6 \right) \right)^2$$

is a transcendental number

Alternate forms

$$-\Phi + 276 + \frac{1}{81} \left(9 - 16 \log \left(27512614111 \sqrt{22} (4 + \sqrt{5})^6 \right) \right)^2$$

$$-\Phi + 276 + \frac{1}{729} \left(48 \log \left(27512614111 \sqrt{22(1729418921 + 773419248\sqrt{5})} \right) - 27 \right)^2$$

$$-\Phi + 276 + \frac{1}{729} \left(48 \left(\frac{\log(22)}{2} + \log(31) + 6(\log(6 + \sqrt{5}) + \log(17 + 2(1 + \sqrt{5}))) \right) - 27 \right)^2$$

$$-\Phi + 276 + \frac{1}{81} \left(-9 + 8 \log(11) + 16 \log(31) + \log(256) + 96 \log(6 + \sqrt{5}) + 96 \log(19 + 2\sqrt{5}) \right)^2$$

$$-\Phi + 276 + \frac{1}{729} \left(48 \left(\frac{1}{2} (\log(2) + \log(11)) + \log(31) + 6 \log(6 + \sqrt{5}) + 6 \log(17 + 2(1 + \sqrt{5})) \right) - 27 \right)^2$$

Expanded forms

$$-\Phi + 277 + \frac{256}{81} \log^2(809448619759731 \sqrt{22} + 361625799874984 \sqrt{110}) - \frac{32}{9} \log(809448619759731 \sqrt{22} + 361625799874984 \sqrt{110})$$

$$-\Phi + 277 + \frac{256}{81} \log^2(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6) - \frac{32}{9} \log(31 \sqrt{22} (6 + \sqrt{5})^6 (17 + 2(1 + \sqrt{5}))^6)$$

We have that, from:

$$\left(\frac{1}{\sqrt{t}}\right)^5 + \left(\frac{\sqrt{5}-1}{2} \sqrt{t}\right)^5 = \frac{1}{\sqrt{x}} \sqrt{\frac{(1-x)(1-x^2)}{(1-x^5)(1-x^{10})}} \times \left\{ \left(1 - \frac{\sqrt{5}-1}{2}x + x^2\right) \left(1 - \frac{\sqrt{5}-1}{2}x^2 + x^4\right) \right\}^6$$

$$\left(\frac{1}{\sqrt{t}}\right)^5 + \left(\frac{\sqrt{5}-1}{2} \sqrt{t}\right)^5 = \frac{1}{\sqrt{x}} \sqrt{\frac{(1-x)(1-x^2)}{(1-x^5)(1-x^{10})}} \times \frac{1}{\left\{ \left(1 - \frac{\sqrt{5}-1}{2}x + x^2\right) \left(1 - \frac{\sqrt{5}-1}{2}x^2 + x^4\right) \right\}^6}$$

For $x = 2.14843522327\dots$ that is equal to:

$$\frac{11}{7} (2\zeta(3) - \zeta(5))$$

$\zeta(s)$ is the Riemann zeta function

2.1484352232762864413070313146035034211716869024242763525305506800

...

2.14843522....

Expanded form

$$\frac{22\zeta(3)}{7} - \frac{11\zeta(5)}{7}$$

Alternative representations

$$\frac{1}{7} (2\zeta(3) - \zeta(5)) 11 = \frac{11}{7} (2\zeta(3, 1) - \zeta(5, 1))$$

$$\frac{1}{7} (2\zeta(3) - \zeta(5)) 11 = \frac{11}{7} (2S_{2,1}(1) - S_{4,1}(1))$$

$$\frac{1}{7} (2\zeta(3) - \zeta(5)) 11 = \frac{11}{7} \left(-\frac{2\text{Li}_3(-1)}{\frac{3}{4}} + \frac{\text{Li}_5(-1)}{1 - \frac{1}{2^4}} \right)$$

$\zeta(s, a)$ is the generalized Riemann zeta function

$S_{n,p}(x)$ is the Nielsen generalized polylogarithm function

$\text{Li}_n(x)$ is the polylogarithm function

Series representations

$$\frac{1}{7} (2\zeta(3) - \zeta(5))_{11} = \sum_{k=1}^{\infty} \frac{11(-1+2k^2)}{7k^5}$$

$$\frac{1}{7} (2\zeta(3) - \zeta(5))_{11} = \sum_{k=0}^{\infty} \frac{176(17+124k+124k^2)}{1519(1+2k)^5}$$

$$\frac{1}{7} (2\zeta(3) - \zeta(5))_{11} = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n \frac{11(-1)^k(3+8k+4k^2) \binom{n}{k}}{28(1+k)^4(1+n)} \right)$$

Integral representations

$$\frac{1}{7} (2\zeta(3) - \zeta(5))_{11} = \int_0^{\infty} -\frac{22t^2(-30+t^2)}{315(1+e^t)} dt$$

$$\frac{1}{7} (2\zeta(3) - \zeta(5))_{11} = \int_0^{\infty} -\frac{11t^2(-24+t^2)}{168(-1+e^t)} dt$$

$$\frac{1}{7} (2\zeta(3) - \zeta(5))_{11} = \int_0^{\infty} \frac{22t^2(186-7t^2) \operatorname{csch}(t)}{4557} dt$$

Thence, we have:

$$\frac{((1/\sqrt{(11/7(2\zeta(3) - \zeta(5))}))) \sqrt{((1 - ((11/7(2\zeta(3) - \zeta(5))))(1 - ((11/7(2\zeta(3) - \zeta(5)))^2)))/((1 - ((11/7(2\zeta(3) - \zeta(5))))^5)(1 - ((11/7(2\zeta(3) - \zeta(5))))^{10}))}}{1}$$

Input

$$\frac{1}{\sqrt{\frac{11}{7}(2\zeta(3) - \zeta(5))}} \sqrt{\frac{\left(1 - \frac{11}{7}(2\zeta(3) - \zeta(5))\right)\left(1 - \left(\frac{11}{7}(2\zeta(3) - \zeta(5))\right)^2\right)}{\left(1 - \left(\frac{11}{7}(2\zeta(3) - \zeta(5))\right)^5\right)\left(1 - \left(\frac{11}{7}(2\zeta(3) - \zeta(5))\right)^{10}\right)}}$$

$\zeta(s)$ is the Riemann zeta function

Exact result

$$\sqrt{\frac{7\left(\frac{11}{7}(2\zeta(3) - \zeta(5)) - 1\right)\left(\frac{121}{49}(2\zeta(3) - \zeta(5))^2 - 1\right)}{11\left(\frac{161051(2\zeta(3) - \zeta(5))^5}{16807} - 1\right)\left(\frac{25937424601(2\zeta(3) - \zeta(5))^{10}}{282475249} - 1\right)(2\zeta(3) - \zeta(5))}}$$

Decimal approximation

0.0045402107109401188055067420539909708552137971639323868012781585

...

0.00454021071....

Alternate forms

$$\sqrt{\left(\left(7\left(\frac{22\zeta(3)}{7} - \frac{11\zeta(5)}{7} - 1\right)\left(\frac{484\zeta(3)^2}{49} - \frac{484\zeta(3)\zeta(5)}{49} + \frac{121\zeta(5)^2}{49} - 1\right)\right) / \right. \\ \left. \left(11(2\zeta(3) - \zeta(5)) \right. \right. \\ \left. \left(\frac{5153632\zeta(3)^5}{16807} - \frac{12884080\zeta(3)^4\zeta(5)}{16807} + \frac{12884080\zeta(3)^3\zeta(5)^2}{16807} - \right. \right. \\ \left. \left. \frac{6442040\zeta(3)^2\zeta(5)^3}{16807} + \frac{1610510\zeta(3)\zeta(5)^4}{16807} - \frac{161051\zeta(5)^5}{16807} - 1 \right) \right. \\ \left. \left(\frac{26559922791424\zeta(3)^{10}}{282475249} - \frac{132799613957120\zeta(3)^9\zeta(5)}{282475249} + \right. \right. \\ \left. \frac{298799131403520\zeta(3)^8\zeta(5)^2}{282475249} - \right. \\ \left. \frac{398398841871360\zeta(3)^7\zeta(5)^3}{282475249} + \frac{49799855233920\zeta(3)^6\zeta(5)^4}{40353607} - \right. \\ \left. \frac{29879913140352\zeta(3)^5\zeta(5)^5}{40353607} + \frac{12449963808480\zeta(3)^4\zeta(5)^6}{40353607} - \right. \\ \left. \frac{24899927616960\zeta(3)^3\zeta(5)^7}{282475249} + \frac{4668736428180\zeta(3)^2\zeta(5)^8}{282475249} - \right. \\ \left. \frac{518748492020\zeta(3)\zeta(5)^9}{282475249} + \frac{25937424601\zeta(5)^{10}}{282475249} - 1 \right) \left. \right) \left. \right)$$

(117649

$$\sqrt{(7/(11(4802\zeta(3) - 15092\zeta(3)^2 + 47432\zeta(3)^3 - 149072\zeta(3)^4 + 468512\zeta(3)^5 - 2401\zeta(5) + 15092\zeta(3)\zeta(5) - 71148\zeta(3)^2\zeta(5) + 298144\zeta(3)^3\zeta(5) - 1171280\zeta(3)^4\zeta(5) - 3773\zeta(5)^2 + 35574\zeta(3)\zeta(5)^2 - 223608\zeta(3)^2\zeta(5)^2 + 1171280\zeta(3)^3\zeta(5)^2 - 5929\zeta(5)^3 + 74536\zeta(3)\zeta(5)^3 - 585640\zeta(3)^2\zeta(5)^3 - 9317\zeta(5)^4 + 146410\zeta(3)\zeta(5)^4 - 14641\zeta(5)^5)))))/$$

$$(7546\zeta(3) + 23716\zeta(3)^2 + 74536\zeta(3)^3 +$$

$$234256$$

$$\zeta(3)^4 - 3773$$

$$\zeta(5) - 23716$$

$$\zeta(3)$$

$$\zeta(5) - 111804$$

$$\zeta(3)^2$$

$$\zeta(5) -$$

$$468512\zeta(3)^3\zeta(5) + 5929\zeta(5)^2 +$$

$$55902\zeta(3)\zeta(5)^2 +$$

$$351384\zeta(3)^2\zeta(5)^2 -$$

$$9317\zeta(5)^3 -$$

$$117128\zeta(3)\zeta(5)^3 +$$

$$14641\zeta(5)^4 +$$

$$2401)$$

117649

$$\sqrt{(7/(11(2\zeta(3) - \zeta(5))(234256\zeta(3)^4 - 3773\zeta(5) + 5929\zeta(5)^2 - 9317\zeta(5)^3 + 14641\zeta(5)^4 - 10648\zeta(3)^3(44\zeta(5) - 7) + 484\zeta(3)^2(-231\zeta(5) + 726\zeta(5)^2 + 49) - 22\zeta(3)(1078\zeta(5) - 2541\zeta(5)^2 + 5324\zeta(5)^3 - 343) + 2401) (54875873536\zeta(3)^8 - 219503494144\zeta(3)^7\zeta(5) + 14235529\zeta(5)^2 + 35153041\zeta(5)^4 + 86806489\zeta(5)^6 + 214358881\zeta(5)^8 + 793659328\zeta(3)^6(484\zeta(5)^2 + 7) - 793659328\zeta(3)^5\zeta(5)(484\zeta(5)^2 + 21) - 3279584\zeta(3)^3\zeta(5)(4235\zeta(5)^2 + 29282\zeta(5)^4 + 343) + 1639792\zeta(3)^4(12705\zeta(5)^2 + 146410\zeta(5)^4 + 343) + 3388\zeta(3)^2(249018\zeta(5)^2 + 1537305\zeta(5)^4 + 7086244\zeta(5)^6 + 16807) - 484\zeta(3)\zeta(5)(581042\zeta(5)^2 + 2152227\zeta(5)^4 + 7086244\zeta(5)^6 + 117649) + 5764801)))))$$

Alternative representations

$$\frac{\sqrt{\frac{\left(1-\frac{11}{7}(2\zeta(3)-\zeta(5))\right)\left(1-\left(\frac{11}{7}(2\zeta(3)-\zeta(5))\right)^2\right)}{\left(1-\left(\frac{11}{7}(2\zeta(3)-\zeta(5))\right)^5\right)\left(1-\left(\frac{11}{7}(2\zeta(3)-\zeta(5))\right)^{10}\right)}}}{\sqrt{\frac{11}{7}(2\zeta(3)-\zeta(5))}} =$$

$$\frac{\sqrt{\frac{\left(1-\frac{11}{7}(2\zeta(3,1)-\zeta(5,1))\right)\left(1-\left(\frac{11}{7}(2\zeta(3,1)-\zeta(5,1))\right)^2\right)}{\left(1-\left(\frac{11}{7}(2\zeta(3,1)-\zeta(5,1))\right)^5\right)\left(1-\left(\frac{11}{7}(2\zeta(3,1)-\zeta(5,1))\right)^{10}\right)}}}{\sqrt{\frac{11}{7}(2\zeta(3,1)-\zeta(5,1))}}$$

$$\frac{\sqrt{\frac{\left(1-\frac{11}{7}(2\zeta(3)-\zeta(5))\right)\left(1-\left(\frac{11}{7}(2\zeta(3)-\zeta(5))\right)^2\right)}{\left(1-\left(\frac{11}{7}(2\zeta(3)-\zeta(5))\right)^5\right)\left(1-\left(\frac{11}{7}(2\zeta(3)-\zeta(5))\right)^{10}\right)}}}{\sqrt{\frac{11}{7}(2\zeta(3)-\zeta(5))}} =$$

$$\frac{\sqrt{\frac{\left(1-\frac{11}{7}(2S_{2,1}(1)-S_{4,1}(1))\right)\left(1-\left(\frac{11}{7}(2S_{2,1}(1)-S_{4,1}(1))\right)^2\right)}{\left(1-\left(\frac{11}{7}(2S_{2,1}(1)-S_{4,1}(1))\right)^5\right)\left(1-\left(\frac{11}{7}(2S_{2,1}(1)-S_{4,1}(1))\right)^{10}\right)}}}{\sqrt{\frac{11}{7}(2S_{2,1}(1)-S_{4,1}(1))}}$$

$$\frac{\sqrt{\frac{\left(1-\frac{11}{7}(2\zeta(3)-\zeta(5))\right)\left(1-\left(\frac{11}{7}(2\zeta(3)-\zeta(5))\right)^2\right)}{\left(1-\left(\frac{11}{7}(2\zeta(3)-\zeta(5))\right)^5\right)\left(1-\left(\frac{11}{7}(2\zeta(3)-\zeta(5))\right)^{10}\right)}}}{\sqrt{\frac{11}{7}(2\zeta(3)-\zeta(5))}} =$$

$$\frac{\sqrt{\frac{\left(1-\frac{11}{7}\left(-\frac{2\psi^{(2)}(1)}{2!}-\frac{\psi^{(4)}(1)(-1)^5}{4!}\right)\right)\left(1-\left(\frac{11}{7}\left(-\frac{2\psi^{(2)}(1)}{2!}-\frac{\psi^{(4)}(1)(-1)^5}{4!}\right)\right)^2\right)}{\left(1-\left(\frac{11}{7}\left(-\frac{2\psi^{(2)}(1)}{2!}-\frac{\psi^{(4)}(1)(-1)^5}{4!}\right)\right)^5\right)\left(1-\left(\frac{11}{7}\left(-\frac{2\psi^{(2)}(1)}{2!}-\frac{\psi^{(4)}(1)(-1)^5}{4!}\right)\right)^{10}\right)}}}{\sqrt{\frac{11}{7}\left(-\frac{2\psi^{(2)}(1)}{2!}-\frac{\psi^{(4)}(1)(-1)^5}{4!}\right)}}$$

$\zeta(s, a)$ is the generalized Riemann zeta function

$S_{n,p}(x)$ is the Nielsen generalized polylogarithm function

$\psi^{(n)}(x)$ is the n^{th} derivative of the digamma function

$n!$ is the factorial function

In conclusion, we obtain:

$$1/((0.004540210710940118 \times 1/[(1 - ((\sqrt{5}-1)/2)((11/7(2\zeta(3)-\zeta(5)))) + ((11/7(2\zeta(3)-\zeta(5)))^2)(1 - ((\sqrt{5}-1)/2)((11/7(2\zeta(3)-\zeta(5))))^2 + ((11/7(2\zeta(3)-\zeta(5)))^4)]^6))$$

Input interpretation

$$1 / \left(0.004540210710940118 \times \frac{1}{\left(\left(1 - \left(\frac{1}{2} (\sqrt{5} - 1) \right) \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right) + \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right)^2 \right) \left(1 - \left(\frac{1}{2} (\sqrt{5} - 1) \right) \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right)^2 + \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right)^4 \right) \right)^6} \right)$$

$\zeta(s)$ is the Riemann zeta function

Result

$$7.418355067475220... \times 10^{13}$$

$$7.418355067475220... \times 10^{13} \text{ (possible instanton value)}$$

Alternative representations

$$\frac{1}{0.0045402107109401180000} \left(\left(1 - \frac{(\sqrt{5}-1)(11(2\zeta(3)-\zeta(5)))}{2 \times 7} + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 \right) \left(1 - \frac{1}{2} (\sqrt{5}-1) \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^4 \right) \right)^6$$

$$= 1/0.0045402107109401180000 /$$

$$\left(\left(1 + \left(\frac{11}{7} (2\zeta(3, 1) - \zeta(5, 1)) \right)^4 - \frac{1}{2} \left(\frac{11}{7} (2\zeta(3, 1) - \zeta(5, 1)) \right)^2 (-1 + \sqrt{5}) \right) \left(1 + \left(\frac{11}{7} (2\zeta(3, 1) - \zeta(5, 1)) \right)^2 - \frac{11(-1 + \sqrt{5})(2\zeta(3, 1) - \zeta(5, 1))}{2 \times 7} \right) \right)^6$$

$$\begin{aligned}
& \left(\left(1 - \frac{(\sqrt{5}-1)(11(2\zeta(3)-\zeta(5)))}{2 \times 7} + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 \right) \left(1 - \frac{1}{2} (\sqrt{5}-1) \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^4 \right) \right)^6 \\
& = 1/0.0045402107109401180000 / \\
& \left(\left(1 + \left(\frac{11}{7} (2S_{2,1}(1) - S_{4,1}(1)) \right)^2 - \frac{11(2S_{2,1}(1) - S_{4,1}(1))(-1 + \sqrt{5})}{2 \times 7} \right) \right. \\
& \quad \left(1 + \left(\frac{11}{7} (2S_{2,1}(1) - S_{4,1}(1)) \right)^4 - \right. \\
& \quad \left. \left. \frac{1}{2} \left(\frac{11}{7} (2S_{2,1}(1) - S_{4,1}(1)) \right)^2 (-1 + \sqrt{5}) \right) \right)^6
\end{aligned}$$

$$\begin{aligned}
& \left(\left(1 - \frac{(\sqrt{5}-1)(11(2\zeta(3)-\zeta(5)))}{2 \times 7} + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 \right) \left(1 - \frac{1}{2} (\sqrt{5}-1) \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^4 \right) \right)^6 \\
& = 1/0.0045402107109401180000 / \\
& \left(\left(1 + \left(\frac{11}{7} \left(-\frac{2\psi^{(2)}(1)}{2!} - \frac{\psi^{(4)}(1)(-1)^5}{4!} \right) \right)^2 - \right. \\
& \quad \left. \frac{11 \left(-\frac{2\psi^{(2)}(1)}{2!} - \frac{\psi^{(4)}(1)(-1)^5}{4!} \right) (-1 + \sqrt{5})}{2 \times 7} \right) \\
& \quad \left(1 + \left(\frac{11}{7} \left(-\frac{2\psi^{(2)}(1)}{2!} - \frac{\psi^{(4)}(1)(-1)^5}{4!} \right) \right)^4 - \right. \\
& \quad \left. \frac{1}{2} \left(\frac{11}{7} \left(-\frac{2\psi^{(2)}(1)}{2!} - \frac{\psi^{(4)}(1)(-1)^5}{4!} \right) \right)^2 (-1 + \sqrt{5}) \right) \right)^6
\end{aligned}$$

$\zeta'(s, a)$ is the generalized Riemann zeta function

$S_{n,p}(x)$ is the Nielsen generalized polylogarithm function

$\psi^{(n)}(x)$ is the n^{th} derivative of the digamma function

$n!$ is the factorial function

Series representations

$$\begin{aligned}
& \frac{1}{0.0045402107109401180000} \\
& \left(\left(1 - \frac{(\sqrt{5}-1)(11(2\zeta(3)-\zeta(5)))}{2 \times 7} + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 \right) \left(1 - \frac{1}{2} (\sqrt{5}-1) \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^4 \right) \right)^6 \\
& = 220.25409472525013293 \\
& \left(1 + \frac{121}{49} \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right)^2 - \frac{11}{14} \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right) \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} \right) \right)^6 \\
& \left(1 + \frac{14641 \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right)^4}{2401} - \frac{121}{98} \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right)^2 \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} \right) \right)^6
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{0.0045402107109401180000} \\
& \left(\left(1 - \frac{(\sqrt{5}-1)(11(2\zeta(3)-\zeta(5)))}{2 \times 7} + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 \right) \left(1 - \frac{1}{2} (\sqrt{5}-1) \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^4 \right) \right)^6 \\
& = 220.25409472525013293 \\
& \left(1 + \frac{121}{49} \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right)^2 - \frac{11}{14} \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right) \right. \\
& \quad \left. \left(-1 + \exp \left(i \pi \left\lfloor \frac{\arg(5-x)}{2\pi} \right\rfloor \right) \sqrt{x} \sum_{k=0}^{\infty} \frac{(-1)^k (5-x)^k x^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)^6 \\
& \left(1 + \frac{14641 \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right)^4}{2401} - \frac{121}{98} \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right)^2 \right. \\
& \quad \left. \left(-1 + \exp \left(i \pi \left\lfloor \frac{\arg(5-x)}{2\pi} \right\rfloor \right) \sqrt{x} \sum_{k=0}^{\infty} \frac{(-1)^k (5-x)^k x^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)^6
\end{aligned}$$

for $(x \in \mathbb{R} \text{ and } x < 0)$

$$\begin{aligned}
& \left(\left(1 - \frac{(\sqrt{5}-1)(11(2\zeta(3)-\zeta(5)))}{2 \cdot 7} + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 \right) \left(1 - \frac{1}{2} (\sqrt{5}-1) \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^2 + \left(\frac{11}{7} (2\zeta(3)-\zeta(5)) \right)^4 \right) \right)^6 \\
& = 220.25409472525013293 \\
& \left(1 + \frac{121}{49} \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right)^2 - \frac{11}{14} \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right) \left(-1 + \left(\frac{1}{z_0} \right)^{1/2 \lfloor \arg(5-z_0)/(2\pi) \rfloor} \right. \right. \\
& \quad \left. \left. z_0^{1/2(1+\lfloor \arg(5-z_0)/(2\pi) \rfloor)} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k (5-z_0)^k z_0^{-k}}{k!} \right) \right)^6 \\
& \left(1 + \frac{14641 \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right)^4}{2401} - \frac{121}{98} \left(\sum_{k=1}^{\infty} \frac{-1+2k^2}{k^5} \right)^2 \left(-1 + \left(\frac{1}{z_0} \right)^{1/2 \lfloor \arg(5-z_0)/(2\pi) \rfloor} \right. \right. \\
& \quad \left. \left. z_0^{1/2(1+\lfloor \arg(5-z_0)/(2\pi) \rfloor)} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k (5-z_0)^k z_0^{-k}}{k!} \right) \right)^6
\end{aligned}$$

$\binom{n}{m}$ is the binomial coefficient

$\arg(z)$ is the complex argument

$\lfloor x \rfloor$ is the floor function

$(a)_n$ is the Pochhammer symbol (rising factorial)

i is the imaginary unit

$\mathbb{R}$ is the set of real numbers

We note the following mathematical connection:

$$\begin{aligned}
\mathfrak{L}^{\text{eff}}(z) = & 2^{10+3N^f} \pi^{6+2N^f} g^{-8} \int \rho^{3N^f-5} d\rho \exp\left\{-\frac{8\pi^2}{[g_R^D(\mu)]^2}\right. \\
& + \log(\mu\rho) \left[\frac{22}{3} - \frac{1}{6} \sum_i N^s(t) C(t) - \frac{2}{3} N^f \right] + A - \sum_i N^s(t) A(t) - N^f B \Big\} \\
& \times \left\langle \prod_{s=1}^{N^f} (\bar{\psi}_s \omega)(\bar{\omega} \psi_s) \right\rangle + \text{h.c.} , \tag{B13}
\end{aligned}$$

Considering:

$$\left\langle \prod_{s=1}^{N^f} (\bar{\psi}_s \omega)(\bar{\omega} \psi_s) \right\rangle + \text{h.c.}$$

$$= 1/2 , \text{ and } x = 1.2 ,$$

we obtain:

$$\begin{aligned}
\mathfrak{L}^{\text{eff}}(z) = & 2^{10+3N^f} \pi^{6+2N^f} g^{-8} \int \rho^{3N^f-5} d\rho \exp\left\{-\frac{8\pi^2}{[g_R^D(\mu)]^2}\right. \\
& + \log(\mu\rho) \left[\frac{22}{3} - \frac{1}{6} \sum_i N^s(t) C(t) - \frac{2}{3} N^f \right] + A - \sum_i N^s(t) A(t) - N^f B \Big\} \\
& \times \left\langle \prod_{s=1}^{N^f} (\bar{\psi}_s \omega)(\bar{\omega} \psi_s) \right\rangle + \text{h.c.} ,
\end{aligned}$$

$$\left(1.2^2 \times \frac{1}{2}\right) \times \frac{2^{16} \pi^{10}}{0.30282212^8} \int \exp\left(\frac{-8\pi^2}{768^2} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 4 - \frac{2}{3} \times 2\right) + 7.05399103 - (8 \times 1.09457662 + 2 \times (-0.35952290))\right) x dx$$

$$= 7.41835 \times 10^{13} x^2 \rightarrow 7.41835 * 10^{13}$$

And:

$$\frac{1}{\sqrt{\frac{11}{7} (2\zeta(3) - \zeta(5))}} \sqrt{\frac{\left(1 - \frac{11}{7} (2\zeta(3) - \zeta(5))\right) \left(1 - \left(\frac{11}{7} (2\zeta(3) - \zeta(5))\right)^2\right)}{\left(1 - \left(\frac{11}{7} (2\zeta(3) - \zeta(5))\right)^5\right) \left(1 - \left(\frac{11}{7} (2\zeta(3) - \zeta(5))\right)^{10}\right)}}$$

$$= 0.004540210710940118$$

$$1 / \left(0.004540210710940118 \times \frac{1}{\left(\left(1 - \left(\frac{1}{2} (\sqrt{5} - 1) \right) \right) \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right) + \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right)^2 \right) \left(1 - \left(\frac{1}{2} (\sqrt{5} - 1) \right) \right) \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right)^2 + \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right)^4 \right)^6} \right)$$

$$= 7.418355067475220... \times 10^{13} = 7.41835506... * 10^{13}$$

We have also:

$$\left(1.2^2 \times \frac{1}{2}\right) \times \frac{2^{16} \pi^{10}}{0.30282212^8} \int \exp\left(\frac{-8\pi^2}{768^2} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 - \frac{2}{3} \times 2\right) + 7.05399103 - (8 \times 0.30869069 + 2 \times (-0.35952290))\right) x dx$$

$\log(x)$ is the natural logarithm

Result

$$2.61357 \times 10^{21} x^2$$

$$2.61357 * 10^{21}$$

$$\frac{1}{((1/\sqrt{5} * \sqrt{((1-3.225558)(1-3.225558^2))/((1-3.225558^5)(1-3.225558^{10})))}) * 1/[(1-((\sqrt{5}-1)/2)*3.225558+3.225558^2)(1-((\sqrt{5}-1)/2)*3.225558^2+3.225558^4)]^6)}$$

Input interpretation

$$\frac{1}{\left(\frac{1}{\sqrt{5}} \sqrt{\frac{(1-3.225558)(1-3.225558^2)}{(1-3.225558^5)(1-3.225558^{10})}} \times \frac{1}{\left(\left(1 + \left(\frac{1}{2}(\sqrt{5}-1) \right) \times (-3.225558) + 3.225558^2 \right) \left(1 - \left(\frac{1}{2}(\sqrt{5}-1) \right) \times 3.225558^2 + 3.225558^4 \right) \right)^6} \right)}$$

Result

$$2.61357920891648959967863254093457001293968768615983444419274... \times 10^{21}$$

$$2.6135792... * 10^{21} \text{ (possible instanton value)}$$

And again:

$$\frac{1}{\sqrt{5} * \sqrt{((1-((7 + \log(3) \log(8) - \log(9))/\log(9)))/\log(9)))(1-((7 + \log(3) \log(8) - \log(9))/\log(9))^2))/((1-((7 + \log(3) \log(8) - \log(9))/\log(9))^5)(1-((7 + \log(3) \log(8) - \log(9))/\log(9))^10)))}$$

Input

$$\frac{1}{\sqrt{5}} \sqrt{\frac{\left(1 - \frac{7+\log(3) \log(8) - \log(9)}{\log(9)} \right) \left(1 - \left(\frac{7+\log(3) \log(8) - \log(9)}{\log(9)} \right)^2 \right)}{\left(1 - \left(\frac{7+\log(3) \log(8) - \log(9)}{\log(9)} \right)^5 \right) \left(1 - \left(\frac{7+\log(3) \log(8) - \log(9)}{\log(9)} \right)^{10} \right)}}$$

Exact result

$$\sqrt{\frac{\left(\frac{(7+\log(3)\log(8)-\log(9))^2}{\log^2(9)}-1\right)\left(\frac{7+\log(3)\log(8)-\log(9)}{\log(9)}-1\right)}{5\left(\frac{(7+\log(3)\log(8)-\log(9))^{10}}{\log^{10}(9)}-1\right)\left(\frac{(7+\log(3)\log(8)-\log(9))^5}{\log^5(9)}-1\right)}}$$

Decimal approximation

0.0003140393501792731335507126917038542424641666929497904934829324

...

0.000314039350179.....

Alternate forms

$$\begin{aligned} & (64 \log^6(3)) / \left((2401 + 16 \log^4(3) + 81 \log^4(2) \log^4(3) - \right. \\ & \quad 112 \log^3(3) - 54 \log^3(2) \log^3(3) (3 \log(3) - 14) + \\ & \quad 784 \log^2(3) + 18 \log^2(2) \log^2(3) (147 + 8 \log^2(3) - 63 \log(3)) - \\ & \quad \left. 6 \log(2) \log(3) (-686 + 8 \log^3(3) - 112 \log^2(3) + 441 \log(3)) - 2058 \log(3) \right) \\ & \sqrt{5 \left(2401 + 80 \log^4(3) + 81 \log^4(2) \log^4(3) - 560 \log^3(3) - \right. \\ & \quad 54 \log^3(2) \log^3(3) (5 \log(3) - 14) + 1960 \log^2(3) + \\ & \quad 18 \log^2(2) \log^2(3) (147 + 20 \log^2(3) - 105 \log(3)) - 6 \log(2) \log(3) \\ & \quad \left. (-686 + 40 \log^3(3) - 280 \log^2(3) + 735 \log(3)) - 3430 \log(3) \right) \end{aligned}$$

$$\begin{aligned} & \log^6(9) / \\ & \left((2401 + \log^4(3) \log^4(8) + \log^4(9) + 28 \log^3(3) \log^3(8) - 3 \log^3(3) \log^3(8) \log(9) - \right. \\ & \quad 14 \log^3(9) - 2 \log(3) \log(8) \log^3(9) + 294 \log^2(3) \log^2(8) - \\ & \quad 63 \log^2(3) \log^2(8) \log(9) + 196 \log^2(9) + \\ & \quad 56 \log(3) \log(8) \log^2(9) + 4 \log^2(3) \log^2(8) \log^2(9) + \\ & \quad \left. 1372 \log(3) \log(8) - 1029 \log(9) - 441 \log(3) \log(8) \log(9) \right) \\ & \sqrt{5 \left(2401 + \log^4(3) \log^4(8) + 5 \log^4(9) + 28 \log^3(3) \log^3(8) - \right. \\ & \quad 5 \log^3(3) \log^3(8) \log(9) - 70 \log^3(9) - 10 \log(3) \log(8) \log^3(9) + \\ & \quad 294 \log^2(3) \log^2(8) - 105 \log^2(3) \log^2(8) \log(9) + 490 \log^2(9) + \\ & \quad 140 \log(3) \log(8) \log^2(9) + 10 \log^2(3) \log^2(8) \log^2(9) + \\ & \quad \left. 1372 \log(3) \log(8) - 1715 \log(9) - 735 \log(3) \log(8) \log(9) \right) \end{aligned}$$

$$\sqrt{\frac{(7 + \log(3) (\log(8) - 4)) \left(\frac{(7 + \log(3) \log(8) - \log(9))^2}{4 \log^2(3)} - 1 \right)}{5 \left(\frac{(7 + \log(3) \log(8) - \log(9))^5}{32 \log^5(3)} - 1 \right) \left(\frac{(7 + \log(3) \log(8) - \log(9))^{10}}{1024 \log^{10}(3)} - 1 \right) \log(9)}}$$

$$\sqrt{\frac{\left(\frac{(7 - 2 \log(3) + 3 \log(2) \log(3))}{2 \log(3)} - 1 \right) \left(\frac{(7 - 2 \log(3) + 3 \log(2) \log(3))^2}{4 \log^2(3)} - 1 \right)}{5 \left(\frac{(7 - 2 \log(3) + 3 \log(2) \log(3))^5}{32 \log^5(3)} - 1 \right) \left(\frac{(7 - 2 \log(3) + 3 \log(2) \log(3))^{10}}{1024 \log^{10}(3)} - 1 \right)}}$$

Alternative representations

$$\sqrt{\frac{\left(1 - \frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right) \left(1 - \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 \right)}{\left(1 - \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^5 \right) \left(1 - \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^{10} \right)}} =$$

$$\sqrt{5}$$

$$\sqrt{\frac{\left(1 - \frac{7 + \text{Li}_1(-8) + \text{Li}_1(-7) \text{Li}_1(-2)}{\text{Li}_1(-8)} \right) \left(1 - \left(\frac{7 + \text{Li}_1(-8) + \text{Li}_1(-7) \text{Li}_1(-2)}{\text{Li}_1(-8)} \right)^2 \right)}{\left(1 - \left(\frac{7 + \text{Li}_1(-8) + \text{Li}_1(-7) \text{Li}_1(-2)}{\text{Li}_1(-8)} \right)^5 \right) \left(1 - \left(\frac{7 + \text{Li}_1(-8) + \text{Li}_1(-7) \text{Li}_1(-2)}{\text{Li}_1(-8)} \right)^{10} \right)}} =$$

$$\sqrt{5}$$

$$\sqrt{\frac{\left(1 - \frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right) \left(1 - \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 \right)}{\left(1 - \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^5 \right) \left(1 - \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^{10} \right)}} =$$

$$\sqrt{5}$$

$$\sqrt{\frac{\left(1 - \frac{7 + \log_e(3) \log_e(8) - \log_e(9)}{\log_e(9)} \right) \left(1 - \left(\frac{7 + \log_e(3) \log_e(8) - \log_e(9)}{\log_e(9)} \right)^2 \right)}{\left(1 - \left(\frac{7 + \log_e(3) \log_e(8) - \log_e(9)}{\log_e(9)} \right)^5 \right) \left(1 - \left(\frac{7 + \log_e(3) \log_e(8) - \log_e(9)}{\log_e(9)} \right)^{10} \right)}} =$$

$$\sqrt{5}$$

$$\sqrt{\frac{\left(1 - \frac{7 + \log(3) \log(8) - \log(9)}{\log(9)}\right) \left(1 - \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)}\right)^2\right)}{\left(1 - \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)}\right)^5\right) \left(1 - \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)}\right)^{10}\right)}}} = \sqrt{\frac{\left(1 - \frac{7 + \log^2(a) \log_{\log(a)}(3) \log_{\log(a)}(8) - \log(a) \log_{\log(a)}(9)}{\log(a) \log_{\log(a)}(9)}\right) \left(1 - \left(\frac{7 + \log^2(a) \log_{\log(a)}(3) \log_{\log(a)}(8) - \log(a) \log_{\log(a)}(9)}{\log(a) \log_{\log(a)}(9)}\right)^2\right)}{\left(1 - \left(\frac{7 + \log^2(a) \log_{\log(a)}(3) \log_{\log(a)}(8) - \log(a) \log_{\log(a)}(9)}{\log(a) \log_{\log(a)}(9)}\right)^5\right) \left(1 - \left(\frac{7 + \log^2(a) \log_{\log(a)}(3) \log_{\log(a)}(8) - \log(a) \log_{\log(a)}(9)}{\log(a) \log_{\log(a)}(9)}\right)^{10}\right)}}}$$

$\text{Li}_n(x)$ is the polylogarithm function

$\log_b(x)$ is the base- b logarithm

From which:

$$1/((0.000314039350179 * 1 / [(1 - ((\sqrt{5} - 1)/2) * ((7 + \ln(3) \ln(8) - \ln(9))/\ln(9)) + ((7 + \ln(3) \ln(8) - \ln(9))/\ln(9))^2] * (1 - ((\sqrt{5} - 1)/2) * ((7 + \ln(3) \ln(8) - \ln(9))/\ln(9))^2 + ((7 + \ln(3) \ln(8) - \ln(9))/\ln(9))^4])^6))$$

Input interpretation

$$1 / \left(0.000314039350179 \times 1 / \left(\left(1 - \left(\frac{1}{2} (\sqrt{5} - 1) \right) \times \frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} + \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 \right) \left(1 - \left(\frac{1}{2} (\sqrt{5} - 1) \right) \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 + \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^4 \right) \right)^6 \right)$$

$\log(x)$ is the natural logarithm

Result

$$2.61313101973... \times 10^{21}$$

2.61313101973*10²¹ (possible instanton value)

Alternative representations

$$\begin{aligned}
 & 1 / 0.0003140393501790000 / \\
 & \left(\left(1 - \frac{(\sqrt{5} - 1)(7 + \log(3) \log(8) - \log(9))}{2 \log(9)} + \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 \right) \right. \\
 & \quad \left(1 - \frac{1}{2} (\sqrt{5} - 1) \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 + \right. \\
 & \quad \left. \left. \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^4 \right) \right)^6 = \\
 & 1 / 0.0003140393501790000 / \left(\left(1 + \left(- \frac{7 + \text{Li}_1(-8) + \text{Li}_1(-7) \text{Li}_1(-2)}{\text{Li}_1(-8)} \right)^2 - \right. \right. \\
 & \quad \left. \frac{(7 + \text{Li}_1(-8) + \text{Li}_1(-7) \text{Li}_1(-2))(-1 + \sqrt{5})}{2(-\text{Li}_1(-8))} \right) \\
 & \quad \left(1 + \left(- \frac{7 + \text{Li}_1(-8) + \text{Li}_1(-7) \text{Li}_1(-2)}{\text{Li}_1(-8)} \right)^4 - \right. \\
 & \quad \left. \left. \frac{1}{2} \left(- \frac{7 + \text{Li}_1(-8) + \text{Li}_1(-7) \text{Li}_1(-2)}{\text{Li}_1(-8)} \right)^2 (-1 + \sqrt{5}) \right) \right)^6
 \end{aligned}$$

$$\begin{aligned}
 & 1 / 0.0003140393501790000 / \\
 & \left(\left(1 - \frac{(\sqrt{5} - 1)(7 + \log(3) \log(8) - \log(9))}{2 \log(9)} + \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 \right) \right. \\
 & \quad \left(1 - \frac{1}{2} (\sqrt{5} - 1) \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 + \right. \\
 & \quad \left. \left. \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^4 \right) \right)^6 = \\
 & 1 / 0.0003140393501790000 / \left(\left(1 + \left(\frac{7 + \log_e(3) \log_e(8) - \log_e(9)}{\log_e(9)} \right)^2 - \right. \right. \\
 & \quad \left. \frac{(7 + \log_e(3) \log_e(8) - \log_e(9))(-1 + \sqrt{5})}{2 \log_e(9)} \right) \\
 & \quad \left(1 + \left(\frac{7 + \log_e(3) \log_e(8) - \log_e(9)}{\log_e(9)} \right)^4 - \right. \\
 & \quad \left. \left. \frac{1}{2} \left(\frac{7 + \log_e(3) \log_e(8) - \log_e(9)}{\log_e(9)} \right)^2 (-1 + \sqrt{5}) \right) \right)^6
 \end{aligned}$$

$$\begin{aligned}
& 1 / 0.0003140393501790000 / \\
& \left(\left(1 - \frac{(\sqrt{5} - 1)(7 + \log(3) \log(8) - \log(9))}{2 \log(9)} + \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 \right) \right. \\
& \quad \left(1 - \frac{1}{2} (\sqrt{5} - 1) \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 + \right. \\
& \quad \quad \left. \left. \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^4 \right) \right)^6 = \\
& 1 / 0.0003140393501790000 / \left(\left(1 + \left(- \frac{7 + S_{0,1}(-8) + S_{0,1}(-7) S_{0,1}(-2)}{S_{0,1}(-8)} \right)^2 - \right. \right. \\
& \quad \left. \left. \frac{(7 + S_{0,1}(-8) + S_{0,1}(-7) S_{0,1}(-2))(-1 + \sqrt{5})}{2(-S_{0,1}(-8))} \right) \right. \\
& \quad \left(1 + \left(- \frac{7 + S_{0,1}(-8) + S_{0,1}(-7) S_{0,1}(-2)}{S_{0,1}(-8)} \right)^4 - \right. \\
& \quad \quad \left. \left. \frac{1}{2} \left(- \frac{7 + S_{0,1}(-8) + S_{0,1}(-7) S_{0,1}(-2)}{S_{0,1}(-8)} \right)^2 (-1 + \sqrt{5}) \right) \right)^6
\end{aligned}$$

$\text{Li}_n(x)$ is the polylogarithm function

$\log_b(x)$ is the base- b logarithm

$S_{n,p}(x)$ is the Nielsen generalized polylogarithm function

Series representations

$$\begin{aligned}
 & 1 / 0.0003140393501790000 / \\
 & \left(\left(1 - \frac{(\sqrt{5} - 1)(7 + \log(3) \log(8) - \log(9))}{2 \log(9)} + \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 \right) \right. \\
 & \quad \left(1 - \frac{1}{2} (\sqrt{5} - 1) \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 + \right. \\
 & \quad \left. \left. \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^4 \right) \right)^6 = \\
 & 3184.314320578003 \left(1 + \frac{(7 + \log(3) \log(8) - \log(9))^4}{\log^4(9)} - \right. \\
 & \quad \left. \frac{(7 + \log(3) \log(8) - \log(9))^2 \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} \right)}{2 \log^2(9)} \right)^6 \\
 & \left(1 + \frac{(7 + \log(3) \log(8) - \log(9))^2}{\log^2(9)} - \right. \\
 & \quad \left. \frac{(7 + \log(3) \log(8) - \log(9)) \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} \right)}{2 \log(9)} \right)^6
 \end{aligned}$$

$$\begin{aligned}
& 1 / 0.0003140393501790000 / \\
& \left(\left(1 - \frac{(\sqrt{5} - 1)(7 + \log(3) \log(8) - \log(9))}{2 \log(9)} + \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 \right) \right. \\
& \quad \left(1 - \frac{1}{2}(\sqrt{5} - 1) \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 + \right. \\
& \quad \quad \left. \left. \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^4 \right) \right)^6 = 3184.314320578003 \\
& \left(1 + \frac{(7 + \log(3) \log(8) - \log(9))^4}{\log^4(9)} - \frac{1}{2 \log^2(9)} (7 + \log(3) \log(8) - \log(9))^2 \right. \\
& \quad \left. \left(-1 + \exp \left(i \pi \left\lfloor \frac{\arg(5 - x)}{2 \pi} \right\rfloor \right) \sqrt{x} \sum_{k=0}^{\infty} \frac{(-1)^k (5 - x)^k x^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)^6 \\
& \left(1 + \frac{(7 + \log(3) \log(8) - \log(9))^2}{\log^2(9)} - \frac{1}{2 \log(9)} (7 + \log(3) \log(8) - \log(9)) \right. \\
& \quad \left. \left(-1 + \exp \left(i \pi \left\lfloor \frac{\arg(5 - x)}{2 \pi} \right\rfloor \right) \sqrt{x} \sum_{k=0}^{\infty} \frac{(-1)^k (5 - x)^k x^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)^6 \\
& \text{for } (x \in \mathbb{R} \text{ and } x < 0)
\end{aligned}$$

$$\begin{aligned}
& 1 / 0.0003140393501790000 / \\
& \left(\left(1 - \frac{(\sqrt{5} - 1)(7 + \log(3) \log(8) - \log(9))}{2 \log(9)} + \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 \right) \right. \\
& \quad \left(1 - \frac{1}{2} (\sqrt{5} - 1) \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^2 + \right. \\
& \quad \quad \left. \left. \left(\frac{7 + \log(3) \log(8) - \log(9)}{\log(9)} \right)^4 \right) \right)^6 = \\
& 3184.314320578003 \left(1 + \frac{(7 + \log(3) \log(8) - \log(9))^4}{\log^4(9)} - \right. \\
& \quad \frac{1}{2 \log^2(9)} (7 + \log(3) \log(8) - \log(9))^2 \left(-1 + \left(\frac{1}{z_0} \right)^{1/2 [\arg(5 - z_0)/(2\pi)]} \right. \\
& \quad \quad \left. z_0^{1/2 (1 + [\arg(5 - z_0)/(2\pi)])} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (5 - z_0)^k z_0^{-k}}{k!} \right)^6 \\
& \quad \left(1 + \frac{(7 + \log(3) \log(8) - \log(9))^2}{\log^2(9)} - \frac{1}{2 \log(9)} (7 + \log(3) \log(8) - \log(9)) \right. \\
& \quad \quad \left(-1 + \left(\frac{1}{z_0} \right)^{1/2 [\arg(5 - z_0)/(2\pi)]} z_0^{1/2 (1 + [\arg(5 - z_0)/(2\pi)])} \right. \\
& \quad \quad \quad \left. \left. \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (5 - z_0)^k z_0^{-k}}{k!} \right) \right)^6
\end{aligned}$$

We have also:

$$R = \frac{1}{12} (\log 2\pi + \gamma) + \frac{1}{2\pi^2} \sum_2^{\infty} \frac{\log s}{s^2} = 0.248754477.$$

From $((\sqrt{(10-2\sqrt{5})}-2))/(\sqrt{5}-1)) = \kappa$, after some calculations, we obtain:

$$\begin{aligned}
& 1/2((2(11+1/\sqrt{3})((((\sqrt{(10-2\sqrt{5})}-2))/(\sqrt{5}-1))) * 0.248754477)) + \\
& 2(11+1/\sqrt{2})((((\sqrt{(10-2\sqrt{5})}-2))/(\sqrt{5}-1))) * 0.248754477))
\end{aligned}$$

Input interpretation

$$\frac{1}{2} \left(2 \left(11 + \frac{1}{\sqrt{3}} \right) \left(\frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} \times 0.248754477 \right) + \right. \\ \left. 2 \left(11 + \frac{1}{\sqrt{2}} \right) \left(\frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} \times 0.248754477 \right) \right)$$

Result

1.64541790...

1.64541790.... $\approx \zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$ (trace of the instanton shape)

And:

$$((2(11+1/\sqrt{2})((((\sqrt{(10-2\sqrt{5})}-2))/(\sqrt{5}-1)))\times 0.248754477))))$$

Input interpretation

$$2 \left(11 + \frac{1}{\sqrt{2}} \right) \left(\frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} \times 0.248754477 \right)$$

Result

1.6545872697262925632541080963586485848029382391805230514062031935

...

1.654587269..... result very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

From:

D-instantons in Type IIA string theory on Calabi-Yau threefolds

Sergei Alexandrov, Ashoke Sen, Bogdan Stefanski, jr. - arXiv:2108.04265v1 [hep-th]

9 Aug 2021

We have:

$$ds_{\text{inst}}^2 = \sum_{\gamma} \frac{\Omega_{\gamma} \Sigma_{\gamma}}{16r \sqrt{2\pi \mathcal{T}_{\gamma}^{\text{R}}}} \left[\frac{|Z_{\gamma}|^2}{rK} \left(d\sigma + \tilde{\zeta}_{\Lambda} d\zeta^{\Lambda} - \zeta^{\Lambda} d\tilde{\zeta}_{\Lambda} - \frac{4i\sqrt{rK}}{|Z_{\gamma}|} \mathcal{C}_{\gamma} + 8r \operatorname{Im} \partial \log \frac{Z_{\gamma}}{K} \right)^2 \right. \\ \left. - \frac{1}{\pi^2} \left(d\mathcal{T}_{\gamma} - 8\pi |Z_{\gamma}| \sqrt{\frac{r}{K}} d \log \frac{|Z_{\gamma}|^2}{K} - \frac{16\pi i}{K} \operatorname{Im} \left[\bar{Z}_{\gamma} (z^{\Lambda} d\tilde{\zeta}_{\Lambda} - F_{\Lambda} d\zeta^{\Lambda}) \right] \right) d\mathcal{T}_{\gamma} \right], \quad (3.5)$$

$$\mathcal{K} = -\log K, \quad K = i \int_{\mathfrak{Y}} \bar{\Omega} \wedge \Omega = z^{\Lambda} N_{\Lambda\Sigma} \bar{z}^{\Sigma}, \\ N_{\Lambda\Sigma} \equiv -2 \operatorname{Im} F_{\Lambda\Sigma}, \quad F_{\Lambda\Sigma} \equiv \partial_{z^{\Lambda}} \partial_{z^{\Sigma}} F,$$

For: $K = -2*i*i*F$; we consider $F = 8$; $K = 16$

$$r = \frac{V}{2(2\pi)^6 g_s^2} = \frac{\pi V}{2\kappa^2},$$

$$\mathcal{T}_{\gamma} = 8\pi \sqrt{\frac{r}{K}} |Z_{\gamma}| + 2\pi i \Theta_{\gamma}.$$

$$= 6.05892*10^{-8}$$

$$-1.55982... = z = \Theta_{\gamma}$$

$$|Z_{\gamma}| = -0.5 \quad V = -0.5$$

From:

$$\mathcal{T}_\gamma = 8\pi\sqrt{\frac{r}{K}}|Z_\gamma| + 2\pi i\Theta_\gamma.$$

$$8\pi\sqrt{\frac{\pi x}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2\times\frac{1}{16}}}y+2\pi i z$$

Input

$$8\pi\sqrt{\frac{\pi x}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2\times\frac{1}{16}}}y+2\pi i z$$

i is the imaginary unit

Exact result

$$\frac{\sqrt{2}\left(\sqrt{5}-1\right)\pi^{3/2}\sqrt{x}y}{\sqrt{10-2\sqrt{5}}-2}+2i\pi z$$

Alternate forms

$$\sqrt{2\left(4+\sqrt{5}+2\sqrt{5+2\sqrt{5}}\right)}\pi^{3/2}\sqrt{x}y+2i\pi z$$

$$\pi\left(\frac{\left(\sqrt{5}-1\right)\sqrt{2\pi}\sqrt{x}y}{\sqrt{10-2\sqrt{5}}-2}+2i z\right)$$

Expanded form

$$\frac{\sqrt{10}\pi^{3/2}\sqrt{x}y}{\sqrt{10-2\sqrt{5}}-2}-\frac{\sqrt{2}\pi^{3/2}\sqrt{x}y}{\sqrt{10-2\sqrt{5}}-2}+2i\pi z$$

Root for the variable z

$$z = \frac{i(\sqrt{5}-1)\sqrt{\frac{\pi}{2}}\sqrt{x}y}{\sqrt{2(5-\sqrt{5})}-2}$$

$$z \approx (4.41185 i) \sqrt{x} y$$

Partial derivatives

$$\frac{\partial}{\partial x} \left(\frac{\sqrt{2}(\sqrt{5}-1)\pi^{3/2}\sqrt{x}y}{\sqrt{10-2\sqrt{5}}-2} + 2i\pi z \right) = \frac{(\sqrt{5}-1)\pi^{3/2}y}{\sqrt{2}(\sqrt{10-2\sqrt{5}}-2)\sqrt{x}}$$

$$\frac{\partial}{\partial y} \left(\frac{\sqrt{2}(\sqrt{5}-1)\pi^{3/2}\sqrt{x}y}{\sqrt{10-2\sqrt{5}}-2} + 2i\pi z \right) = \frac{\sqrt{2}(\sqrt{5}-1)\pi^{3/2}\sqrt{x}}{\sqrt{10-2\sqrt{5}}-2}$$

$$\frac{\partial}{\partial z} \left(\frac{\sqrt{2}(\sqrt{5}-1)\pi^{3/2}\sqrt{x}y}{\sqrt{10-2\sqrt{5}}-2} + 2i\pi z \right) = 2i\pi$$

Indefinite integral

$$\int \left(8\pi \sqrt{\frac{\pi x}{\left(2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2\right)^{16}}} y + 2\pi i z \right) dx =$$

$$\pi \left(\frac{2(\sqrt{5}-1)\sqrt{2\pi}x^{3/2}y}{3(\sqrt{10-2\sqrt{5}}-2)} + 2ixz \right) + \text{constant}$$

From

$$z = \frac{i(\sqrt{5}-1)\sqrt{\frac{\pi}{2}}\sqrt{x}y}{\sqrt{2(5-\sqrt{5})}-2}}$$

$$z \approx (4.41185 i) \sqrt{x} y$$

we obtain:

$$8\pi \sqrt{\frac{\pi x}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}} \times \frac{1}{16} y + 2\pi i((4.41185 i) \sqrt{x} y)$$

Input interpretation

$$8\pi \sqrt{\frac{\pi x}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}} \times \frac{1}{16} y + 2\pi i((4.41185 i) \sqrt{x} y)$$

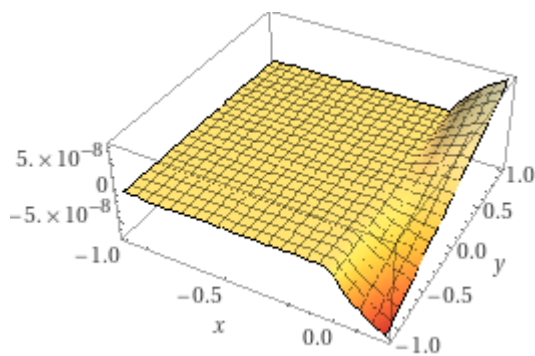
i is the imaginary unit

Result

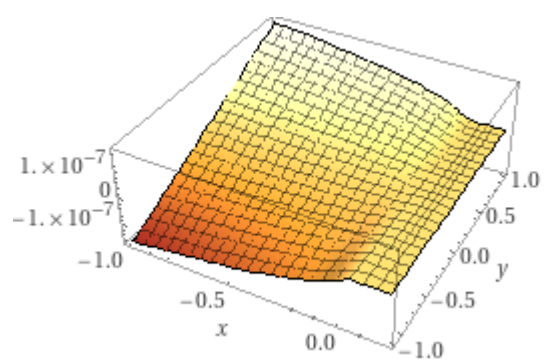
$$(1.71372 \times 10^{-7} + 0 i) \sqrt{x} y$$

3D plots

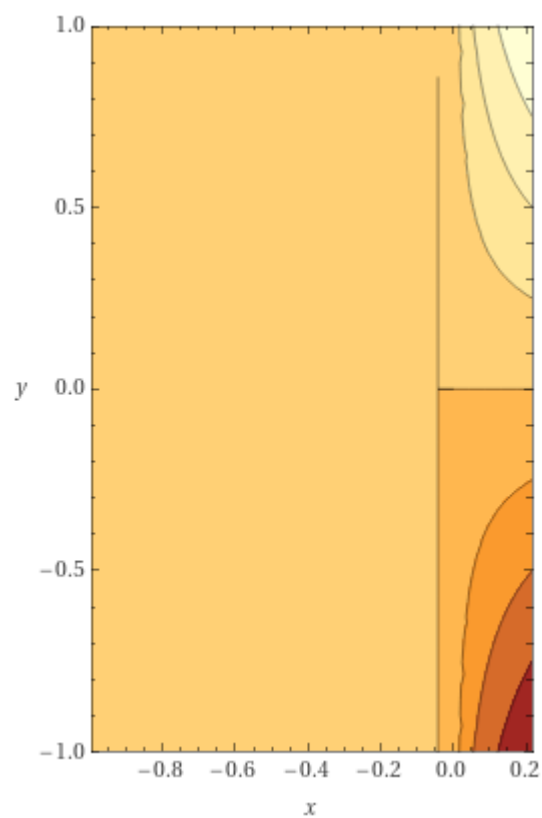
Real part (figures that can be related to the D-branes/Instantons)



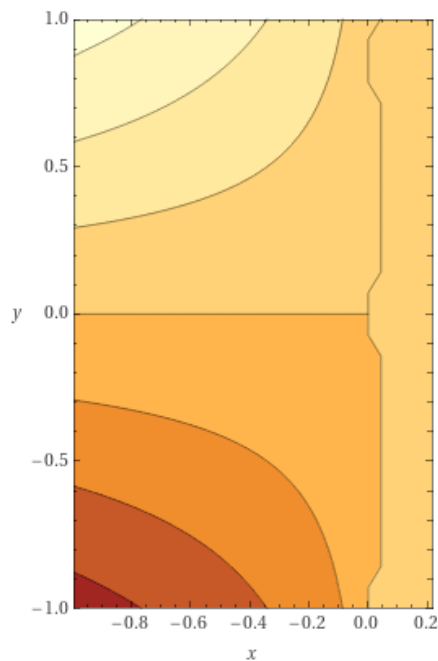
Imaginary part



Contour plots Real part



Imaginary part



Expanded form

$$1.71372 \times 10^{-7} \sqrt{x} \ y$$

Roots

$$x \neq 0, \quad y = 0$$

$$x = 0$$

Root for the variable y

$$y = 0$$

Partial derivatives

$$\frac{\partial}{\partial x} (1.71372 \times 10^{-7} \sqrt{x} \ y) = \frac{8.56861 \times 10^{-8} \ y}{\sqrt{x}}$$

$$\frac{\partial}{\partial y}(1.71372 \times 10^{-7} \sqrt{x} y) = 1.71372 \times 10^{-7} \sqrt{x}$$

Indefinite integral

$$\int \left(8 \pi \sqrt{\frac{\pi x}{\left(2 \left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1} \right)^2 \right) 16}} y + 2 \pi i \left((4.41185 i) \sqrt{x} y \right) \right) dx =$$

$$1.14248 \times 10^{-7} x^{3/2} y + \text{constant}$$

Series representations

$$8 \pi \sqrt{\frac{\pi x}{16 \left(2 \left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1} \right)^2 \right)}} y + (2 (\sqrt{x} y) \pi i) 4.41185 i = \sum_{k=0}^{\infty} \pi y \binom{\frac{1}{2}}{k}$$

$$\left(8.8237 i^2 (-1+x)^{-k} \sqrt{-1+x} + 8 \left(-1 + \frac{\pi x (-1+\sqrt{5})^2}{32 (-2+\sqrt{10-2\sqrt{5}})^2} \right)^{-k} \right.$$

$$\left. \sqrt{-1 + \frac{\pi x (-1+\sqrt{5})^2}{32 (-2+\sqrt{10-2\sqrt{5}})^2}} \right)$$

$$\text{for } \left(|-1+x| > 1 \text{ and } \left| -1 + \frac{\pi x (-1+\sqrt{5})^2}{32 (-2+\sqrt{10-2\sqrt{5}})^2} \right| > 1 \right)$$

$$8\pi \sqrt{\frac{\pi x}{16\left(2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2\right)}} y + (2(\sqrt{x} y)\pi i) 4.41185 i =$$

$$\sum_{k=0}^{\infty} \frac{8(-1)^k \pi y \left(-\frac{1}{2}\right)_k \left(1.10296 i^2 (-1+x)^k + \left(-1 + \frac{\pi x (-1+\sqrt{5})^2}{32(-2+\sqrt{10-2\sqrt{5}})^2}\right)^k\right)}{k!}$$

for $\left(-1+x\right) < 1$ and $\left|-1 + \frac{\pi x (-1+\sqrt{5})^2}{32(-2+\sqrt{10-2\sqrt{5}})^2}\right| < 1$

$$8\pi \sqrt{\frac{\pi x}{16\left(2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2\right)}} y + (2(\sqrt{x} y)\pi i) 4.41185 i = \sum_{k=0}^{\infty} \frac{1}{k!} (-1)^k \pi y \left(-\frac{1}{2}\right)_k$$

$$\left(8.8237 i^2 (-1+x)^{-k} \sqrt{-1+x} + 8 \left(-1 + \frac{\pi x (-1+\sqrt{5})^2}{32(-2+\sqrt{10-2\sqrt{5}})^2}\right)^{-k} \sqrt{-1 + \frac{\pi x (-1+\sqrt{5})^2}{32(-2+\sqrt{10-2\sqrt{5}})^2}}\right)$$

for $\left(-1+x\right) > 1$ and $\left|-1 + \frac{\pi x (-1+\sqrt{5})^2}{32(-2+\sqrt{10-2\sqrt{5}})^2}\right| > 1$

From

$$(1.71372 \times 10^{-7} + 0i) \sqrt{x} y$$

$$1.71372 \times 10^{-7} \sqrt{-0.5} (-0.5)$$

Input interpretation

$$1.71372 \times 10^{-7} \sqrt{-0.5} \times (-0.5)$$

Result

$$-6.05892... \times 10^{-8} i$$

Polar coordinates

$$r = 6.05892 \times 10^{-8} \text{ (radius), } \theta = -1.5708 \text{ (angle)}$$

$$6.05892 \times 10^{-8}$$

From:

$$z \approx (4.41185 i) \sqrt{x} y$$

$$((4.41185 i) \sqrt{x}) y$$

Input interpretation

$$(4.41185 i) \sqrt{-0.5} \times 0.5$$

i is the imaginary unit

Result

$$-1.55982...$$

$$-1.55982... = z$$

From:

$$\Sigma_{\gamma} = \sum_{k=1}^{\infty} \frac{\sigma_{k\gamma}}{\sqrt{k}} e^{-kT_{\gamma}}$$

For:

$$\sigma_{k\gamma} = 1$$

$$k = 2$$

$$1/\text{sqrt}2 * e^{(-2*6.05892*10^{-8})}$$

Input interpretation

$$\frac{1}{\sqrt{2}} \exp(-2 \times 6.05892 \times 10^{-8})$$

Result

0.7071066955005...

0.707106695.....

From:

$$\Omega = z^\Lambda \alpha_\Lambda - F_\Lambda \beta^\Lambda,$$

$$-1.55982*64 - 8*8$$

Input interpretation

$$-1.55982 \times 64 - 8 \times 8$$

Result

-163.82848

-163.82848

From:

$$\sum_{\gamma} \frac{\Omega_{\gamma} \Sigma_{\gamma}}{16r \sqrt{2\pi \mathcal{T}_{\gamma}^{\text{R}}}} \left[\frac{|Z_{\gamma}|^2}{rK} \left(d\sigma + \tilde{\zeta}_{\Lambda} d\zeta^{\Lambda} - \zeta^{\Lambda} d\tilde{\zeta}_{\Lambda} - \frac{4i\sqrt{rK}}{|Z_{\gamma}|} \mathcal{C}_{\gamma} + 8r \operatorname{Im} \partial \log \frac{Z_{\gamma}}{K} \right)^2 \right. \\ \left. - \frac{1}{\pi^2} \left(d\mathcal{T}_{\gamma} - 8\pi |Z_{\gamma}| \sqrt{\frac{r}{K}} d \log \frac{|Z_{\gamma}|^2}{K} - \frac{16\pi i}{K} \operatorname{Im} \left[\bar{Z}_{\gamma} (z^{\Lambda} d\tilde{\zeta}_{\Lambda} - F_{\Lambda} d\zeta^{\Lambda}) \right] \right) d\mathcal{T}_{\gamma} \right],$$

$$(-163.82848 \times 1/\sqrt{2}) \times 1/(16 \times (((-0.5\pi)/(2 \times (((\sqrt{10-2\sqrt{5}}-2))/(\sqrt{5}-1))))^2))) \times \sqrt{2\pi \times 6.05892 \times 10^{-8}}$$

Input interpretation

$$\left(-163.82848 \times \frac{1}{\sqrt{2}}\right) \times \frac{1}{16 \times \frac{-0.5\pi}{2 \left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2} \sqrt{2\pi \times 6.05892 \times 10^{-8}}}$$

Result

1205.74...

1205.74....

$$(-0.5)^2 / (16 \times (((-0.5\pi)/(2 \times (((\sqrt{10-2\sqrt{5}}-2))/(\sqrt{5}-1))))^2))) \times [(1-4i \times \sqrt{16 \times (((-0.5\pi)/(2 \times (((\sqrt{10-2\sqrt{5}}-2))/(\sqrt{5}-1))))^2)))/(-0.5)] + 8 \times (((-0.5\pi)/(2 \times (((\sqrt{10-2\sqrt{5}}-2))/(\sqrt{5}-1))))^2) \times i \times \ln(-0.5/16)]^2$$

Input

$$\frac{(-0.5)^2}{16 \times \frac{-0.5\pi}{2 \left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}} \left(\sqrt{16 \times \frac{-0.5\pi}{2 \left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}} + 8 \times \frac{-0.5\pi}{2 \left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2} \left(i \log\left(-\frac{0.5}{16}\right)\right) \right)^2$$

$\log(x)$ is the natural logarithm

i is the imaginary unit

Result

- 33.1046... -

157.126... i

Polar coordinates

$r = 160.576$ (radius), $\theta = -1.77845$ (angle)

160.576

Polar forms

$$160.576 (\cos(-1.77845) + i \sin(-1.77845))$$

$$160.576 e^{-1.77845 i}$$

Alternative representations

$$\frac{\left(1 - i \left(4 \sqrt{\frac{16(-0.5\pi)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}}\right) + \frac{8(-0.5\pi)\left(i \log\left(-\frac{0.5}{16}\right)\right)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}\right)^2 (-0.5)^2}{\frac{16(-0.5\pi)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}} =$$

$$- \frac{(-0.5)^2 \left(1 - \frac{4i\pi \log\left(-\frac{0.5}{16}\right)}{2\left(\frac{-2+\sqrt{10-2\sqrt{5}}}{-1+\sqrt{5}}\right)^2} - 4i \sqrt{\frac{-8\pi}{-0.5\left(2\left(\frac{-2+\sqrt{10-2\sqrt{5}}}{-1+\sqrt{5}}\right)^2\right)}}\right)^2}{\frac{8\pi}{2\left(\frac{-2+\sqrt{10-2\sqrt{5}}}{-1+\sqrt{5}}\right)^2}}$$

$$\begin{aligned}
& \frac{\left(1 - i \left(4 \sqrt{\frac{16(-0.5\pi)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}}\right) + \frac{8(-0.5\pi)\left(i \log\left(-\frac{0.5}{16}\right)\right)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}\right)^2 (-0.5)^2}{\frac{16(-0.5\pi)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}} = \\
& - \frac{(-0.5)^2 \left(1 - \frac{4i\pi \log(a) \log_a\left(-\frac{0.5}{16}\right)}{2\left(\frac{-2+\sqrt{10-2\sqrt{5}}}{-1+\sqrt{5}}\right)^2} - 4i \sqrt{\frac{-8\pi}{-0.5\left(2\left(\frac{-2+\sqrt{10-2\sqrt{5}}}{-1+\sqrt{5}}\right)^2\right)}}\right)^2}{\frac{8\pi}{2\left(\frac{-2+\sqrt{10-2\sqrt{5}}}{-1+\sqrt{5}}\right)^2}}
\end{aligned}$$

$$\begin{aligned}
& \frac{\left(1 - i \left(4 \sqrt{\frac{16(-0.5\pi)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}}\right) + \frac{8(-0.5\pi)\left(i \log\left(-\frac{0.5}{16}\right)\right)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}\right)^2 (-0.5)^2}{\frac{16(-0.5\pi)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}} = \\
& - \frac{(-0.5)^2 \left(1 + \frac{4i\pi \operatorname{Li}_1\left(1+\frac{0.5}{16}\right)}{2\left(\frac{-2+\sqrt{10-2\sqrt{5}}}{-1+\sqrt{5}}\right)^2} - 4i \sqrt{\frac{-8\pi}{-0.5\left(2\left(\frac{-2+\sqrt{10-2\sqrt{5}}}{-1+\sqrt{5}}\right)^2\right)}}\right)^2}{\frac{8\pi}{2\left(\frac{-2+\sqrt{10-2\sqrt{5}}}{-1+\sqrt{5}}\right)^2}}
\end{aligned}$$

Integral representation

$$\frac{\left(1-i\left(4\sqrt{\frac{16(-0.5\pi)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}}+ \frac{8(-0.5\pi)\left(i\log\left(-\frac{0.5}{16}\right)\right)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}\right)(-0.5)^2\right)^2}{\frac{16(-0.5\pi)}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}} =$$

$$-\frac{1}{\pi(-1+\sqrt{5})^2}0.0625\left(-2+\sqrt{10-2\sqrt{5}}\right)^2$$

$$\left(1-\frac{2i\pi(-1+\sqrt{5})^2}{\left(-2+\sqrt{10-2\sqrt{5}}\right)^2}\int_1^{-0.03125}\frac{1}{t}dt-4i\sqrt{\frac{8\pi(-1+\sqrt{5})^2}{\left(-2+\sqrt{10-2\sqrt{5}}\right)^2}}\right)^2$$

$$-1/(\pi)^2 [6.05892*10^{-8} -8\pi(-0.5)*\sqrt{(((1/16*(((0.5\pi)/(2*(((\sqrt{(10-2\sqrt{5}}-2))/(\sqrt{5}-1))))^2)))))*\ln(-0.5^2/16)-1/16*16\pi*i*i(-0.5(-163.82848))*(6.05892*10^{-8})}]$$

Input interpretation

$$-\frac{1}{\pi^2}\left(6.05892\times 10^{-8}-8\pi\left(-0.5\sqrt{\frac{\frac{1}{16}\times\frac{-0.5\pi}{2\left(\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}\right)^2}\log\left(-\frac{0.5^2}{16}\right)}\right)-\frac{1}{16}\times 16(\pi i i)\left((-0.5\times(-163.82848))\times 6.05892\times 10^{-8}\right)\right)$$

$\log(x)$ is the natural logarithm

i is the imaginary unit

Result

3.11965... +

4.12983... i

Polar coordinates

$r = 5.17569$ (radius), $\theta = 0.923853$ (angle)

5.17569

In conclusion, we obtain:

$$1205.74 \cdot (160.576 - 5.17569)$$

Input interpretation

1205.74 (160.576 – 5.17569)

Result

187372.3697794

187372.3697794

We note that, from the following Table:

Table 46
Phi Λ (n/7) scale (octave = 4)

#	Phi Λ (n/7)	Frequency (Hz)
1	1.0000000	306.342
2	1.0711625	328.142
3	1.1473892	351.494
4	1.2290403	376.508
5	1.3165020	403.300
6	1.4101876	432.000
7	1.5105401	462.742
8	1.6180340	495.672
9	1.7331774	530.945
10	1.8565147	568.729
11	1.9886290	609.201
		612.684

Note. Author's calculation with data
From Lange, Nardelli, & Bini (2013,
p.3). ©

Table of Frequency System based on Phi

If we calculate the square root of

$$1205.74 (160.576 - 5.17569)$$

after some calculations, we obtain:

$$495.672 * 1 / (\sqrt{1205.74 * (160.576 - 5.17569)}) * 1 / 1.4101876$$

where the two values highlighted in red are in the above table

Input interpretation

$$495.672 \times \frac{1}{\sqrt{1205.74 (160.576 - 5.17569)}} \times \frac{1}{1.4101876}$$

Result

1.6147991179794231231866849771498385989419683480036261824881476990

...

1.61479911797..... result that is a very good approximation to the value of the golden ratio 1.618033988749...

We obtain also:

$$((1205.74 * (160.576 - 5.17569))^{1 / (\ln(40448921875))})$$

where $\ln(40448921875) = 24.4233$, that is a value of a black hole entropy

Input interpretation

$$\log(40448921875) \sqrt{1205.74 (160.576 - 5.17569)}$$

$\log(x)$ is the natural logarithm

Result

1.6439487812452945609174398975125712106762684760397763844703196435

...

1.64394878.... $\approx \zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$ (trace of the instanton shape)

Alternative representations

$$\log(40\,448\,921\,875)\sqrt{1205.74\,(160.576-5.17569)} = \frac{\log_e(40\,448\,921\,875)}{\sqrt{187\,372.}}$$

$$\log(40\,448\,921\,875)\sqrt{1205.74\,(160.576-5.17569)} = \frac{\log(a)\log_q(40\,448\,921\,875)}{\sqrt{187\,372.}}$$

$$\log(40\,448\,921\,875)\sqrt{1205.74\,(160.576-5.17569)} = 187\,372.^{-1/\text{Li}_1(-40\,448\,921\,874)}$$

Series representations

$$\log(40\,448\,921\,875)\sqrt{1205.74\,(160.576-5.17569)} = \frac{\log(40\,448\,921\,874)-\sum_{k=1}^{\infty}\frac{\left(-\frac{1}{40\,448\,921\,874}\right)^k}{k}}{\sqrt{187\,372.}}$$

$$\log(40\,448\,921\,875)\sqrt{1205.74\,(160.576-5.17569)} = \frac{2\,i\,\pi\left[\frac{\arg(40\,448\,921\,875-x)}{2\,\pi}\right]+\log(x)-\sum_{k=1}^{\infty}\frac{(-1)^k(40\,448\,921\,875-x)^kx^{-k}}{k}}{\sqrt{187\,372.}} \text{ for } x < 0$$

$$\log(40\,448\,921\,875)\sqrt{1205.74\,(160.576-5.17569)} = \frac{\log(z_0)+\left[\frac{\arg(40\,448\,921\,875-z_0)}{2\,\pi}\right]\left(\log\left(\frac{1}{z_0}\right)+\log(z_0)\right)-\sum_{k=1}^{\infty}\frac{(-1)^k(40\,448\,921\,875-z_0)^kz_0^{-k}}{k}}{\sqrt{187\,372.}}$$

Integral representations

$$\log(40\,448\,921\,875)\sqrt{1205.74\,(160.576-5.17569)} = \int_1^{40\,448\,921\,875}\frac{1}{t}\,dt\sqrt{187\,372.}$$

$$\log(40448921875) \sqrt{1205.74(160.576 - 5.17569)} =$$

$$\exp\left(\frac{24.2817 i \pi}{\int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{40448921874^{-s} \Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds}\right) \text{ for } -1 < \gamma < 0$$

We note that, if the square root of 187372.3697794 gives a result close to the fundamental value of the table of the frequency system based on ϕ , i.e. 432 and, consequently, a result close to the golden ratio, and another, close to the value of $\zeta(2)$, this reinforces our proposal that ϕ and $\zeta(2)$ are mathematical constants at the base of the forms of the microcosm and the macrocosm

From the already previously analyzed expression

$$(((1.2^2 \times \frac{1}{2})(2^{16} * \pi^{10} * (0.30282212)^{-8}) \text{ integrate } [\text{Exp}((((- 8\pi^2)/(768^2)) + \ln(16) (22/3 - 1/6 * 8 * 4 - 2/3 * 2) + 7.05399103 - (8 * 1.09457662 - 2 * 0.35952290)))]x))$$

Input interpretation

$$\left(1.2^2 \times \frac{1}{2}\right) \times \frac{2^{16} \pi^{10}}{0.30282212^8}$$

$$\int \exp\left(\frac{-8\pi^2}{768^2} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 4 - \frac{2}{3} \times 2\right) + 7.05399103 - (8 \times 1.09457662 + 2 \times (-0.35952290))\right) x dx$$

$\log(x)$ is the natural logarithm

Result

$$7.41835 \times 10^{13} x^2$$

7.41835*10¹³ (possible instanton value)

dividing it by the last one:

$1/((1205.74*(160.576-5.17569)))*(((1.2^2*1/2)(2^{16} * \pi^{10} * (0.30282212)^{-8})$
 integrate [Exp(((((-8Pi^2)/(768^2))+ln(16) (22/3-1/6*8*4-2/3*2)+7.05399103-
 (8*1.09457662-2*0.35952290))))x))

Input interpretation

$$\frac{1}{1205.74 (160.576 - 5.17569)} \left(\left(1.2^2 \times \frac{1}{2} \right) \times \frac{2^{16} \pi^{10}}{0.30282212^8} \int \exp \left(\frac{-8 \pi^2}{768^2} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 4 - \frac{2}{3} \times 2 \right) + 7.05399103 - (8 \times 1.09457662 + 2 \times (-0.35952290)) \right) x dx \right)$$

$\log(x)$ is the natural logarithm

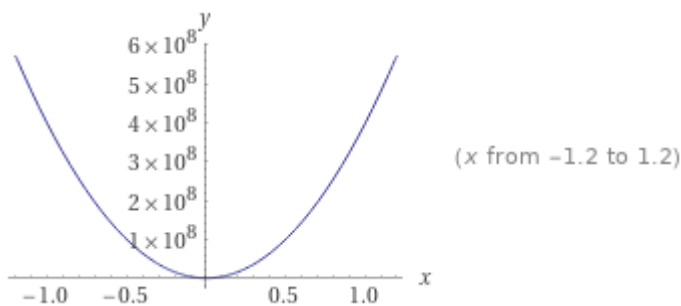
Result

$$3.95915 \times 10^8 x^2$$

For $x = 1$:

$$3.95915 \times 10^8$$

Plot



Alternate form assuming x is real

$$3.95915 \times 10^8 x^2 + 0$$

Indefinite integral assuming all variables are real

$$1.31972 \times 10^8 x^3 + \text{constant}$$

From the following expression:

$$1/(((2^{11/17} e \log^{25/17}(2))^*1/(\log^{9/17}(27))))$$

Input

$$\frac{1}{(2^{11/17} e \log^{25/17}(2)) \times \frac{1}{\log^{9/17}(27)}}$$

$\log(x)$ is the natural logarithm

Exact result

$$\frac{\log^{9/17}(27)}{2^{11/17} e \log^{25/17}(2)}$$

Decimal approximation

0.7572141949135438099180060969730828236866789484742118720130592380
...

0.7572141949....

Alternate form

$$\frac{(3 \log(3))^{9/17}}{2^{11/17} e \log^{25/17}(2)}$$

Alternative representations

$$\frac{1}{2^{11/17} (e \log^{25/17}(2)) \log^{9/17}(27)} = \frac{1}{e 2^{11/17} \log_e^{25/17}(2) \log_e^{9/17}(27)}$$

$$\frac{1}{2^{11/17} (e \log^{25/17}(2)) \log^{9/17}(27)} = \frac{1}{e 2^{11/17} (\log(a) \log_a(2))^{25/17} (\log(a) \log_a(27))^{9/17}}$$

$$\frac{1}{\frac{2^{11/17} (e \log^{25/17}(2))}{\log^{9/17}(27)}} = \frac{1}{\frac{e 2^{11/17} (-\text{Li}_1(-1))^{25/17}}{(-\text{Li}_1(-26))^{9/17}}}$$

Series representations

$$\frac{1}{\frac{2^{11/17} (e \log^{25/17}(2))}{\log^{9/17}(27)}} = \frac{\left(2 i \pi \left\lfloor \frac{\arg(27-x)}{2 \pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (27-x)^k x^{-k}}{k}\right)^{9/17}}{2^{11/17} e \left(2 i \pi \left\lfloor \frac{\arg(2-x)}{2 \pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k}\right)^{25/17}}$$

for $x < 0$

$$\frac{1}{\frac{2^{11/17} (e \log^{25/17}(2))}{\log^{9/17}(27)}} = \frac{\left(2 i \pi \left\lfloor \frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2 \pi} \right\rfloor + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (27-z_0)^k z_0^{-k}}{k}\right)^{9/17}}{2^{11/17} e \left(2 i \pi \left\lfloor \frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2 \pi} \right\rfloor + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k}\right)^{25/17}}$$

$$\frac{1}{\frac{2^{11/17} (e \log^{25/17}(2))}{\log^{9/17}(27)}} = \left(\left\lfloor \frac{\arg(27-z_0)}{2 \pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(27-z_0)}{2 \pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (27-z_0)^k z_0^{-k}}{k} \right)^{9/17} /$$

$$\left(2^{11/17} e \left(\left\lfloor \frac{\arg(2-z_0)}{2 \pi} \right\rfloor \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left\lfloor \frac{\arg(2-z_0)}{2 \pi} \right\rfloor \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k} \right)^{25/17} \right)$$

Integral representations

$$\frac{1}{\frac{2^{11/17} (e \log^{25/17}(2))}{\log^{9/17}(27)}} = \frac{\left(\int_1^{27} \frac{1}{t} dt\right)^{9/17}}{2^{11/17} e \left(\int_1^2 \frac{1}{t} dt\right)^{25/17}}$$

$$\frac{1}{\frac{2^{11/17} (e \log^{25/17}(2))}{\log^{9/17}(27)}} = \frac{2^{5/17} \pi^{16/17} \left(-i \int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{\Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds\right)^{9/17} \left(-i \int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{26^{-s} \Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds\right)^{9/17}}{e \left(\int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{\Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds\right)^2}$$

for $-1 < \gamma < 0$

From which, in conclusion:

$$1/(((2^{11/17} e \log^{25/17}(2)) * 1/(\log^{9/17}(27)))) (3.95915 \times 10^8)$$

Input interpretation

$$\frac{1}{(2^{11/17} e \log^{25/17}(2)) \times \frac{1}{\log^{9/17}(27)}} \times 3.95915 \times 10^8$$

$\log(x)$ is the natural logarithm

Result

$$2.99792... \times 10^8$$

$$2.99792... * 10^8 \approx c$$

Below, the four analyzed expressions:

$$\left(\frac{1}{\sqrt{t}}\right)^5 + \left(\frac{\sqrt{5}-1}{2} \sqrt{t}\right)^5 = \frac{1}{\sqrt{x}} \sqrt{\frac{(1-x)(1-x^2)}{(1-x^5)(1-x^{10})}} \times \frac{1}{\left\{\left(1-\frac{\sqrt{5}-1}{2}x+x^2\right)\left(1-\frac{\sqrt{5}-1}{2}x^2+x^4\right)\right\}^6}$$

$$= 7.41835506... * 10^{13}$$

$$2^{14} \pi^6 g^{-8} \rho^{-5} d^4 z d\rho \exp \left\{ -\frac{8\pi^2}{g_B^2(n)} + \left(\ln \rho + \frac{1}{4-n} \right) \left[\frac{22}{3} - \frac{1}{6} \sum_t N^s(t) C(t) \right] + A - \sum_t N^s(t) A(t) \right\},$$

$$\begin{aligned} \mathfrak{L}^{\text{eff}}(z) = & 2^{10+3N^f} \pi^{6+2N^f} g^{-8} \int \rho^{3N^f-5} d\rho \exp \left\{ -\frac{8\pi^2}{[g_R^D(\mu)]^2} \right. \\ & + \log(\mu\rho) \left[\frac{22}{3} - \frac{1}{6} \sum_i N^s(t) C(t) - \frac{2}{3} N^f \right] + A - \sum_i N^s(t) A(t) - N^f B \Big\} \\ & \times \left\langle \prod_{s=1}^{N^f} (\bar{\psi}_s \omega)(\bar{\omega} \psi_s) \right\rangle + \text{h.c.}, \end{aligned}$$

$$\begin{aligned} & \left(1.2^2 \times \frac{1}{2}\right) \times \frac{2^{16} \pi^{10}}{0.30282212^8} \\ & \int \exp \left(\frac{-8\pi^2}{768^2} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 4 - \frac{2}{3} \times 2 \right) + 7.05399103 - \right. \\ & \quad \left. (8 \times 1.09457662 + 2 \times (-0.35952290)) \right) x \, dx \end{aligned}$$

$$= 7.41835*10^{13} \, \text{x}^2 \rightarrow 7.41835 * 10^{13}$$

And:

$$\frac{1}{\sqrt{\frac{11}{7} (2 \zeta(3) - \zeta(5))}} \sqrt{\frac{\left(1 - \frac{11}{7} (2 \zeta(3) - \zeta(5))\right) \left(1 - \left(\frac{11}{7} (2 \zeta(3) - \zeta(5))\right)^2\right)}{\left(1 - \left(\frac{11}{7} (2 \zeta(3) - \zeta(5))\right)^5\right) \left(1 - \left(\frac{11}{7} (2 \zeta(3) - \zeta(5))\right)^{10}\right)}}$$

$$= 0.004540210710940118$$

$$1 / \left(0.004540210710940118 \times \right. \\ \left. 1 / \left(\left(1 - \left(\frac{1}{2} (\sqrt{5} - 1) \right) \right) \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right) + \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right)^2 \right) \right. \\ \left. \left(1 - \left(\frac{1}{2} (\sqrt{5} - 1) \right) \right) \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right)^2 + \left(\frac{11}{7} (2\zeta(3) - \zeta(5)) \right)^4 \right) \right)^6$$

$$= 7.418355067475220... \times 10^{13} = 7.41835506... * 10^{13}$$

$$ds_{\text{inst}}^2 = \sum_{\gamma} \frac{\Omega_{\gamma} \Sigma_{\gamma}}{16r \sqrt{2\pi \mathcal{T}_{\gamma}^{\text{R}}}} \left[\frac{|Z_{\gamma}|^2}{rK} \left(d\sigma + \tilde{\zeta}_{\Lambda} d\zeta^{\Lambda} - \zeta^{\Lambda} d\tilde{\zeta}_{\Lambda} - \frac{4i\sqrt{rK}}{|Z_{\gamma}|} \mathcal{C}_{\gamma} + 8r \operatorname{Im} \partial \log \frac{Z_{\gamma}}{K} \right)^2 \right. \\ \left. - \frac{1}{\pi^2} \left(d\mathcal{T}_{\gamma} - 8\pi |Z_{\gamma}| \sqrt{\frac{r}{K}} d \log \frac{|Z_{\gamma}|^2}{K} - \frac{16\pi i}{K} \operatorname{Im} \left[\bar{Z}_{\gamma} (z^{\Lambda} d\tilde{\zeta}_{\Lambda} - F_{\Lambda} d\zeta^{\Lambda}) \right] \right) d\mathcal{T}_{\gamma} \right], \quad (3.5)$$

$$1205.74*(160.576-5.17569)$$

$$1205.74\,(160.576-5.17569)$$

$$= 187372.3697794$$

Mathematical connections with some sectors of String Theory

From:

Modular equations and approximations to π - Srinivasa Ramanujan
 Quarterly Journal of Mathematics, XLV, 1914, 350 – 372

We have that:

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

Again

$$G_{37} = (6 + \sqrt{37})^{\frac{1}{4}},$$

$$\begin{aligned} 64G_{37}^{24} &= e^{\pi\sqrt{37}} + 24 + 276e^{-\pi\sqrt{37}} + \dots, \\ 64G_{37}^{-24} &= 4096e^{-\pi\sqrt{37}} - \dots, \end{aligned}$$

so that

$$64(G_{37}^{24} + G_{37}^{-24}) = e^{\pi\sqrt{37}} + 24 + 4372e^{-\pi\sqrt{37}} - \dots = 64\{(6 + \sqrt{37})^6 + (6 - \sqrt{37})^6\}.$$

Hence

$$e^{\pi\sqrt{37}} = 199148647.999978\dots$$

Similarly, from

$$g_{58} = \sqrt{\left(\frac{5 + \sqrt{29}}{2}\right)},$$

we obtain

$$64(g_{58}^{24} + g_{58}^{-24}) = e^{\pi\sqrt{58}} - 24 + 4372e^{-\pi\sqrt{58}} + \dots = 64\left\{\left(\frac{5 + \sqrt{29}}{2}\right)^{12} + \left(\frac{5 - \sqrt{29}}{2}\right)^{12}\right\}.$$

Hence

$$e^{\pi\sqrt{58}} = 24591257751.99999982\dots$$

From:

An Update on Brane Supersymmetry Breaking

J. Mourad and A. Sagnotti - arXiv:1711.11494v1 [hep-th] 30 Nov 2017

From the following vacuum equations:

$$T e^{\gamma_E \phi} = - \frac{\beta_E^{(p)} h^2}{\gamma_E} e^{-2(8-p)C + 2\beta_E^{(p)} \phi}$$

$$16 k' e^{-2C} = \frac{h^2 \left(p + 1 - \frac{2\beta_E^{(p)}}{\gamma_E} \right) e^{-2(8-p)C + 2\beta_E^{(p)} \phi}}{(7-p)}$$

$$(A')^2 = k e^{-2A} + \frac{h^2}{16(p+1)} \left(7-p + \frac{2\beta_E^{(p)}}{\gamma_E} \right) e^{-2(8-p)C + 2\beta_E^{(p)} \phi}$$

we have obtained, from the results almost equals of the equations, putting

$4096 e^{-\pi\sqrt{18}}$ instead of

$$e^{-2(8-p)C + 2\beta_E^{(p)} \phi}$$

a new possible mathematical connection between the two exponentials. Thence, also the values concerning p , C , β_E and ϕ correspond to the exponents of e (i.e. of exp). Thence we obtain for $p = 5$ and $\beta_E = 1/2$:

$$e^{-6C+\phi} = 4096 e^{-\pi\sqrt{18}}$$

Therefore, with respect to the exponentials of the vacuum equations, the Ramanujan's exponential has a coefficient of 4096 which is equal to 642 , while $-6C+\phi$ is equal to $-\pi\sqrt{18}$. From this it follows that it is possible to establish mathematically, the dilaton value.

For

$\exp((- \pi \sqrt{18}))$ we obtain:

Input:

$$\exp\left(-\pi \sqrt{18}\right)$$

Exact result:

$$e^{-3 \sqrt{2} \pi}$$

Decimal approximation:

$$1.6272016226072509292942156739117979541838581136954016... \times 10^{-6}$$

$$1.6272016... * 10^{-6}$$

Property:

$e^{-3 \sqrt{2} \pi}$ is a transcendental number

Series representations:

$$e^{-\pi \sqrt{18}} = e^{-\pi \sqrt{17} \sum_{k=0}^{\infty} 17^{-k} \binom{1/2}{k}}$$

$$e^{-\pi \sqrt{18}} = \exp\left(-\pi \sqrt{17} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{17}\right)^k \left(-\frac{1}{2}\right)_k}{k!}\right)$$

$$e^{-\pi \sqrt{18}} = \exp\left(-\frac{\pi \sum_{j=0}^{\infty} \operatorname{Res}_{s=-\frac{1}{2}+j} 17^{-s} \Gamma\left(-\frac{1}{2}-s\right) \Gamma(s)}{2 \sqrt{\pi}}\right)$$

Now, we have the following calculations:

$$e^{-6C+\phi} = 4096e^{-\pi\sqrt{18}}$$

$$e^{-\pi\sqrt{18}} = 1.6272016... * 10^{-6}$$

from which:

$$\frac{1}{4096} e^{-6C+\phi} = 1.6272016... * 10^{-6}$$

$$0.000244140625 e^{-6C+\phi} = e^{-\pi\sqrt{18}} = 1.6272016... * 10^{-6}$$

Now:

$$\ln(e^{-\pi\sqrt{18}}) = -13.328648814475 = -\pi\sqrt{18}$$

And:

$$(1.6272016 * 10^{-6}) * 1 / (0.000244140625)$$

Input interpretation:

$$\frac{1.6272016}{10^6} \times \frac{1}{0.000244140625}$$

Result:

0.0066650177536

0.006665017...

Thence:

$$0.000244140625 e^{-6C+\phi} = e^{-\pi\sqrt{18}}$$

Dividing both sides by 0.000244140625, we obtain:

$$\frac{0.000244140625}{0.000244140625} e^{-6C+\phi} = \frac{1}{0.000244140625} e^{-\pi\sqrt{18}}$$

$$e^{-6C+\phi} = 0.0066650177536$$

$$((((\exp((-Pi*\sqrt{18})))))))*1/0.000244140625$$

Input interpretation:

$$\exp\left(-\pi\sqrt{18}\right)\times\frac{1}{0.000244140625}$$

Result:

0.00666501785...

0.00666501785...

Series representations:

$$\frac{\exp(-\pi\sqrt{18})}{0.000244141} = 4096 \exp\left(-\pi\sqrt{17} \sum_{k=0}^{\infty} 17^{-k} \left(\frac{1}{2}\right)_k\right)$$

$$\frac{\exp(-\pi\sqrt{18})}{0.000244141} = 4096 \exp\left(-\pi\sqrt{17} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{17}\right)^k \left(-\frac{1}{2}\right)_k}{k!}\right)$$

$$\frac{\exp(-\pi\sqrt{18})}{0.000244141} = 4096 \exp\left(-\frac{\pi \sum_{j=0}^{\infty} \operatorname{Res}_{s=-\frac{1}{2}+j} 17^{-s} \Gamma\left(-\frac{1}{2}-s\right) \Gamma(s)}{2\sqrt{\pi}}\right)$$

Now:

$$e^{-6\mathcal{C}+\phi} = 0.0066650177536$$

$$\exp\left(-\pi\sqrt{18}\right)\times\frac{1}{0.000244140625} =$$

$$e^{-\pi\sqrt{18}}\times\frac{1}{0.000244140625}$$

$$= 0.00666501785\dots$$

From:

$$\ln(0.00666501784619)$$

Input interpretation:

$$\log(0.00666501784619)$$

Result:

$$-5.010882647757\dots$$

$$-5.010882647757\dots$$

Alternative representations:

$$\log(0.006665017846190000) = \log_e(0.006665017846190000)$$

$$\log(0.006665017846190000) = \log(a) \log_a(0.006665017846190000)$$

$$\log(0.006665017846190000) = -\operatorname{Li}_1(0.993334982153810000)$$

Series representations:

$$\log(0.006665017846190000) = -\sum_{k=1}^{\infty} \frac{(-1)^k (-0.993334982153810000)^k}{k}$$

$$\log(0.006665017846190000) = 2 i \pi \left[\frac{\arg(0.006665017846190000 - x)}{2 \pi} \right] +$$

$$\log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (0.006665017846190000 - x)^k x^{-k}}{k} \quad \text{for } x < 0$$

$$\log(0.006665017846190000) = \left[\frac{\arg(0.006665017846190000 - z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) +$$

$$\log(z_0) + \left[\frac{\arg(0.006665017846190000 - z_0)}{2 \pi} \right] \log(z_0) -$$

$$\sum_{k=1}^{\infty} \frac{(-1)^k (0.006665017846190000 - z_0)^k z_0^{-k}}{k}$$

Integral representation:

$$\log(0.006665017846190000) = \int_1^{0.006665017846190000} \frac{1}{t} dt$$

In conclusion:

$$-6C + \phi = -5.010882647757 \dots$$

and for C = 1, we obtain:

$$\phi = -5.010882647757 + 6 = \mathbf{0.989117352243} = \phi$$

Note that the values of n_s (spectral index) 0.965, of the average of the Omega mesons Regge slope 0.987428571 and of the dilaton 0.989117352243, are also connected to the following two Rogers-Ramanujan continued fractions:

$$\frac{e^{-\frac{\pi}{5}}}{\sqrt{(\phi-1)\sqrt{5}} - \phi + 1} = 1 - \frac{e^{-\pi}}{1 + \frac{e^{-2\pi}}{1 + \frac{e^{-3\pi}}{1 + \frac{e^{-4\pi}}{1 + \dots}}}} \approx 0.9568666373$$

$$\frac{\frac{e^{-\frac{\pi}{\sqrt{5}}}}{\sqrt{5}}}{1 + \sqrt[5]{\sqrt{\phi^5 4 \sqrt{5^3}} - 1}} = 1 - \frac{e^{-\pi\sqrt{5}}}{1 + \frac{e^{-2\pi\sqrt{5}}}{1 + \frac{e^{-3\pi\sqrt{5}}}{1 + \frac{e^{-4\pi\sqrt{5}}}{1 + \dots}}}} \approx 0.9991104684$$

(<http://www.bitman.name/math/article/102/109/>)

The mean between the two results of the above Rogers-Ramanujan continued fractions is 0.97798855285, value very near to the ψ Regge slope 0.979:

Ψ		3		$m_c = 1500$		0.979		-0.09
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Also performing the 512th root of the inverse value of the Pion meson rest mass 139.57, we obtain:

$$((1/(139.57)))^{1/512}$$

Input interpretation:

$$\sqrt[512]{\frac{1}{139.57}}$$

Result:

0.990400732708644027550973755713301415460732796178555551684...

0.99040073.... result very near to the dilaton value **0.989117352243** = ϕ and to the value of the following Rogers-Ramanujan continued fraction:

$$\frac{e^{-\frac{\pi}{\sqrt{5}}}}{\sqrt{5}} = 1 - \frac{e^{-\pi\sqrt{5}}}{1 + \frac{e^{-2\pi\sqrt{5}}}{1 + \frac{e^{-3\pi\sqrt{5}}}{1 + \frac{e^{-4\pi\sqrt{5}}}{1 + \dots}}}} \approx 0.9991104684$$

From

AdS Vacua from Dilaton Tadpoles and Form Fluxes - *J. Mourad and A. Sagnotti*
- arXiv:1612.08566v2 [hep-th] 22 Feb 2017 - March 27, 2018

We have:

$$e^{2C} = \frac{2\xi e^{\frac{\phi}{2}}}{1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}}$$

$$\frac{h^2}{32} = \frac{\xi^7 e^{4\phi}}{\left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right)^7} \left[\frac{42}{\xi} \left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right) + 5 T e^{2\phi} \right]. \quad (2.7)$$

For

$$T = \frac{16}{\pi^2}$$

$$\xi = 1$$

we obtain:

$$(2 \cdot e^{(0.989117352243/2)}) / (1 + \sqrt{((1 - 1/3 \cdot 16/(\pi)^2 \cdot e^{2 \cdot 0.989117352243})))})$$

Input interpretation:

$$\frac{2 e^{0.989117352243/2}}{1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}}$$

Result:

0.83941881822... -
1.4311851867... i

Polar coordinates:

$r = 1.65919106525$ (radius), $\theta = -59.607521917^\circ$ (angle)

1.65919106525..... result very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Series representations:

$$\frac{2 e^{0.9891173522430000/2}}{1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}}} = \frac{2 e^{0.4945586761215000}}{1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \binom{\frac{1}{2}}{k}}$$

$$\frac{2 e^{0.9891173522430000/2}}{1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}}} = \frac{2 e^{0.4945586761215000}}{1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!}}$$

$$\frac{2 e^{0.9891173522430000/2}}{1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}}} = \frac{2 e^{0.4945586761215000}}{1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} z_0\right)^k z_0^{-k}}{k!}}$$

for (not ($z_0 \in \mathbb{R}$ and $-\infty < z_0 \leq 0$))

From

$$\frac{h^2}{32} = \frac{\xi^7 e^{4\phi}}{\left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right)^7} \left[\frac{42}{\xi} \left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right) + 5 T e^{2\phi} \right]$$

we obtain:

$$e^{(4 \times 0.989117352243) / (((1 + \sqrt{1 - 1/3 \times 16 / (\pi)^2 \times e^{(2 \times 0.989117352243)}}))^{1/7} [42(1 + \sqrt{1 - 1/3 \times 16 / (\pi)^2 \times e^{(2 \times 0.989117352243)}}) + 5 \times 16 / (\pi)^2 \times e^{(2 \times 0.989117352243)}]}$$

Input interpretation:

$$\frac{e^{4 \times 0.989117352243}}{\left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}\right)^7} \left(42 \left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}\right) + 5 \times \frac{16}{\pi^2} e^{2 \times 0.989117352243} \right)$$

Result:

50.84107889... –
20.34506335... i

Polar coordinates:

$r = 54.76072411$ (radius), $\theta = -21.80979492^\circ$ (angle)

54.76072411.....

Series representations:

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right)^7 = \\
& \left(2 \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right. \right. \\
& \quad \left. \left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)_k \right) \right) / \\
& \quad \left(\pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)_k \right) \right)^7 \Bigg) \\
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right)^7 = \\
& \left(2 \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right. \right. \\
& \quad \left. \left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right) / \\
& \quad \left(\pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)^7 \Bigg)
\end{aligned}$$

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right)^7 = \\
& \left(2 \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \right. \right. \\
& \quad \left. \left. \pi^2 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right) \right) / \\
& \left(\pi^2 \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

From which:

$$\begin{aligned}
& e^{(4 \times 0.989117352243)} / (((1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{(2 \times 0.989117352243)}}))^7 \\
& [42(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{(2 \times 0.989117352243)}}) + 5 \times \frac{16}{\pi^2} e^{(2 \times 0.989117352243)}] \times \frac{1}{34}
\end{aligned}$$

Input interpretation:

$$\begin{aligned}
& \frac{e^{4 \times 0.989117352243}}{\left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}} \right)^7} \\
& \left(42 \left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}} \right) + 5 \times \frac{16}{\pi^2} e^{2 \times 0.989117352243} \right) \times \frac{1}{34}
\end{aligned}$$

Result:

1.495325850... -
0.5983842161... *i*

Polar coordinates:

$r = 1.610609533$ (radius), $\theta = -21.80979492^\circ$ (angle)

1.610609533.... result that is a good approximation to the value of the golden ratio
1.618033988749...

Series representations:

$$\left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) e^{4 \times 0.9891173522430000} \right) / \left(34 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) \right)^7 =$$

$$\left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right.$$

$$\left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)^k \right) /$$

$$\left(17 \pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)^k \right) \right)^7$$

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(34 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) \right)^7 = \\
& \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right. \\
& \quad \left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) / \\
& \quad \left(17 \pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right)^7 \\
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(34 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) \right)^7 = \\
& \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \right. \\
& \quad \left. \pi^2 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right) / \\
& \quad \left(17 \pi^2 \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right) \right)^7
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

Now, we have:

$$e^{2C} = \frac{2\xi e^{-\frac{\phi}{2}}}{1 + \sqrt{1 + \frac{\xi\Lambda}{3} e^{2\phi}}}, \quad (2.9)$$

$$\frac{h^2}{32} = \frac{e^{-4\phi}}{\left[1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}}\right]^7} \left[42 \left(1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}}\right) - 13\Lambda e^{2\phi}\right]. \quad (2.10)$$

For:

$$\xi = 1$$

$$\Lambda \simeq \frac{4\pi^2}{25}$$

$$\phi = 0.989117352243$$

From

$$e^{2C} = \frac{2\xi e^{-\frac{\phi}{2}}}{1 + \sqrt{1 + \frac{\xi\Lambda}{3} e^{2\phi}}},$$

we obtain:

$$\frac{((2 * e^{(-0.989117352243/2)}))}{((((1 + \sqrt{((1 + 1/3 * (4\pi^2)/25 * e^{(2 * 0.989117352243)}))}))))}$$

Input interpretation:

$$\frac{2 e^{-0.989117352243/2}}{1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}}}$$

Result:

0.382082347529...

0.382082347529....

Series representations:

$$\frac{2 e^{-0.9891173522430000/2}}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} = 2 / \left(e^{0.4945586761215000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right)_k \right) \right)$$

$$\frac{2 e^{-0.9891173522430000/2}}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} = 2 / \left(e^{0.4945586761215000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)$$

$$\frac{2 e^{-0.9891173522430000/2}}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} = \frac{2}{e^{0.4945586761215000} \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} z_0 \right)^k z_0^{-k}}{k!} \right)}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

From which:

$$1 + 1 / (((4 * (2 * e^{(-0.989117352243/2)})) / (((1 + \sqrt{((1 + 1/3 * (4 \pi^2)/25 * e^{(2 * 0.989117352243))}))))))))))$$

Input interpretation:

$$1 + \frac{1}{4 \times \frac{2 e^{-0.989117352243/2}}{1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}}}}$$

Result:

1.65430921270...

1.6543092..... We note that, the result 1.6543092... is very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Indeed:

$$G_{505} = P^{-1/4} Q^{1/6} = (\sqrt{5} + 2)^{1/2} \left(\frac{\sqrt{5} + 1}{2} \right)^{1/4} (\sqrt{101} + 10)^{1/4} \\ \times \left((130\sqrt{5} + 29\sqrt{101}) + \sqrt{169440 + 7540\sqrt{505}} \right)^{1/6}.$$

Thus, it remains to show that

$$(130\sqrt{5} + 29\sqrt{101}) + \sqrt{169440 + 7540\sqrt{505}} = \left(\sqrt{\frac{113 + 5\sqrt{505}}{8}} + \sqrt{\frac{105 + 5\sqrt{505}}{8}} \right)^3,$$

which is straightforward. $\square$

$$\sqrt[14]{\left(\sqrt{\frac{113 + 5\sqrt{505}}{8}} + \sqrt{\frac{105 + 5\sqrt{505}}{8}} \right)^3} = 1.65578 \dots$$

Series representations:

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} = \frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{1 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \binom{\frac{1}{2}}{k}}$$

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} = \frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{1 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!}}$$

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} = 1 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000} \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0 \right)^k}{k!} z_0^{-k}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

And from

$$\frac{h^2}{32} = \frac{e^{-4\phi}}{\left[1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}}\right]^7} \left[42 \left(1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}}\right) - 13 \Lambda e^{2\phi}\right].$$

we obtain:

$$e^{(-4 \times 0.989117352243) / [1 + \sqrt{((1 + 1/3 \times (4\pi^2)/25 \times e^{(2 \times 0.989117352243))})}]^7 \times [42(1 + \sqrt{((1 + 1/3 \times (4\pi^2)/25 \times e^{(2 \times 0.989117352243))})} - 13 \times (4\pi^2)/25 \times e^{(2 \times 0.989117352243)})]$$

Input interpretation:

$$\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2)\right) e^{2 \times 0.989117352243}}\right)^7 \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2)\right) e^{2 \times 0.989117352243}}\right) - 13 \left(\frac{1}{25} (4 \pi^2)\right) e^{2 \times 0.989117352243}\right)}$$

Result:

-0.034547055658...

-0.034547055658...

Series representations:

$$\left(\left(42 \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \frac{1}{25} (4\pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right. \right. \\ \left. \left. e^{-4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \right. \\ \left. - \left(42 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\ \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\ \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right)_k \right) \right) / \left(25 e^{5.934704113458000} \right. \\ \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right)_k \right) \right)^7 \right) \right)$$

$$\left(\left(42 \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \frac{1}{25} (4\pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right. \right. \\ \left. \left. e^{-4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \right. \\ \left. - \left(42 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\ \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\ \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right) / \left(25 e^{5.934704113458000} \right. \\ \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)^7 \right) \right)$$

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \frac{1}{25} (4\pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right. \right. \\
& \quad \left. \left. e^{-4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \right. \\
& \quad - \left(\left(42 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - 25 e^{1.978234704486000} \right. \right. \right. \\
& \quad \left. \left. \left. \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) \right) / \left(25 \right. \right. \\
& \quad \left. \left. \left. e^{5.934704113458000} \right. \right. \right. \\
& \quad \left. \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \right) \right) \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

From which:

$$\begin{aligned}
& 47 * 1 / (((-1 / (((e^{(-4 * 0.989117352243)} / \\
& [1 + \sqrt{((1 + 1/3 * (4\pi^2)/25 * e^{(2 * 0.989117352243)}))])^7 * \\
& [42(1 + \sqrt{((1 + 1/3 * (4\pi^2)/25 * e^{(2 * 0.989117352243)}))}) - \\
& 13 * (4\pi^2)/25 * e^{(2 * 0.989117352243)}])])])
\end{aligned}$$

Input interpretation:

$$\begin{aligned}
& 47 \left(- \left(1 / 1 / \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4\pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \right. \right. \right. \\
& \quad \left. \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4\pi^2) \right) e^{2 \times 0.989117352243}} - \right. \right. \right. \\
& \quad \left. \left. \left. 13 \left(\frac{1}{25} (4\pi^2) \right) e^{2 \times 0.989117352243} \right) \right) \right) \right)
\end{aligned}$$

Result:

1.6237116159...

1.6237116159.... result that is an approximation to the value of the golden ratio
1.618033988749...

Series representations:

$$\begin{aligned}
 & - \left(47 / 1 / \left(e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right. \right. \right. \right. \\
 & \quad \left. \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) \right) / \\
 & \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
 & \left(1974 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \\
 & \quad \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
 & \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right) \right) \right) / \left(25 e^{5.934704113458000} \right. \\
 & \quad \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right) \right) \right)^7
 \end{aligned}$$

$$\begin{aligned}
& - \left(47 / 1 / \left(e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = \\
& \left(1974 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \\
& \quad 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right) / \left(25 e^{5.934704113458000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^7 \right) \right) \\
& - \left(47 / 1 / \left(e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = \\
& \left(1974 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - 25 e^{1.978234704486000} \right. \right. \\
& \quad \left. \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) \Bigg) / \left(25 e^{5.934704113458000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \right) \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

And again:

$$32((((e^{(-4*0.989117352243)} / [1+\sqrt{((1+1/3*(4\pi^2)/25*e^{(2*0.989117352243)})})}]^7 * [42(1+\sqrt{((1+1/3*(4\pi^2)/25*e^{(2*0.989117352243)})})})-13*(4\pi^2)/25*e^{(2*0.989117352243)})])])])$$

Input interpretation:

$$32 \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243} \right) \right)$$

Result:

-1.1055057810...

-1.1055057810....

We note that the result -1.1055057810.... is very near to the value of Cosmological Constant, less 10^{-52} , thence 1.1056, with minus sign

Series representations:

$$\begin{aligned}
 & \left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \\
 & \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
 & \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
 & - \left(\left(1344 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\
 & \quad \left. \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \right. \\
 & \quad \left. \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \binom{\frac{1}{2}}{k} \right) \right) / \left(25 e^{5.934704113458000} \right. \right. \\
 & \quad \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \binom{\frac{1}{2}}{k} \right)^7 \right) \right)
 \end{aligned}$$

$$\begin{aligned}
& \left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
& - \left(\left(1344 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\
& \quad \left. \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right) / \left(25 e^{5.934704113458000} \right. \right. \\
& \quad \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right)^7 \right) \right)
\end{aligned}$$

$$\begin{aligned}
& \left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
& - \left(\left(1344 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - 25 e^{1.978234704486000} \right. \right. \right. \\
& \quad \left. \left. \left. \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) \right) / \left(25 e^{5.934704113458000} \right. \right. \\
& \quad \left. \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \right)
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

And:

$$- [32 \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \right. \\ \left. [42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}] \right)^5$$

Input interpretation:

$$- \left[32 \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \right. \right. \\ \left. \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - \right. \right. \\ \left. \left. \left. 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243} \right) \right) \right)^5$$

Result:

1.651220569...

1.651220569.... result very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Series representations:

$$\begin{aligned}
& - \left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg)^5 = \\
& \left(4385270057140224 \left(-25 + 52 e^{1.978234704486000} \pi^2 - 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)^5 \right) \right) / \\
& \quad \left(9765625 e^{19.78234704486000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)^{35} \right) \right)
\end{aligned}$$

$$\begin{aligned}
& - \left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg)^5 = \\
& \quad \left(4385270057140224 \left(-25 + 52 e^{1.978234704486000} \pi^2 - 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^5 \right) / \\
& \quad \left(9765625 e^{19.78234704486000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^{35} \right)
\end{aligned}$$

$$\begin{aligned}
& - \left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg)^5 = \\
& \quad \left(4385270057140224 \left(-25 + 52 e^{1.978234704486000} \pi^2 - \right. \right. \\
& \quad \left. \left. 25 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^5 \right) / \\
& \quad \left(9765625 e^{19.78234704486000} \right. \\
& \quad \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^{35} \right)
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

We obtain also:

$$-\left[32\left(\frac{e^{-4 \times 0.989117352243}}{\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)^7}\right)}\right)^7 \times \left[42\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)}\right)-13\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}\right)\right]\right)^{\frac{1}{2}}$$

Input interpretation:

$$-\sqrt{\left(32\left(\frac{e^{-4 \times 0.989117352243}}{\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)^7}\right)}\right)^7 \times \left[42\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)}\right)-13\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}\right)\right]\right)^{\frac{1}{2}}}\right)$$

Result:

$$-0.10514303501 \dots i$$

Polar coordinates:

$$r = 1.05143035007 \text{ (radius), } \theta = -90^\circ \text{ (angle)}$$

$$1.05143035007$$

Series representations:

$$\begin{aligned}
 & - \sqrt{\left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\
 & \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
 & \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = -\frac{8}{5} \sqrt{21} \\
 & \sqrt{\left(\left(25 - 52 e^{1.978234704486000} \pi^2 + 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
 & \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right) \right) \right) / \left(e^{3.956469408972000} \right. \\
 & \quad \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right) \right)^7 \right) \Bigg)
 \end{aligned}$$

$$\begin{aligned}
& - \sqrt{\left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = -\frac{8}{5} \sqrt{21} \\
& \sqrt{\left(\left(25 - 52 e^{1.978234704486000} \pi^2 + 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) / \right. \\
& \quad \left. \left(e^{3.956469408972000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \right. \\
& \quad \left. \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^7 \right) \right) \Bigg) \\
& - \sqrt{\left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = \\
& -\frac{8}{5} \sqrt{21} \sqrt{\left(\left(25 - 52 e^{1.978234704486000} \pi^2 + \right. \right. \\
& \quad \left. \left. 25 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) / \right. \\
& \quad \left(e^{3.956469408972000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \right) \Bigg)
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

$$\frac{1}{\left[\frac{32(e^{-4 \cdot 0.989117352243})}{1 + \sqrt{\left(\left(1 + \frac{1}{3} \cdot (4\pi^2)/25 \cdot e^{(2 \cdot 0.989117352243)} \right) \right)^7} \cdot \left(42 \left(1 + \sqrt{\left(\left(1 + \frac{1}{3} \cdot (4\pi^2)/25 \cdot e^{(2 \cdot 0.989117352243)} \right) \right) - \frac{13 \cdot (4\pi^2)}{25} \cdot e^{(2 \cdot 0.989117352243)} \right) \right)} \right]^{1/2}}$$

Input interpretation:

$$- \left(1 / \sqrt{\left(32 \frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \right.} \right.$$

Result:

 $0.95108534763... i$

Polar coordinates:

$$r = 0.95108534763 \text{ (radius)}, \quad \theta = 90^\circ \text{ (angle)}$$

0.95108534763

We know that the primordial fluctuations are consistent with Gaussian purely adiabatic scalar perturbations characterized by a power spectrum with a spectral index $n_s = 0.965 \pm 0.004$, consistent with the predictions of slow-roll, single-field, inflation.

Thence 0.95108534763 is a result very near to the spectral index n_s , to the mesonic Regge slope, to the inflaton value at the end of the inflation 0.9402 and to the value of the following Rogers-Ramanujan continued fraction:

$$\frac{e^{-\frac{\pi}{5}}}{\sqrt{(\varphi-1)\sqrt{5}}-\varphi+1} = 1 - \frac{e^{-\pi}}{1 + \frac{e^{-2\pi}}{1 + \frac{e^{-3\pi}}{1 + \frac{e^{-4\pi}}{1 + \dots}}}} \approx 0.9568666373$$

Series representations:

$$\begin{aligned} & - \left[1 / \left(\sqrt{\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \right. \\ & \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \right. \\ & \quad \left. \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \right) \right) = \\ & - \left[5 / \left(8 \sqrt{21} \sqrt{\left(25 - 52 e^{1.978234704486000} \pi^2 + 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \right. \\ & \quad \left. \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right) \right) \right) / \right. \\ & \quad \left. \left(e^{3.956469408972000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right) \right. \right. \\ & \quad \left. \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right) \right) \right) \right) \right) \right) \end{aligned}$$

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$$1+1/(((4((2*e^{(-0.989117352243/2)})) / (((1+sqrt(((1+1/3*(4\pi^2)/25*e^{(2*0.989117352243)})))))))))) + (-0.034547055658)$$

Input interpretation:

$$1 + \frac{1}{4 \times \frac{2 e^{-0.989117352243/2}}{1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}}}} - 0.034547055658$$

Result:

1.61976215705...

1.61976215705..... result that is a very good approximation to the value of the golden ratio 1.618033988749...

Series representations:

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} - 0.0345470556580000 = \frac{1}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} \\ 0.9654529443420000 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000} \\ \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)_k}$$

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} - 0.0345470556580000 = \frac{1}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} \\ 0.9654529443420000 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000} \\ \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!}}$$

$$\begin{aligned}
& 1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} - 0.0345470556580000 = \\
& \frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{0.9654529443420000 + \frac{e^{0.4945586761215000}}{8} +} \\
& \frac{1}{8} e^{0.4945586761215000} \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

From

Properties of Nilpotent Supergravity

E. Dudas, S. Ferrara, A. Kehagias and A. Sagnotti - arXiv:1507.07842v2 [hep-th] 14 Sep 2015

We have that:

Cosmological inflation with a tiny tensor-to-scalar ratio r , consistently with PLANCK data, may also be described within the present framework, for instance choosing

$$\alpha(\Phi) = i M \left(\Phi + b \Phi e^{ik\Phi} \right). \quad (4.35)$$

This potential bears some similarities with the Kähler moduli inflation of [32] and with the poly-instanton inflation of [33]. One can verify that $\chi = 0$ solves the field equations, and that the potential along the $\chi = 0$ trajectory is now

$$V = \frac{M^2}{3} \left(1 - a \phi e^{-\gamma \phi} \right)^2. \quad (4.36)$$

We analyzing the following equation:

$$V = \frac{M^2}{3} \left(1 - a \phi e^{-\gamma \phi} \right)^2.$$

$$\phi = \varphi - \frac{\sqrt{6}}{k},$$

$$a = \frac{b\gamma}{e} < 0, \quad \gamma = \frac{k}{\sqrt{6}} < 0.$$

We have:

$$(M^2)/3*[1-(b/\text{euler number} * k/\text{sqrt6}) * (\varphi - \text{sqrt6}/k) * \exp(-(k/\text{sqrt6})(\varphi - \text{sqrt6}/k))]^2$$

i.e.

$$V = (M^2)/3*[1-(b/\text{euler number} * k/\text{sqrt6}) * (\varphi - \text{sqrt6}/k) * \exp(-(k/\text{sqrt6})(\varphi - \text{sqrt6}/k))]^2$$

For $k = 2$ and $\varphi = 0.9991104684$, that is the value of the scalar field that is equal to the value of the following Rogers-Ramanujan continued fraction:

$$\frac{\frac{e^{-\frac{\pi}{\sqrt{5}}}}{\sqrt{5}}}{1 + \sqrt[5]{\sqrt{\varphi^5 \sqrt[4]{5^3}} - 1} - \varphi + 1} = 1 - \frac{e^{-\pi\sqrt{5}}}{1 + \frac{e^{-2\pi\sqrt{5}}}{1 + \frac{e^{-3\pi\sqrt{5}}}{1 + \frac{e^{-4\pi\sqrt{5}}}{1 + \dots}}}} \approx 0.9991104684$$

we obtain:

$$V = (M^2)/3*[1-(b/\text{euler number} * 2/\text{sqrt6}) * (0.9991104684 - \text{sqrt6}/2) * \exp(-(2/\text{sqrt6})(0.9991104684 - \text{sqrt6}/2))]^2$$

Input interpretation:

$$V = \frac{M^2}{3} \left(1 - \left(\frac{b}{e} \times \frac{2}{\sqrt{6}} \right) \left(0.9991104684 - \frac{\sqrt{6}}{2} \right) \exp \left(- \frac{2}{\sqrt{6}} \left(0.9991104684 - \frac{\sqrt{6}}{2} \right) \right) \right)^2$$

Result:

$$V = \frac{1}{3} (0.0814845 b + 1)^2 M^2$$

Solutions:

$$b = \frac{225.913 \left(-0.054323 M^2 \pm 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} \quad (M \neq 0)$$

Alternate forms:

$$V = 0.00221324 (b + 12.2723)^2 M^2$$

$$V = 0.00221324 (b^2 M^2 + 24.5445 b M^2 + 150.609 M^2)$$

$$-0.00221324 b^2 M^2 - 0.054323 b M^2 - \frac{M^2}{3} + V = 0$$

Expanded form:

$$V = 0.00221324 b^2 M^2 + 0.054323 b M^2 + \frac{M^2}{3}$$

Alternate form assuming b, M, and V are positive:

$$V = 0.00221324 (b + 12.2723)^2 M^2$$

Alternate form assuming b, M, and V are real:

$$V = 0.00221324 b^2 M^2 + 0.054323 b M^2 + 0.333333 M^2 + 0$$

Derivative:

$$\frac{\partial}{\partial b} \left(\frac{1}{3} (0.0814845 b + 1)^2 M^2 \right) = 0.054323 (0.0814845 b + 1) M^2$$

Implicit derivatives:

$$\frac{\partial b(M, V)}{\partial V} = \frac{154317775011120075}{36961748 (226802245 + 18480874 b) M^2}$$

$$\frac{\partial b(M, V)}{\partial M} = - \frac{\frac{226802245}{18480874} + b}{M}$$

$$\frac{\partial M(b, V)}{\partial V} = \frac{154317775011120075}{2 (226802245 + 18480874 b)^2 M}$$

$$\frac{\partial M(b, V)}{\partial b} = - \frac{18480874 M}{226802245 + 18480874 b}$$

$$\frac{\partial V(b, M)}{\partial M} = \frac{2 (226802245 + 18480874 b)^2 M}{154317775011120075}$$

$$\frac{\partial V(b, M)}{\partial b} = \frac{36961748 (226802245 + 18480874 b) M^2}{154317775011120075}$$

Global minimum:

$$\min\left\{\frac{1}{3}(0.0814845b+1)^2M^2\right\}=0 \text{ at } (b, M) = (-16, 0)$$

Global minima:

$$\min\left\{\frac{1}{3}M^2\left(1-\frac{(b+2)\left(0.9991104684-\frac{\sqrt{6}}{2}\right)\exp\left(-\frac{2\left(0.9991104684-\frac{\sqrt{6}}{2}\right)}{\sqrt{6}}\right)}{e\sqrt{6}}\right)^2\right\}=0$$

for $b = -\frac{226802245}{18480874}$

$$\min\left\{\frac{1}{3}M^2\left(1-\frac{(b+2)\left(0.9991104684-\frac{\sqrt{6}}{2}\right)\exp\left(-\frac{2\left(0.9991104684-\frac{\sqrt{6}}{2}\right)}{\sqrt{6}}\right)}{e\sqrt{6}}\right)^2\right\}=0$$

for $M = 0$

From:

$$b = \frac{225.913\left(-0.054323M^2 \pm 6.58545 \times 10^{-10}\sqrt{M^4}\right)}{M^2} \quad (M \neq 0)$$

we obtain

$$(225.913 (-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4}))/M^2$$

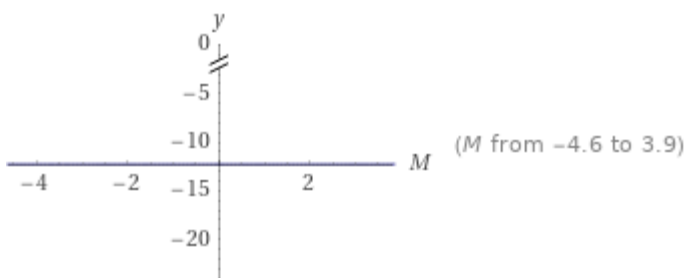
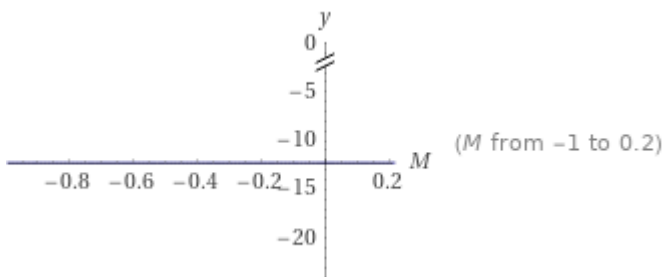
Input interpretation:

$$\frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2}$$

Result:

$$\frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2}$$

Plots:



Alternate form assuming M is real:

$$-12.2723$$

-12.2723 result very near to the black hole entropy value $12.1904 = \ln(196884)$

Alternate forms:

$$- \frac{12.2723 \left(M^2 - 1.21228 \times 10^{-8} \sqrt{M^4} \right)}{M^2}$$

$$\frac{1.48774 \times 10^{-7} \sqrt{M^4} - 12.2723 M^2}{M^2}$$

Expanded form:

$$\frac{1.48774 \times 10^{-7} \sqrt{M^4}}{M^2} - 12.2723$$

Property as a function:**Parity**

even

Series expansion at $M = 0$:

$$\left(\frac{1.48774 \times 10^{-7} \sqrt{M^4}}{M^2} - 12.2723 \right) + O(M^6)$$

(generalized Puiseux series)

Series expansion at $M = \infty$:

$$-12.2723$$

Derivative:

$$\frac{d}{dM} \left(\frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2} \right) = \frac{3.55271 \times 10^{-15}}{M}$$

Indefinite integral:

$$\int \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} dM =$$

$$\frac{1.48774 \times 10^{-7} \sqrt{M^4}}{M} - 12.2723 M + \text{constant}$$

Global maximum:

$$\max \left\{ \frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2} \right\} =$$

$$-\frac{140119826723990341497649}{11417594849251000000000} \text{ at } M = -1$$

Global minimum:

$$\min \left\{ \frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2} \right\} =$$

$$-\frac{140119826723990341497649}{11417594849251000000000} \text{ at } M = -1$$

Limit:

$$\lim_{M \rightarrow \pm\infty} \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} = -12.2723$$

Definite integral after subtraction of diverging parts:

$$\int_0^\infty \left(\frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} - -12.2723 \right) dM = 0$$

From b that is equal to

$$\frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2}$$

From:

$$V = \frac{1}{3} (0.0814845 b + 1)^2 M^2$$

we obtain:

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4}))/M^2) + 1)^2 M^2$$

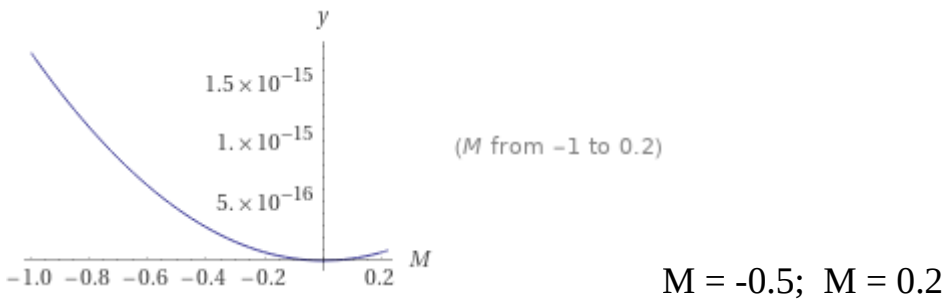
Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

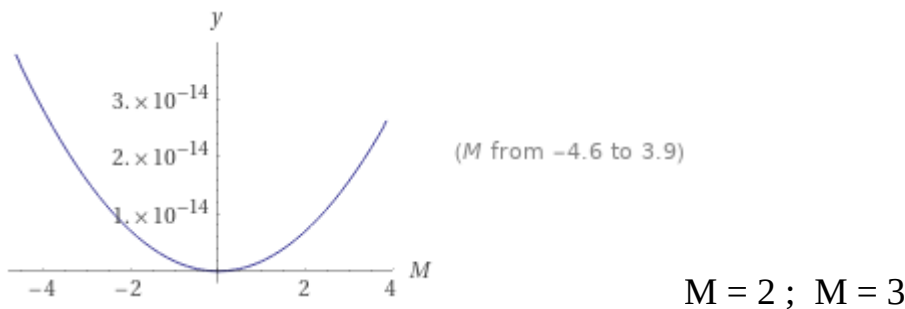
Result:

0

Plots: (possible mathematical connection with an open string)



(possible mathematical connection with an open string)



Root:

$$M = 0$$

Property as a function:

Parity

even

Series expansion at $M = 0$:

$$O(M^{62194})$$

(Taylor series)

Series expansion at $M = \infty$:

$$1.75541 \times 10^{-15} M^2 + O\left(\left(\frac{1}{M}\right)^{62194}\right)$$

(Taylor series)

Definite integral after subtraction of diverging parts:

$$\int_0^\infty \left(\frac{1}{3} M^2 \left(1 + \frac{18.4084 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} \right)^2 - 1.75541 \times 10^{-15} M^2 \right) dM = 0$$

For $M = -0.5$, we obtain:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 (-0.5)^2 + 6.58545 \times 10^{-10} \sqrt{(-0.5)^4}))/(-0.5)^2 + 1)^2 * (-0.5^2)$$

Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 (-0.5)^2 + 6.58545 \times 10^{-10} \sqrt{(-0.5)^4} \right)}{(-0.5)^2} + 1 \right)^2$$

(-0.5^2)

For M = 3:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 3^2 + 6.58545 \times 10^{-10} \sqrt{3^4}))/3^2) + 1)^2 3^2$$

Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 \times 3^2 + 6.58545 \times 10^{-10} \sqrt{3^4} \right)}{3^2} + 1 \right)^2 \times 3^2$$

Result:

$$1.579864841810872363256294820161116875 \times 10^{-14}$$

$$1.57986484181 \times 10^{-14}$$

For M = 2:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 2^2 + 6.58545 \times 10^{-10} \sqrt{2^4}))/2^2) + 1)^2 2^2$$

Result:

$$5.9776991059... \times 10^{-8}$$

$5.9776991059 \times 10^{-8}$ result very near to the Planck's electric flow 5.975498×10^{-8} that is equal to the following formula:

$$\phi_P^E = \mathbf{E}_P l_P^2 = \phi_P l_P = \sqrt{\frac{\hbar c}{\epsilon_0}}$$

We note that:

$$\frac{1}{55} * (((1/[(7.021621519 \times 10^{-15} + 1.57986484181 \times 10^{-14} + 7.021621519 \times 10^{-17} - 4.38851344947 \times 10^{-16})]))^{1/7} - ((\log^{5/8}(2))/(2 \cdot 2^{1/8} \cdot 3^{1/4} \cdot e \log^{3/2}(3))))))$$

Input interpretation:

$$\frac{1}{55} \left(\left(1 / (7.021621519 \times 10^{-15} + 1.57986484181 \times 10^{-14} + 7.021621519 \times 10^{-17} - 4.38851344947 \times 10^{-16}) \right)^{(1/7)} - \frac{\log^{5/8}(2)}{2 \sqrt[8]{2} \sqrt[4]{3} e \log^{3/2}(3)} \right)$$

$\log(x)$ is the natural logarithm

Result:

$$1.6181818182...$$

$1.6181818182...$ result that is a very good approximation to the value of the golden ratio $1.618033988749...$

From the Planck units:

Planck Length

$$l_P = \sqrt{\frac{4\pi\hbar G}{c^3}}$$

$5.729475 \cdot 10^{-35}$ Lorentz-Heaviside value

Planck's Electric field strength

$$\mathbf{E}_P = \frac{F_P}{q_P} = \sqrt{\frac{c^7}{16\pi^2 \epsilon_0 \hbar G^2}}$$

$1.820306 \cdot 10^{61}$ V*m Lorentz-Heaviside value

Planck's Electric flux

$$\phi_P^E = \mathbf{E}_P l_P^2 = \phi_P l_P = \sqrt{\frac{\hbar c}{\epsilon_0}}$$

$5.975498 \cdot 10^{-8}$ V*m Lorentz-Heaviside value

Planck's Electric potential

$$\phi_P = V_P = \frac{E_P}{q_P} = \sqrt{\frac{c^4}{4\pi\epsilon_0 G}}$$

$1.042940 \cdot 10^{27}$ V Lorentz-Heaviside value

Relationship between Planck's Electric Flux and Planck's Electric Potential

$$E_P * I_P = (1.820306 * 10^{61}) * 5.729475 * 10^{-35}$$

Input interpretation:

$$\frac{(1.820306 \times 10^{61}) \times 5.729475}{10^{35}}$$

Result:

1 042 939 771 935 000 000 000 000 000

Scientific notation:

$$1.042939771935 \times 10^{27}$$

$$1.042939771935 * 10^{27} \approx 1.042940 * 10^{27}$$

Or:

$$E_P * I_P^2 / I_P = (5.975498 * 10^{-8}) * 1 / (5.729475 * 10^{-35})$$

Input interpretation:

$$5.975498 \times 10^{-8} \times \frac{1}{\frac{5.729475}{10^{35}}}$$

Result:

1.04293988541707573556041347592929544155441816222254220500133... × 10²⁷

$$1.042939885417 * 10^{27} \approx 1.042940 * 10^{27}$$

Observations

We note that, from the number 8, we obtain as follows:

$$8^2$$

$$64$$

$$8^2 \times 2 \times 8$$

$$1024$$

$$8^4 = 8^2 \times 2^6$$

True

$$8^4 = 4096$$

$$8^2 \times 2^6 = 4096$$

$$2^{13} = 2 \times 8^4$$

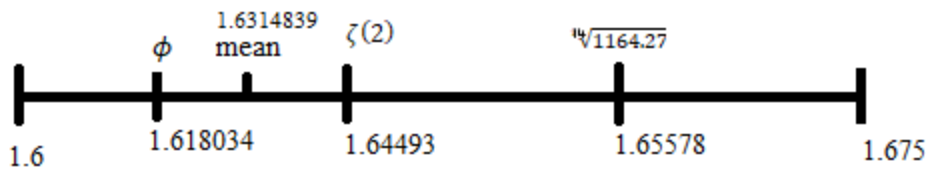
True

$$2^{13} = 8192$$

$$2 \times 8^4 = 8192$$

We notice how from the numbers 8 and 2 we get 64, 1024, 4096 and 8192, and that 8 is the fundamental number. In fact $8^2 = 64$, $8^3 = 512$, $8^4 = 4096$. We define it "fundamental number", since 8 is a Fibonacci number, which by rule, divided by the previous one, which is 5, gives 1.6, a value that tends to the golden ratio, as for all numbers in the Fibonacci sequence

“Golden” Range



Finally we note how $8^2 = 64$, multiplied by 27, to which we add 1, is equal to 1729, the so-called "Hardy-Ramanujan number". Then taking the 15th root of 1729, we obtain a value close to $\zeta(2)$ that 1.6438 ..., which, in turn, is included in the range of what we call "golden numbers"

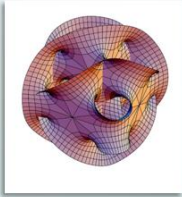
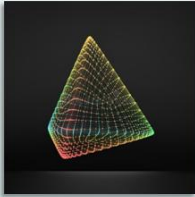
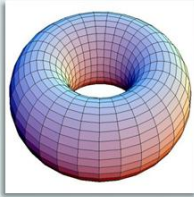
Furthermore for all the results very near to 1728 or 1729, adding $64 = 8^2$, one obtain values about equal to 1792 or 1793. These are values almost equal to the Planck multipole spectrum frequency 1792.35 and to the hypothetical Gluino mass

Appendix

Outlook

Remarkably rich (apparently **UNIQUE**) framework

BUT :



Why a given **“shape” of the extra dimensions** ?
[**CRUCIAL**, it determines the predictions for α , ...]

A. Sagnotti – AstronomiAmo, 23.4.202021

From: A. Sagnotti – AstronomiAmo, 23.04.2020

In the above figure, it is said that: “why a given shape of the extra dimensions? Crucial, it determines the predictions for α ”.

We propose that whatever shape the compactified dimensions are, their geometry must be based on the values of the golden ratio and $\zeta(2)$, (the latter connected to 1728 or 1729, whose fifteenth root provides an excellent approximation to the above mentioned value) which are recurrent as solutions of the equations that we are going to develop. It is important to specify that the initial conditions are **always** values belonging to a fundamental chapter of the work of S. Ramanujan "Modular equations and Approximations to Pi" (see references). These values are some multiples of 8 (64 and 4096), 276, which added to 4096, is equal to 4372, and finally $e^{\pi\sqrt{22}}$

We have, in certain cases, the following connections:

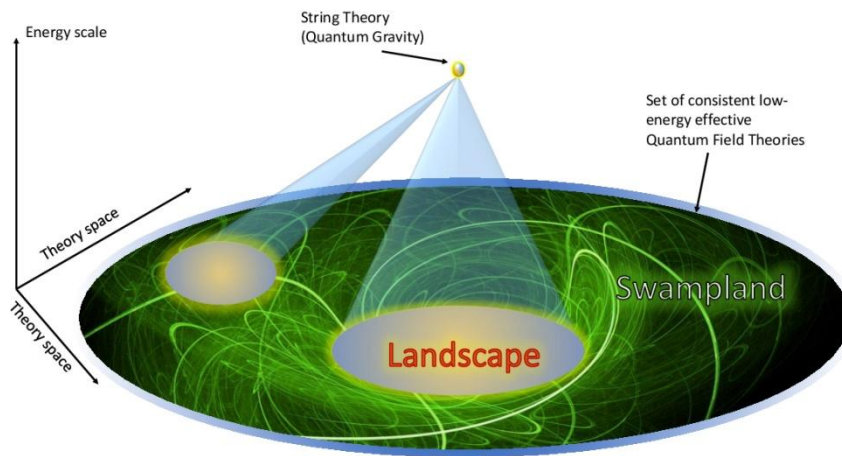


Fig. 1

The String Theory “Landscape”

- Graph axes show only 2 out of hundreds of parameters (“moduli”) that determine the exact Calabi-Yau manifolds and how strings wrap around them

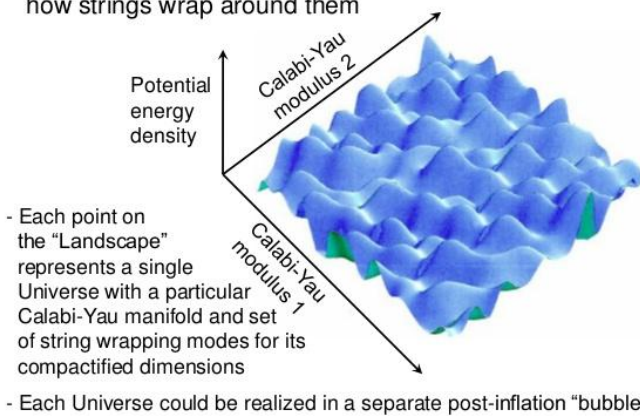


Fig. 2

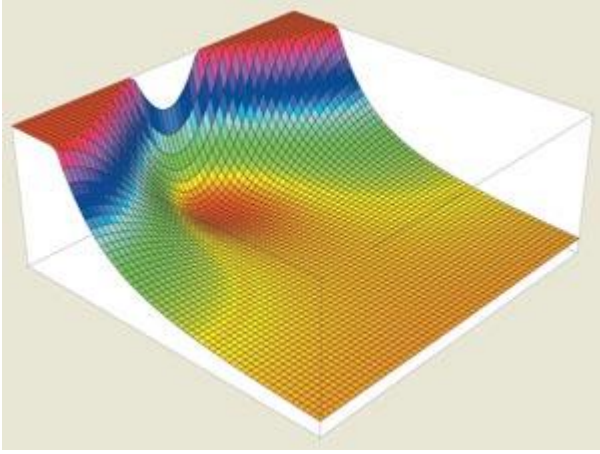


Fig. 3

Stringscape - a small part of the string-theory landscape showing the new de Sitter solution as a local minimum of the energy (vertical axis). The global minimum occurs at the infinite size of the extra dimensions on the extreme right of the figure.

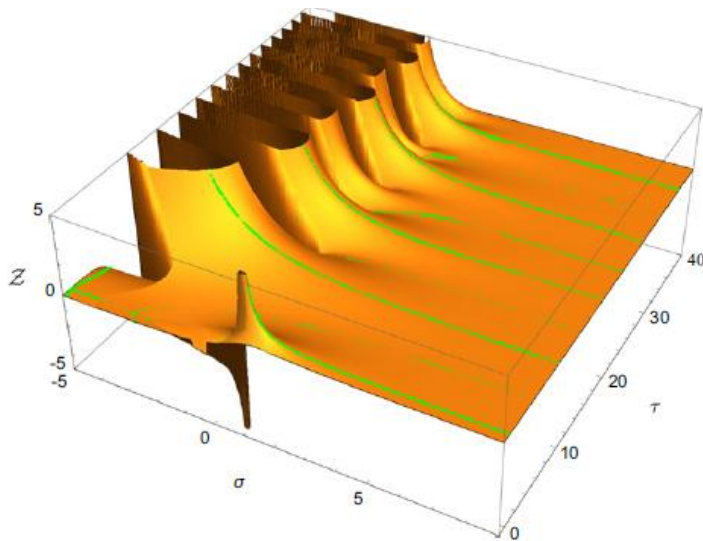


Figure 2. Lines in the complex plane where the Riemann zeta function ζ is real (green) depicted on a relief representing the positive absolute value of ζ for arguments $s \equiv \sigma + i\tau$ where the real part of ζ is positive, and the negative absolute value of ζ where the real part of ζ is negative. This representation brings out most clearly that the lines of constant phase corresponding to phases of integer multiples of 2π run down the hills on the left-hand side, turn around on the right and terminate in the non-trivial zeros. This pattern repeats itself infinitely many times. The points of arrival and departure on the right-hand side of the picture are equally spaced and given by equation (11).

Fig. 4

With regard the Fig. 4 the points of arrival and departure on the right-hand side of the picture are equally spaced and given by the following equation:

$$\tau'_k \equiv k \frac{\pi}{\ln 2},$$

with $k = \dots, -2, -1, 0, 1, 2, \dots$

we obtain:

$$2\pi/(\ln(2))$$

Input:

$$2 \times \frac{\pi}{\log(2)}$$

Exact result:

$$\frac{2\pi}{\log(2)}$$

Decimal approximation:

9.0647202836543876192553658914333336203437229354475911683720330958
...

9.06472028365....

Alternative representations:

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\log_e(2)}$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\log(a) \log_a(2)}$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{2 \coth^{-1}(3)}$$

Series representations:

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{2i\pi \left\lfloor \frac{\arg(2-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k}} \quad \text{for } x < 0$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\log(z_0) + \left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \left(\log\left(\frac{1}{z_0}\right) + \log(z_0) \right) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k}}$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{2i\pi \left\lfloor \frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right\rfloor + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k}}$$

Integral representations:

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\int_1^2 \frac{1}{t} dt}$$

$$\frac{2\pi}{\log(2)} = \frac{4i\pi^2}{\int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{\Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds} \quad \text{for } -1 < \gamma < 0$$

From which:

$$(2\pi/(\ln(2))) \cdot (1/12 \pi \log(2))$$

Input:

$$\left(2 \times \frac{\pi}{\log(2)}\right) \left(\frac{1}{12} \pi \log(2)\right)$$

$\log(x)$ is the natural logarithm

Exact result:

$$\frac{\pi^2}{6}$$

Decimal approximation:

1.6449340668482264364724151666460251892189499012067984377355582293
...

$$1.6449340668\dots = \zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$$

From “**Ramanujan modular equations and approximation to π** ”, we know that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

$$64[(1+\sqrt{2})^{12}+(1-\sqrt{2})^{12}] + 24$$

$$64\left((1+\sqrt{2})^{12}+(1-\sqrt{2})^{12}\right)+24$$

$$2508952$$

$$2058952$$

$$64 \left((1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12} \right) + 24 = 2508952$$

$$(((2508952 / (((1+\text{sqr}2)^{12}+(1-\text{sqr}2)^{12})+ 24))))^{2+5}$$

Input

$$\left(\frac{2508952}{\left((1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12} \right) + 24} \right)^2 + 5$$

Result

$$\frac{1575633383421}{384669769}$$

Decimal approximation

4096.0676153914242218498849593766751137649187087535334756186676057

...

4096.06761539....

And:

$$27(((2508952 / (((1+\text{sqr}2)^{12}+(1-\text{sqr}2)^{12})+ 24)))) + 2^{(1/8)} + 1$$

Input

$$27 \times \frac{2508952}{\left((1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12} \right) + 24} + \sqrt[8]{2} + 1$$

Result

$$1 + \sqrt[8]{2} + \frac{67741704}{24 + (1 - \sqrt{2})^{12} + (1 + \sqrt{2})^{12}}$$

Decimal approximation

1729.0497694468344311665745729868676268593271747523951632315088104

...

1729.049769446....

Alternate forms

$$\frac{33\,890\,465 + 19\,613 \sqrt[8]{2}}{19\,613}$$

$$\frac{33\,890\,465}{19\,613} + \sqrt[8]{2}$$

$$\sqrt[8]{2} + \frac{33\,890\,465}{19\,613}$$

From which:

$$(((27(((2508952 / (((1+\sqrt{2})^{12}+(1-\sqrt{2})^{12})+ 24)))) + 2^{(1/8)} + 1)))^{1/15}$$

Input

$$\sqrt[15]{27 \times \frac{2508952}{((1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}) + 24} + \sqrt[8]{2} + 1}$$

Exact result

$$\sqrt[15]{1 + \sqrt[8]{2} + \frac{67\,741\,704}{24 + (1 - \sqrt{2})^{12} + (1 + \sqrt{2})^{12}}}$$

Decimal approximation

1.6438183831993034781967893292376659600825395477910138131727182622

...

1.6438183831993....

Alternate forms

$$\sqrt[15]{\frac{33890465}{19613} + \sqrt[8]{2}}$$

$$\sqrt[15]{\frac{33890465 + 19613\sqrt[8]{2}}{19613}}$$

Indeed, we have:

$$4096 = 64^2; (27\sqrt[4]{4096})+1 = 10^3 + 9^3 = 12^3 + 1 = 1729;$$

$$\sqrt[15]{1729} = 1.6438152287$$

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